Morgan Stanley | RESEARCH

FOUNDATION

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South Korea | Asia Pacific

Multipolar World Multipliers

We initiate on three defense stocks that we believe are well positioned for a multipolar world. There are growing concerns that the positives are priced in, but we expect further upside as we see a super-cycle ahead for Korea's defense industry.

Key Takeaways

- We prefer front-end value chain owners vs. intermediary suppliers. Our preferred pick within Industrials is Hanwha Aerospace.
- We are also OW Korea Aerospace Industry given our view that KF-21 and FA-50 can satiate growing demand for cost-effective air power. EW on LIG Nex1.

Multipolar World
A Morgan Stanley Research Key Theme of 2025

A breather or tapped out? The recent run was fuelled by new overseas orders and rising defense budgets in multiple regions. This pushed up Korean defense stocks' valuations to 25-40x forward P/E and 4-6x forward P/B, but we do not think this is excessive vs global peers if we assume the Korean players are poised to take market share.

More to come: We expect Korea to become a top 5 global military equipment/ munitions manufacturing powerhouse within the next 5 years. We believe the market is underestimating demand for conventional military equipment, which is Korea's forte, and also the country's position as an advancer of military technology. Key advantages held by Korean players are (1) geopolitical maneuverability, (2) edge in readiness & sustainment costs, (3) compatibility with existing systems (i.e. US/ NATO-based), (4) flexibility on capacity, and (5) strong government support.

We see a super-cycle ahead: Even if conflicts subside, we believe export demand for Korean military equipment will remain strong in key markets due to (1) restocking demand, (2) reinforcement – multipolar world pushing multiple countries/regions, including Korea, to boost defense budgets, and (3) replacement – upgrading of existing equipment/systems + advancement of equipment. Korea's key markets are Eastern Europe, MENA, and ASEAN, but we see potential in new markets such as Canada, LATAM, India, and traditional strongholds such as US and Western Europe.

Korea to push the envelope of military technology due to idiosyncratic factors including the continued threat from North Korea, a multipolar world increasing the need for self-sufficiency, and a shortage of military manpower owing to demographics. We see Korea's military forces undergoing a major transformation, with high tech at the forefront supported by Korea's strong IT, materials, and power generation technology.

Technical factors also look positive including (1) ESG rule loosening for defense stocks, (2) limited alternative stocks within EM, and (3) capital market reform.

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S. KOREA INDUSTRIALS

Asia Pacific
Industry View **Attractive**

Exhibit 1 : Initiating on three South Korea defense stocks

Company Name	Hanwha Aerospace	Korea Aerospace Industries	LIX Nex1
Ticker	012450.KS	047810.KS	079550.KS
Rating	**Overweight**	**Overweight**	**Equal-weight**
Trading Currency	KRW	KRW	KRW
Current Price	941,000	92,600	584,000
Price Target	**1,250,000**	**110,000**	**600,000**
Upside/Downside (%)	**32.8%**	**18.8%**	**2.7%**

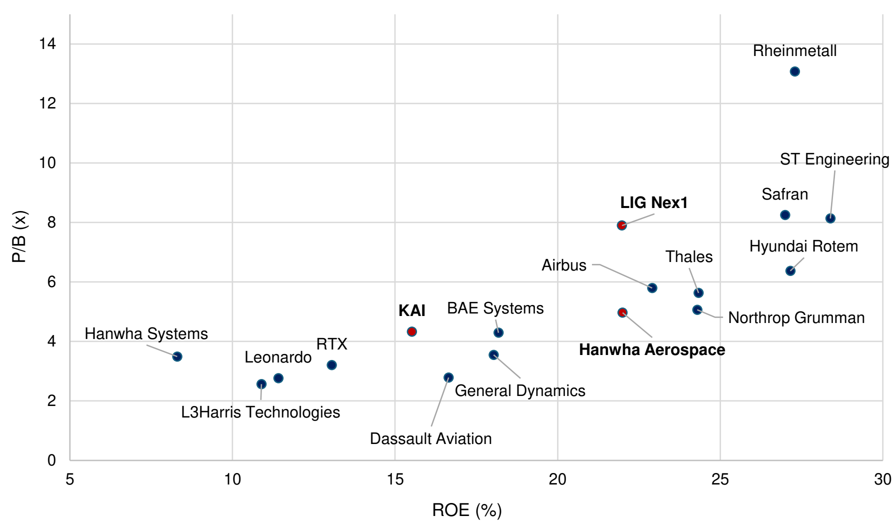
Source: Morgan Stanley Research. Note: Share prices are as of the market close on July 24, 2025.

The Story in Charts

Exhibit 2: Share prices of Korean defense stocks have surged as new orders have picked up and on plans by EU/NATO and other countries to boost defense budgets

Source: Datastream, Morgan Stanley Research.

Exhibit 3: Korean defense stocks are in the middle of the pack in terms of 2025e P/B-ROE. We still see this as attractive, as we expect Korean players to gain share



Source: FactSet, Morgan Stanley Research. Note: Data as of July 24, 2025.

Exhibit 4: When we plot Global Military Expenditure growth vs. the Global Risk Index, we find that generally there is positive linkage. Continued conflicts and trade tensions are leading to a more severe multipolar world as illustrated by the geopolitical risk (GPR) index, which we believe is leading to countries becoming more focused on reinforcing their defense capabilities

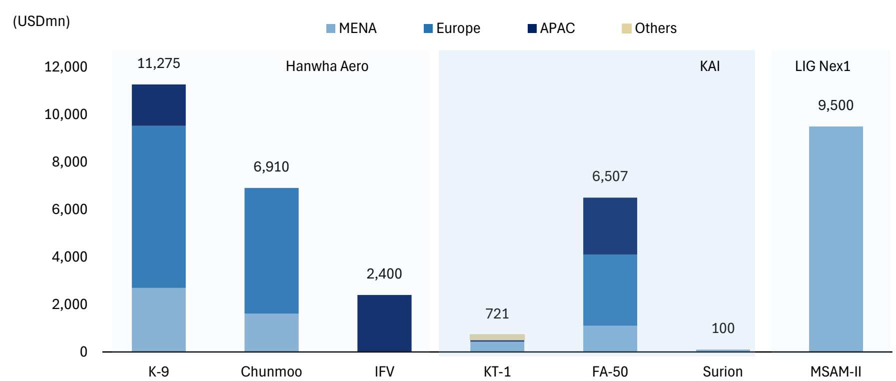
Source: GPR, Stockholm International Peace Research Institute (SIPRI), Morgan Stanley Research.

Exhibit 5: Korea is reducing its dependence on defense imports while growing exports, with 2025 showing a surge in export orders

	Share of global arms exports/imports (%)	
	2020-24	2015-19
Share of global arms exports	**2.2%**	**2.1%**
Key export countries		
#1 export destination	Poland (46%)	UK (17%)
#2	Philippines (14%)	Iraq (14%)
#3	India (7%)	Indonesia (13%
Share of global arms imports	**2.6%**	**3.4%**
Main suppliers		
#1 supplier	US (86%)	US (55%)
#2	Germany (9.7%)	Germany (30%)
#3	UK (2.6%)	Spain (7.8%

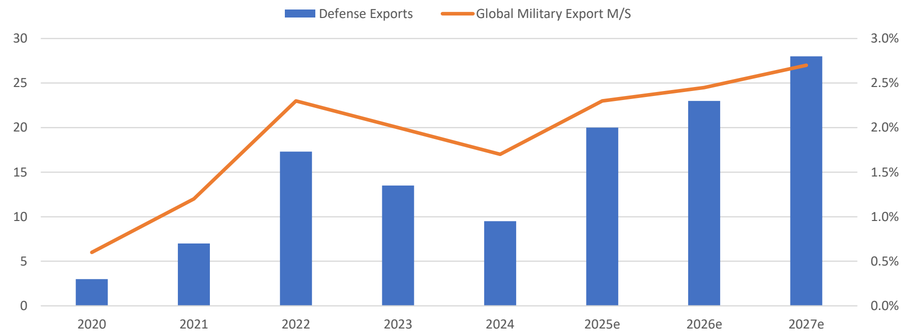
Source: SIPRI, Morgan Stanley Research.

Exhibit 6: Korean defense contractors sell a wide range of military equipment overseas, with artillery (K-9 howitzers), missile systems (MSAM-II, Chunmoo) and combat aircraft (FA-50) being key export items



Source: Company reports, Morgan Stanley Research. Note: Data based on major overseas contracts during 2012-2024.

Exhibit 7: We forecast an extended super-cycle owing to both increased military exports and a continued rise in domestic defense spending



Source: Ministry of Defense, Defense Procurement Agency, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Exhibit 8: S Korea Industrials: Defense – order of preference: Our preferred pick within S Korea Industrials is Hanwha Aerospace, Asia's leading vertically integrated defense player, followed by Korea Aerospace Industries (OW) as we expect the KF-21 to be a competitive combat aircraft. We are EW LIG Nex1, as the company is strong in the missile systems space, but waiting for new orders

	012450.KS Hanwha Aerospace	047810.KS KAI	079550.KS LIG Nex1
Rating	Overweight	Overweight	Equal-weight
TP (W)	**1,250,000**	**110,000**	**600,000**
CP (W)	910,000	92,200	593,000
Upside/Downside (%)	*37.4%*	*19.3%*	*1.2%*
Base Case Top Down Assumption	Global military expenditure to GDP ratio reaches 3% by 2030 with Korea's export market share being 2.5%		
Base Case	- Assume a 9% new order CAGR, 2024-33e, with a 16% sales CAGR, 2024-33e, from core businesses (land systems and aerospace).	- Assume a 12.8% new order CAGR, 2024-33e, with 11.6% sales CAGR, 2024-33e.	- Assume a 10.5% new order CAGR, 2024-33e, with a 11.59% sales CAGR, 2024-33e.
Bull Case Top Down Assumption	Global military expenditure to GDP ratio reaches 3.3% by 2030 with Korea's export market share being 2.7%		
Bull Case	**1,400,000** - Assumea 9% new order CAGR, 2024-33e, with a 16% sales CAGR 2024-33e, from core businesses (land systems and aerospace).	**120,000** - Assume a 14.2% new order CAGR, 2024-33e, with a 12.1% sales CAGR, 2024-33e.	**675,000** - Assume an 11.57% new order CAGR, 2024-33e, with a 12.47% sales CAGR in 2024-33e.
Bear Case Top Down Assumption	Global military expenditure to GDP ratio reaches 2.7% by 2030 with Korea's export market share being 2.2%		
Bear Case	**755,000** - Assume a 8% new order CAGR, 2024-33e, with a 14% sales CAGR 2024-33e, from core businesses (land systems and aerospace).	**75,000** - Assume a 13% new order CAGR, 2024-33e, with 10.3% sales CAGR, 2024-33e.	**450,000** - Assumea 9.8% new order CAGR, 2024-33e, with 9.76% sales CAGR, 2024-33e.
Key Products	- K9 self propelled howitzer - K239 Chunmoo MLRS - Redback IFV	- T-50 trainer jet - FA-50 light combat jet - KUH Surion Helicopter - KF-21 4.5th generation combat jet	- Cheongung II (M-SAM II) - Poniard low cost guided rocket - Haegoong ship to Air Anti-Missile -L-SAM
Key Export Countries	Poland, Romania, India, Saudi Arabia, Vietnam, Malaysia, Norwaym UAE	Poland, Egypt, Peru, Malaysia, Senegal, UAE, Iraq, Indonesia, Thailand	Iraq, UAE, Saudi Arabia
Main Factory Location	Changwon, Yeosu	Sacheon	Yongin, Gimcheon
Mkt.cap (Wtn)	48.5	9.0	12.8
P/E	27.1	37.2	41.8
P/B	5.9	4.8	8.8
EV/EBITDA	15.7	21.8	27.5
Backlog to sales (25E, %)	32%	21%	22%
Export % of total sales (25E)	70%	33%	24%

*Note: Hanwha Aerospace export, backlog only include land system division

Source: FactSet, Morgan Stanley Research. Note: Share prices in this report are as of the market close on July 23, 2025, unless otherwise indicated.

Exhibit 9: Morgan Stanley Defense Coverage: Global Comps Table

Company	Ticker	Currency	Share Price (LCY)	Market Cap. (LCY bn)	Market Cap. (US$ bn)	M. Cap Share %	Rating	Price Target (LCY)	Net Income (LCY bn) 2024	Net Income (LCY bn) 2025e	Net Income (LCY bn) 2026e	ROE 2024	ROE 2025e	ROE 2026e
Hanwha Aerospace	012450.KS	KRW	910,000	46,922.7	34.0	68%	OW	1,250,000	2,277	1,481	2,276	46%	15%	20%
Korea Aerospace Industries	047810.KS	KRW	92,200	8,987.2	6.5	13%	OW	110,000	172	302	429	10%	16%	19%
LIG Nex1	079550.KS	KRW	593,000	13,046.0	9.5	19%	EW	600,000	222	285	402	18%	21%	23%
North American Peers														
Amentum Holdings	AMTM.N	USD	25	6.2	6.2	1%	UW	19	0.5	0.5	0.6	NA	12%	13%
CAE	CAE.TO	CAD	39	12.5	9.1	2%	EW	42	-0.3	0.4	0.4	-7%	10%	9%
Curtiss-Wright	CW.N	USD	479	18.1	18.1	3%	OW	550	0.4	0.5	0.5	17%	19%	18%
General Dynamics	GD.N	USD	314	85.4	85.4	14%	EW	328	3.8	4.1	4.6	18%	19%	19%
L3Harris Technologies	LHX.N	USD	273	51.2	51.2	9%	EW	250	2.5	2.0	2.3	13%	10%	12%
Leonardo DRS	DRS.O	USD	47	12.8	12.8	2%	EW	44	0.2	0.3	0.3	9%	11%	12%
Lockheed Martin	LMT.N	USD	421	98.5	98.5	17%	OW	530	5.3	5.1	6.9	78%	81%	135%
MDA	MDA.TO	CAD	41	5.4	3.9	1%	EW	33	0.1	0.1	0.2	7%	12%	13%
Northrop Grumman	NOC.N	USD	568	81.6	81.6	14%	OW	625	3.8	3.7	4.2	26%	24%	27%
RTX	RTX.N	USD	155	209.8	209.8	35%	OW	180	7.7	8.0	8.9	13%	13%	14%
Textron	TXT.N	USD	81	14.7	14.7	2%	EW	86	1.0	1.1	1.2	15%	16%	16%
V2X	VVX.N	USD	48	1.6	1.6	0%	UW	51	0.1	0.1	0.2	14%	14%	16%
European Peers														
Airbus	AIR.PA	EUR	183	144.1	167.5	28%	OW	223	4.0	5.6	6.1	23%	28%	27%
BAE Systems	BAES.L	GBp	1,850	55.7	74.7	12%	OW	2,158	2.1	2.2	2.4	20%	19%	19%
Dassault Aviation	AM.PA	EUR	267	21.0	24.4	4%	EW	310	1.1	1.1	1.3	18%	17%	19%
Hensoldt	HAGG.DE	EUR	97	10.2	11.8	2%	UW	78	0.2	0.2	0.2	23%	21%	24%
Leonardo	LDOF.MI	EUR	48	27.4	31.9	5%	OW	63	0.9	1.0	1.2	12%	11%	12%
Melrose Industries	MRON.L	GBp	520	6.8	9.2	2%	EW	470	0.4	0.4	0.5	10%	14%	18%
Rheinmetall	RHMG.DE	EUR	1,746	75.8	88.1	15%	OW	2,200	0.9	1.2	1.9	28%	30%	39%
Safran	SAF.PA	EUR	279	117.6	136.7	23%	EW	235	3.1	3.7	4.2	27%	36%	38%
Thales	TCFP.PA	EUR	238	49.0	57.0	9%	EW	250	1.9	1.8	2.2	28%	24%	27%
Other Asian Peers														
Bharat Electronics	BAJE.NS	INR	398	2,911.1	33.7	18%	OW	418	39.8	53.2	56.9	29%	33%	29%
Hindustan Aeronautics	HIAE.NS	INR	4,681	3,130.3	36.2	19%	EW	5,092	64.5	83.8	88.4	27%	29%	25%
Mitsubishi Heavy Industries	7011.T	JPY	3,594	12,079.1	82.2	44%	OW	3,000	297.0	354.0	432.0	13%	14%	16%
Kawasaki Heavy Industries	7012.T	JPY	10,935	1,831.7	12.5	7%	UW	5,100	83.3	77.8	93.7	13%	11%	13%
ST Engineering	STEG.SI	SGD	9	27.7	21.5	12%	OW	9	0.7	0.8	0.9	29%	28%	30%

Company	Ticker	Currency	P/E 2024	P/E 2025e	P/E 2026e	P/B 2024	P/B 2025e	P/B 2026e	EV/EBITDA 2024	EV/EBITDA 2025e	EV/EBITDA 2026e	Div. Yield 2024	Div. Yield 2025e	Div. Yield 2026e
Hanwha Aerospace	012450.KS	KRW	7.2	31.2	20.3	3.27	4.78	3.99	14.4	15.8	12.6	1.1%	0.4%	0.5%
Korea Aerospace Industries	047810.KS	KRW	31.1	29.8	20.9	3.14	4.73	3.97	18.3	23.1	16.7	0.9%	0.7%	0.8%
LIG Nex1	079550.KS	KRW	21.7	45.3	32.2	3.97	9.36	7.54	15.1	28.7	21.6	1.1%	0.5%	0.5%
North American Peers														
Amentum Holdings	AMTM.N	USD	12.8	11.8	10.6	1.40	1.37	1.31	6.9	6.4	6.2	0.0%	0.0%	0.0%
CAE	CAE.TO	CAD	NM	27.3	28.3	2.11	2.32	2.36	64.9	13.1	13.0	0.0%	0.0%	0.0%
Curtiss-Wright	CW.N	USD	33.6	38.6	34.9	5.49	6.44	5.65	21.8	23.7	20.9	0.2%	0.2%	0.2%
General Dynamics	GD.N	USD	19.3	20.7	18.3	3.27	3.53	3.32	13.9	14.8	13.0	2.2%	1.9%	2.1%
L3Harris Technologies	LHX.N	USD	16.0	26.1	22.2	2.04	2.69	2.59	11.3	16.6	14.6	2.2%	1.8%	1.8%
Leonardo DRS	DRS.O	USD	40.6	47.5	39.9	3.40	4.81	4.44	20.9	27.7	23.9	0.0%	0.8%	0.8%
Lockheed Martin	LMT.N	USD	21.8	19.2	14.1	18.13	19.27	16.53	16.0	12.9	10.3	2.7%	3.3%	3.6%
MDA	MDA.TO	CAD	46.9	35.9	29.4	3.32	4.05	3.57	19.6	18.0	15.3	NA	NA	NA
Northrop Grumman	NOC.N	USD	18.0	22.4	19.2	4.45	5.31	5.04	13.8	15.3	13.4	1.7%	1.6%	1.7%
RTX	RTX.N	USD	20.2	26.3	23.2	2.58	3.31	3.24	13.5	16.4	15.1	2.1%	1.7%	1.9%
Textron	TXT.N	USD	14.0	13.2	11.9	1.96	1.99	1.80	9.4	8.7	7.7	0.1%	0.1%	0.1%
V2X	VVX.N	USD	11.0	10.5	8.9	1.49	1.43	1.28	7.0	6.9	6.2	0.0%	0.0%	0.0%
European Peers														
Airbus	AIR.PA	EUR	30.4	25.8	23.7	6.18	6.44	5.50	13.5	12.0	10.7	2.0%	1.6%	1.7%
BAE Systems	BAES.L	GBp	16.7	25.1	23.2	2.97	4.53	4.27	9.5	13.7	13.0	2.9%	2.0%	2.2%
Dassault Aviation	AM.PA	EUR	14.7	19.0	15.8	2.45	3.02	2.70	5.7	7.3	5.7	2.4%	1.9%	2.2%
Hensoldt	HAGG.DE	EUR	21.4	61.4	49.3	4.59	11.83	10.14	11.2	24.7	21.5	1.4%	0.6%	0.7%
Leonardo	LDOF.MI	EUR	16.0	27.1	23.0	1.66	2.76	2.56	9.7	13.6	12.0	1.1%	1.1%	1.3%
Melrose Industries	MRON.L	GBp	20.9	16.6	13.1	2.53	2.46	2.31	13.1	10.5	8.7	1.1%	1.4%	1.7%
Rheinmetall	RHMG.DE	EUR	30.5	62.3	39.3	7.08	15.39	11.71	18.4	31.4	20.8	0.9%	0.5%	0.6%
Safran	SAF.PA	EUR	29.1	31.6	27.2	8.78	10.53	9.24	16.1	17.9	15.6	1.4%	1.3%	1.5%
Thales	TCFP.PA	EUR	15.2	26.8	22.8	3.80	6.06	5.50	10.4	15.7	14.1	2.7%	1.5%	1.8%
Other Asian Peers														
Bharat Electronics	BAJE.NS	INR	37.0	41.4	51.1	9.04	11.03	12.45	26.9	30.7	37.8	1.1%	0.8%	0.8%
Hindustan Aeronautics	HIAE.NS	INR	34.6	33.3	35.4	7.67	7.99	7.69	25.0	27.2	28.2	1.0%	1.0%	1.0%
Mitsubishi Heavy Industries	7011.T	JPY	28.6	34.1	28.0	3.46	4.48	4.03	16.3	20.4	17.2	1.1%	0.9%	1.1%
Kawasaki Heavy Industries	7012.T	JPY	18.0	23.5	19.5	2.15	2.45	2.25	10.9	12.0	10.9	1.6%	1.3%	1.5%
ST Engineering	STEG.SI	SGD	20.7	36.6	32.4	5.45	9.68	8.89	13.1	19.7	18.2	3.6%	2.0%	2.2%

Source: FactSet, Morgan Stanley Research. e = Morgan Stanley Research estimates.

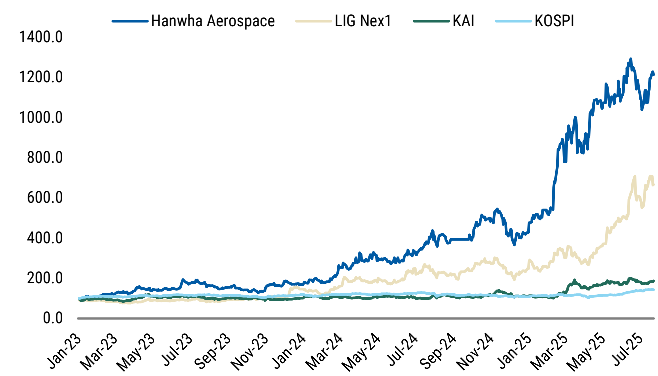
Korea Defense Super-cycle Has Started

Stocks have surged but are not tapped out

The Korean defense sub-sector has been a key outperformer since the beginning of 2024. Hanwha Aerospace and LIG Nex1's share prices are up 183%, and 169% YTD as of July 23, respectively (vs. KOSPI: +33%). Considering the sizable outperformance there is a view that most of the positives have been reflected and valuations are now quite rich.

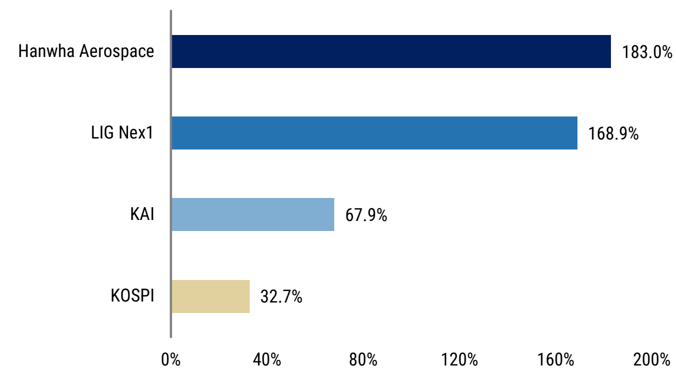
We believe the Korean defense names are only taking a breather after the recent upward climb. In our view, the next step up will be driven less by expectations and newsflow and more by execution, new order flows and technical factors playing out.

Exhibit 10: Performance of Korean defense sub-sector (our coverage only) vs. KOSPI



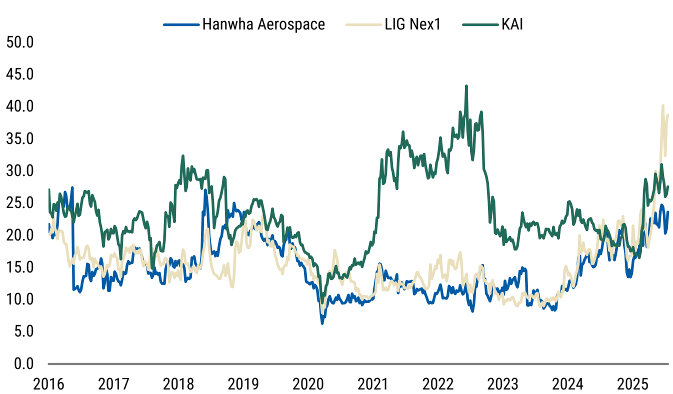
Source: Datastream, Morgan Stanley Research.

Exhibit 11: YTD performance of Korean defense stocks



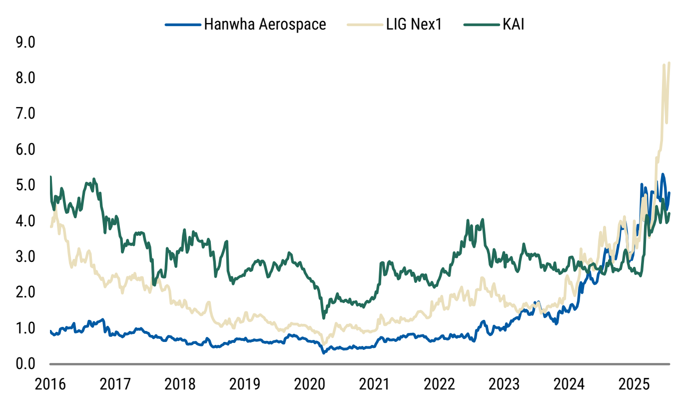
Source: Datastream, Morgan Stanley Research. Note: Data as of July 23, 2025.

Exhibit 12: Fwd P/E of covered defense stocks



Source: WiseFN, Morgan Stanley Research.

Exhibit 13: Fwd P/B of covered defense stocks



Source: WiseFN, Morgan Stanley Research.

More room to run

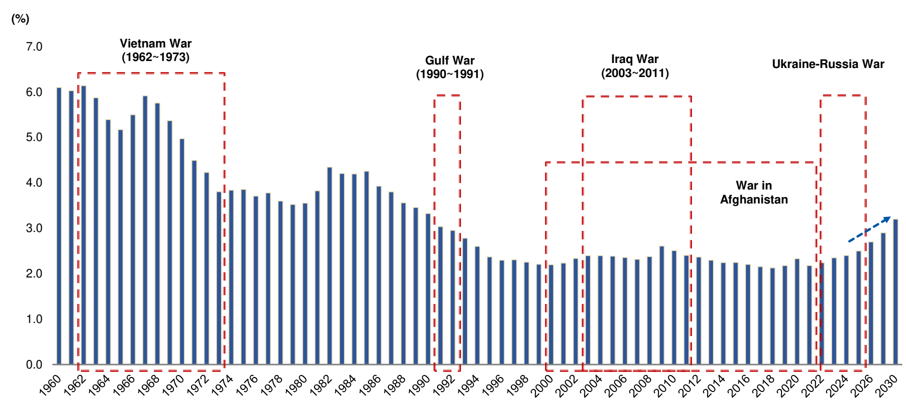
The market has questions about how sizable and how long the current new order cycle will be for the Korean defense sub-sector. We believe Korea is poised to become a top 5 global military equipment/munitions manufacturing powerhouse within the next 5 years. Despite the strong order flows, we believe the market may still be underestimating the demand for Korean conventional arms, and also Korea's position as an advancer of military technology.

We expect further upside ahead, considering (1) a super-cycle is here for the Korean defense sub-sector, (2) the market is underestimating Korea's development of advanced military technology, (3) valuations are not burdensome, and (4) technical factors look favorable.

Competitiveness in conventional arms getting stronger

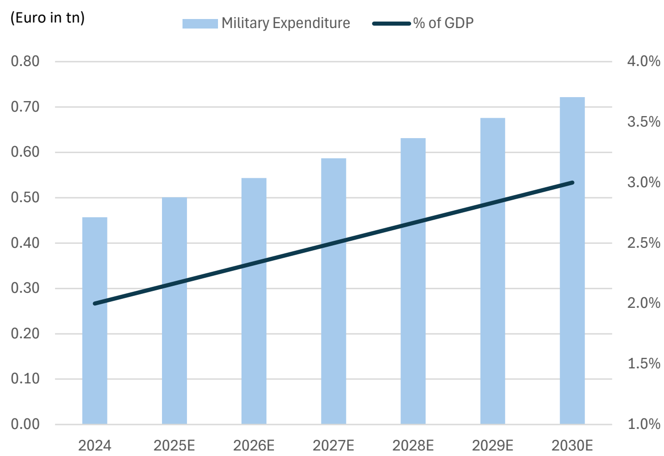
One of our key research thematics is the Multipolar World, with a key sub-theme being the rise in defense spending. Governments are adjusting to a more uncertain security environment and the US is pushing for allies to be more self-reliant on defense. Even if conflicts subside, we believe export demand for Korean military equipment will remain strong in key markets owing to (1) restocking demand (Eastern Europe), (2) reinforcement (Eastern Europe, ASEAN) – multiple countries/regions, including Korea, are increasing defense budgets, and (3) replacement (Eastern Europe, MENA, ASEAN) – upgrading of existing equipment/systems + advancement of equipment.

Exhibit 14: Global defense expenditure vs. GDP



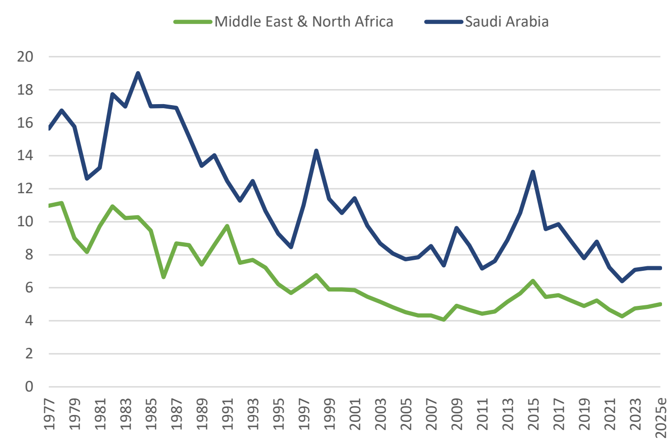
Source: SIPRI, US Government, Morgan Stanley Research forecasts.

Exhibit 15: Military spending by European NATO members



Source: SIPRI, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Exhibit 16: Military spending by MENA and Saudi Arabia



Source: SIPRI, Morgan Stanley Research. e = Morgan Stanley Research estimates.

What advantages does the Korean defense sub-sector possess?

(1) Geopolitical flexibility + strong existing relationships: Considering the Multipolar World is becoming more convoluted with the US tariff issues and extended conflicts across the globe, foreign relations should become an even more important factor when exporting military equipment, in our view. Korea has been exporting an increasing volume of arms to trading partners, but more importantly Korea has built strong partnerships by being a key provider of many other industrial services including infra/downstream/power generation facilities construction, nuclear powerplants, shipbuilding, together with manufacturing platforms across many countries. This has enabled Korea to gain many strategic partners especially in Eastern Europe, MENA, and ASEAN.

(2) Operational + cost effectiveness (purchase + sustainment): Korean military equipment and platforms have been developed with the aim of being more cost-effective than competing equipment/platforms and more importantly to keep costs low. Operational readiness and manageable sustainment costs have been important goals for the Korean armed forces, considering the state of armistice with North Korea and the need to effectively manage its defense budget.

(3) Compatibility and inter-operability with existing systems: This would (i) reduce training time for both operators and support staff, (ii) allow for inter-operable functionality with existing equipment/systems to enhance defense capabilities, and (iii) more efficient maintenance, repair and overhaul (MRO).

Before the Korean military began to procure military equipment from domestic makers, the US was its largest supplier of military equipment, followed by Germany. Even as Korea became more self-sufficient, for the most part its designs, controls, and systems are US/NATO-compatible. In the case of Korea Aerospace Industries' FA-50 and KF-21, the cockpit set-up and avionics are based on existing US warplanes, with the cockpit of the FA-50 being very similar to the F-16, according to a pilot who has tested both planes.

(4) Flexibility on manufacturing capacity: There may be market concerns that new orders may be limited by capacity. While the situation is different for each of the companies, they all believe capacity is not a concern as not only do they have excess capacity remaining, but they also have the flexibility to move their manufacturing platforms to customer countries if needed. For example, Hanwha Aerospace raised W3tn of capital recently, and it has stated that part of the proceeds will be used for facilities investment and shipbuilding capacity.

(5) Strong government support: Government support is imperative in exporting arms as the new order contracts usually involve the buyers' government budget. Prior administrations have been very supportive in boosting defense exports and this should continue to be the case with President Lee's campaign pledge booklet including the vision to make a Korea a top 5 defense manufacturer.

Boosting domestic defense spending linked to US policy path

Korea's defense budget is currently 2.3-2.4% of GDP and the annual increase has been in the range of 3-4%. However, we think Korea may have to increase its defense budget further, as is the case with NATO countries and other US allies. According to Bloomberg (June 27, 2025), US President Trump is demanding Korea increase its defense spending, and the Korean government is currently in talks with the US about this issue. More time will be needed to get clarity on this issue, but it does seem that an increase in Korea's defense budget is likely.

Exhibit 17: Korea's defense budget

(Wbn)	2020	2021	2022	2023	2024	2025p
GDP	2,058,466	2,221,912	2,323,781	2,401,189	2,542,859	2,639,488
Government budget (General)	356,568	380,850	421,416	446,242	449,526	473,659
Defense budget	50,153	52,840	54,611	57,014	59,424	61,246
YoY Increase	7.4%	5.4%	3.4%	4.4%	4.2%	3.1%
vs GDP	2.44%	2.38%	2.35%	2.37%	2.34%	2.32%
vs Government budget	14.1%	13.9%	13.0%	12.8%	13.2%	12.9%

Source: Ministry of National Defense, Morgan Stanley Research. p = Ministry of Defense projection.

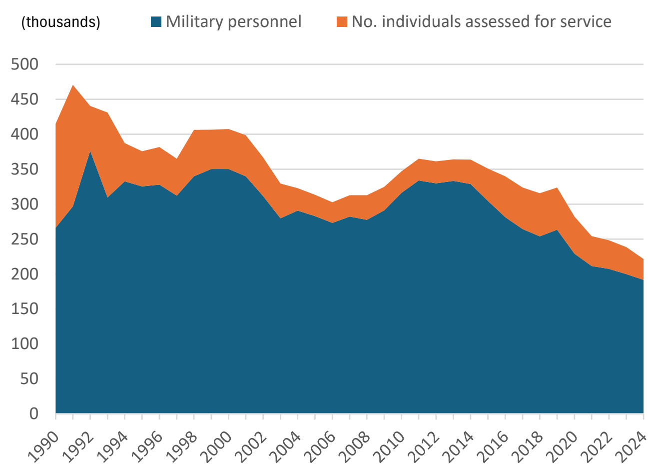
Idiosyncratic factors forcing Korea to push the envelope of military technology; catching up to traditional defense leaders

We believe the market is underestimating Korea's potential to become a leader in military technology despite Korea having an advanced IT manufacturing platform and strong incentives to continue to develop the nation's military capabilities. The key reasons why we think Korea will have to upgrade its military technology are:

(1) Korean peninsula still under an armistice agreement: South Korea is in a distinctive situation as it is officially still in a state of armistice, as a peace treaty with North Korea was never consummated. Relations between the two countries have been tense for many years now and the latter's nuclear+ballistics development continues to advance. Pyongyang is also pushing for the modernization of its military and, according to the Korea Central News Agency, Kim Jong-un said in March 2025 "the field of unmanned equipment and artificial intelligence should be top-prioritised and developed in modernising the armed forces". For more details, see Economics & Strategy: North Korea: Navigating Geopolitical Tensions (23 Aug 2017).

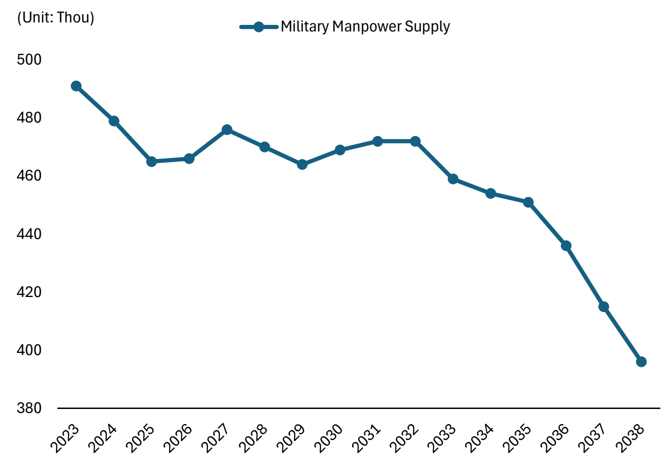
(2) Overcoming super-aging and low fertility rate needs defense technology upgrades: Due to the armistice situation with North Korea, there is a need in South Korea to keep a sizable armed force, and hence there is mandatory military service for all of its male citizens. Since peaking at 365,000 in 2011, Korea's full-time enlisted armed forces personnel slipped to 222,000 in 2024.

Exhibit 18: Korea Military Forces: Full-time personnel



Source: Annual Military Statistics, Morgan Stanley Research.

Exhibit 19: Forecast of future military manpower supply



Source: Korea Institute for Defense Analyses (KIDA), Morgan Stanley Research. Note: KIDA forecasts.

Defense Innovation 4.0, which was implemented in 2022, has the objective of strengthening military capabilities by taking advantage of Korea's competitive prowess in AI and other technologies. We expect to see South Korea's military forces undergo a major transformation as advanced technology and systems are proliferated into the armed forces.

Exhibit 20: Defense Innovation 4.0 – Challenges vs. Opportunities

Challenges to Overcome	Opportunities
Actualization of North Korea's nuclear weapons and missile threats	Highly advanced Korea's cutting-edge S&T - 87.8% compared to the level of US AI technology
Increased instability due to US-China competition for hegemony in North-east Asia	Sharp rise in the global status and competitiveness of Korea's defense industry - Korea's defense industry competitiveness stands at 80% compared to that of the US
Change in warfare paradigm and intensifying competition over technological supremacy	
Decrease in military service recources due to falling demographic - Men aged 20 were reported to be around 257,000 in 2022, and are expected to fall to 230,000 in 2035 and further to 130,000 by 2040	All of these opportunity factors come under the purview of the development of the 4th Industrial Revolution (4IR) S&T

Source: Ministry of Defense, Morgan Stanley Research.

Valuations are reasonable, technical factors are also favorable

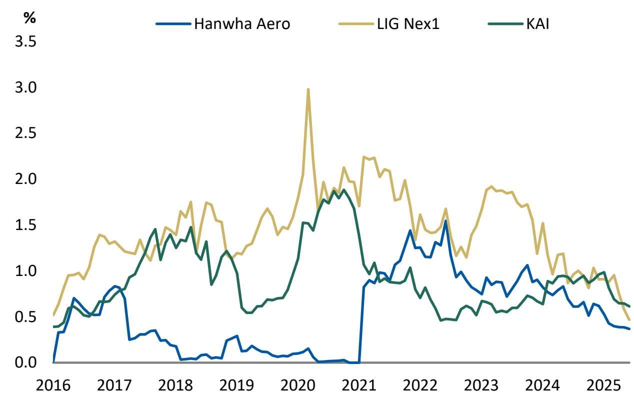
(1) Valuations not expensive considering growth path + limited alternative investment options within EM: Korean defense companies, even after the sharp rally, are trading near the global median level of around 4x forward P/B, but we believe growth prospects look very strong for Korean defense companies in the coming years.

Most listed defense companies belong to developed markets, with US and Europe accounting for more than 80% of total global market cap (based on our Morgan Stanley Defense coverage, as of July 23, 2025).

(2) Minimal impact from US tariffs; could be a negotiating card: Korean defense sub-sector should be minimally affected by the tariff issue since most of Korea's export partners are either in Europe or MENA. Further, US has asked Korea to be a partner in building naval vessels and for MRO services. For more details, see Korea Strategy & Economics: Trade Tensions Playbook v.2 - Political Stability is Key (7 Apr 2025).

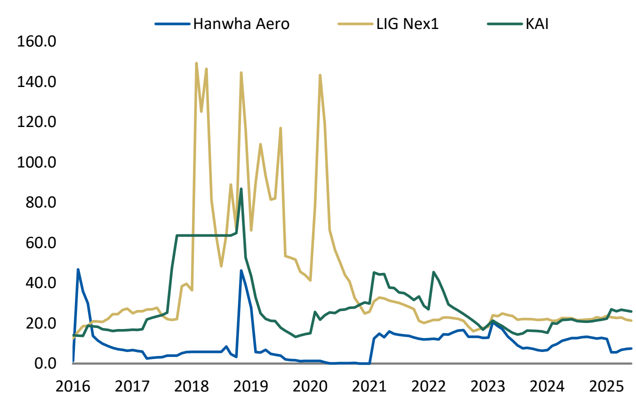
(3) Improving corporate governance + treatment of minority shareholders: Korean conglomerates in general have been seen by investors as needing improved corporate governance. In addition, there should be continued demand from the market for more proactive capital management as operational cash flows improve and capex requirements are satiated. For more details, see Korea Reform Renaissance: Paving the Road to Reform (30 Apr 2025) and Korea Strategy: The Reform Twist Getting Stronger (14 Jul 2025).

Exhibit 21: Korea Defense: Dividend yield (12M trailing)



Source: WiseFN, Morgan Stanley Research.

Exhibit 22: Korea Defense: Dividend payout (12M trailing)



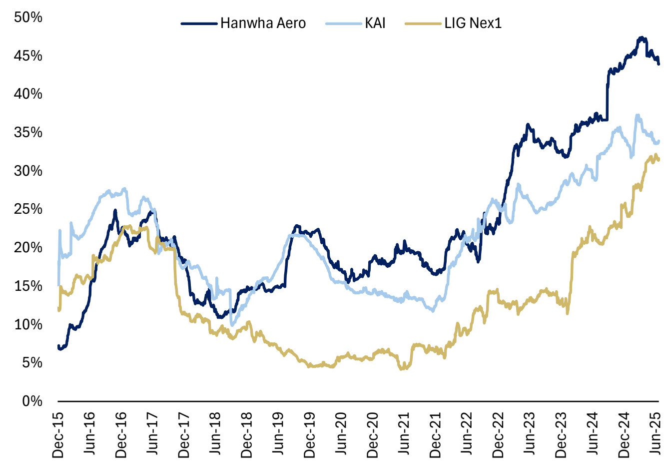
Source: WiseFN, Morgan Stanley Research.

(4) ESG investment limitations being loosened: Investing in defense stocks has had restrictions and hence the available AUM for investment in these stocks has been smaller than for other industrial sub-sectors. However, with EU pushing to boost defense budgets, there has been a gradual move by asset managers to ease investment restrictions.

- Investors have started easing restrictions related to broad-based defense exclusions. While exclusions for Article 9 products remain, the approach to easing within Article 8 products is in line with our expectations. Controversial weapon exclusions have been kept in place, with ESG easing focused on conventional weapons or/and nuclear weapons.
- However, investors are being more nuanced in their approach to conventional weapons, with some investors including exposures to radar systems, military software, and surveillance, but continuing to exclude combat equipment.
- Foreign ownership is highest in Hanwha Aerospace (44.0% as of July 23, 2025), followed by Korea Aerospace Industries (33.9%), and LIG Nex1 (31.7%). Since the end of 2021, the foreign share ratio in all three companies has been rising. In the case of LIG Nex1, foreign ownership, which had previously been in the single digits, began rising after the company secured its first M-SAM export contract to the UAE in 2022, and has continued to climb. Hanwha Aerospace has also seen a significant uptick in foreign shareholding following major export contracts for the K-9 and Chunmoo systems to Poland and other markets. All three companies are currently near record-high levels of foreign ownership.

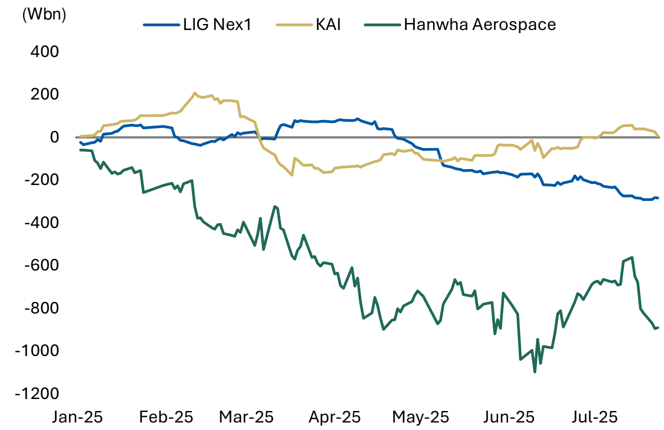
For more details, see Sustainability: Defence in Sustainability: Recent Developments (14 Apr 2025) and Sustainability: Defence in Sustainability: An Evolving Debate (24 Mar 2025).

Exhibit 23: Korea Defense: Foreign ownership



Source: WiseFN, Morgan Stanley Research.

Exhibit 24: Korea Defense: Retail fund flows



Source: WiseFN, Morgan Stanley Research.

Previous Industrials sub-sector cycles; multi-year re-ratings with ebbs and flows

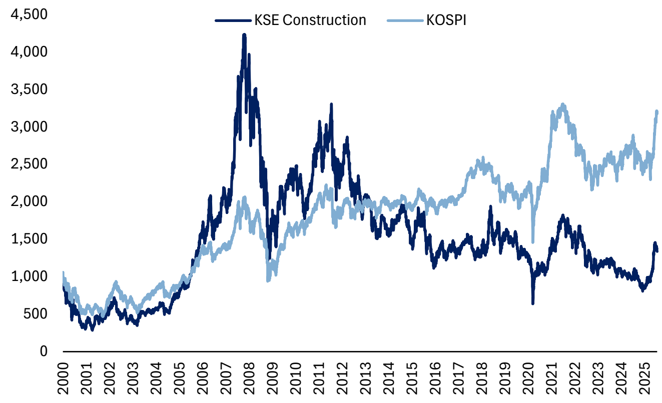
The Industrials sector tends to be more linked to the global macro cycle due to the nature of the businesses for key sub-sectors such as construction, machinery, and shipbuilding. However, some of the Korean sub-sectors including construction and shipbuilding have experienced super-cycles, which led to a significant re-rating of stocks in those sub-sectors.

We believe there is are lessons to learn by looking at the overseas construction super-cycle that took place during 2007-11. There were ebbs and flows on the way up, which is generally the case for super-cycles. The super-cycle did not end well as Korean contractors overextended themselves, which led to sizable losses being taken due to project delays and other problems.

- The overseas construction super-cycle began in 2007 and came to end in 2011 but there was a hitch in the middle due to the Global Financial Crisis (GFC) in 2008-09. This super-cycle was largely driven by the downstream hydrocarbon and infrastructure building boom in the MENA market.
- Korean construction companies took sizable market share away from European contractors that had been dominating the market through pricing competitiveness and also strong relationships with the governments that had been built up since the 1970s.

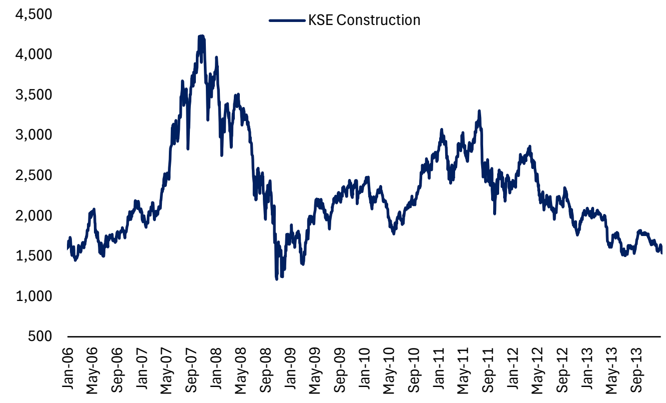
For more details, see Future of UAE: Diversify & Transform (27 Jan 2025) and The Future of Saudi Arabia: Emerging at 'Giga'speed (15 Dec 2022).

Exhibit 25: Korea Construction Sector vs. KOSPI Share Price Trend



Source: WiseFN, Morgan Stanley Research

Exhibit 26: Korea Construction Sector Share Price Trend (2006-13)



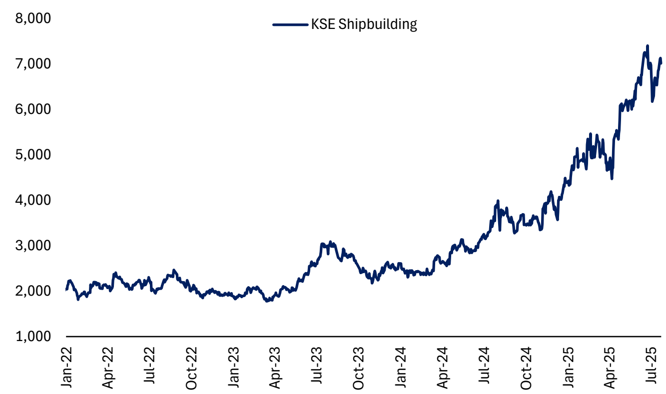
Source: WiseFN, Morgan Stanley Research

In the case of the shipbuilding super-cycle, which was quite recent starting from 2023, this was led by the LNG vessel demand spike due to the ongoing conflict between Ukraine and Russia. Once again, share prices did not go up on a straight line with there being ebbs and flows also but ultimately being pushed up as the story gained strength.

Exhibit 27: Korea Shipbuilders Sector vs. KOSPI Share Price Trend

Source: WiseFN, Morgan Stanley Research

Exhibit 28: Korea Shipbuilding Sector Share Price Trend



Source: WiseFN, Morgan Stanley Research

What lessons can we get from the overseas construction and shipbuilding super-cycles?

(1) Super-cycles tend to last at least three years and share prices have multiple instances of ebbs and flows with the leading companies outperforming the industry – for the construction industry, Samsung E&A outperformed, while for shipbuilders it was Daewoo Shipbuilding & Marine Engineering (now Hanwha Ocean).

(2) For the construction industry, there was a tendency to be overly aggressive on pricing to take market share and the domestic business actually showed better margins than the overseas business. While there is competition for the Korean defense players, the export market has better margins than the domestic business and Korean defense players have some key discernible advantages over its competitors.

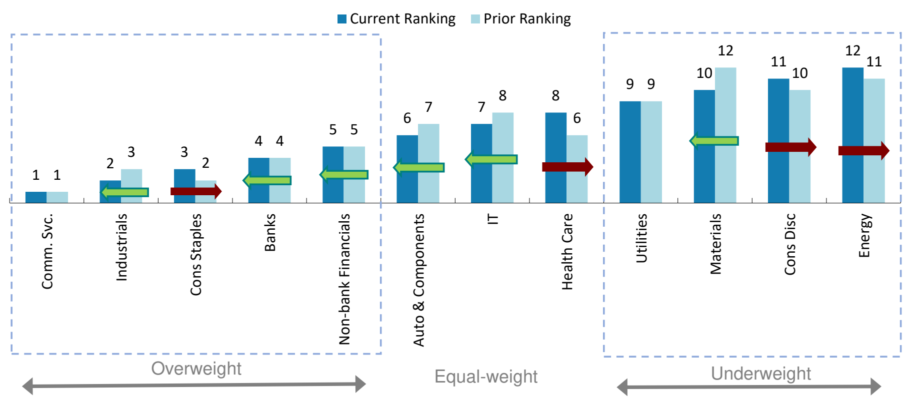
How to be Positioned for the Next Step Up

Overweight on Industrials for strategy sector allocation

We have been positive on Industrials from a strategy sector allocation perspective, as it has the most sub-sectors/stocks that show opportunities for secular growth and should be less impacted by tariff issues. Within Industrials, we see defense, shipyards, and overseas construction as favorably positioned for a multi-polar world. For more details, see South Korea Economics: The Next Breakthrough in Exports (1 Apr 2024), Korea Musings: Cold Front Needs to Recede (16 Jan 2024), and Korea Strategy: The Reform Twist Getting Stronger (14 Jul 2025).

With this report, we assign an Attractive industry view to S Korea Industrials.

Exhibit 29: Our current strategy sector allocation



Source: Morgan Stanley Research. Note: As of July 23, 2025.

Exhibit 30: Sector performance and 2025 cumulative net buying by investor type

As at Jul. 23, 2025	KOSPI	IT	Materials	Industrials	Energy	Cons.Disc.	Financials	Healthcare	Comm.svcs.	Cons.Stp.	Utilities
Index wght	100%	37%	6%	24%	3%	11%	13%	12%	7%	3%	2%
2025 YTD Perf.	32.7%	25.7%	18.5%	68.7%	35.7%	19.0%	44.4%	16.3%	21.3%	26.0%	65.2%
2024 Perf.	-9.6%	-25.3%	-41.2%	11.2%	-21.4%	-5.9%	22.7%	7.6%	-10.3%	8.4%	9.6%
Since 21st Presidential Election (Jun. 3, 2025)	18.0%	18.8%	26.8%	15.3%	20.6%	16.1%	22.6%	12.0%	12.6%	11.0%	24.1%
Since lifting short-sell ban (Mar. 31, 2025)	28.3%	19.9%	19.8%	48.8%	37.6%	21.0%	43.1%	21.0%	20.2%	23.5%	56.6%
Foreigners Net Buying (Wbn)	-8,560.9	-5,486.3	-1,483.6	1,031.7	36.5	-1,389.8	-2,124.7	-2,144.6	-103.8	186.5	650.0
Domestic Retail Net Buying (Wbn)	-5,060.6	1,606.2	-536.5	-1,458.4	-325.6	-127.8	-1,458.3	2,393.6	887.9	108.9	-1,030.6
Domestic Institution Net Buying (Wbn)	5,527.0	-990.3	1,823.6	-86.6	208.4	327.8	3,735.4	-660.7	-355.0	-284.3	538.0
Pension Funds Net Buying (Wbn)	5,197.7	1,638.3	953.7	1,277.9	168.3	444.8	360.1	374.0	174.2	-86.5	263.6

Source: WiseFN, Morgan Stanley Research. Note: Data as of July 23, 2025.

We initiate on three defense stocks, with our preferred stock within S Korea Industrials being Hanwha Aerospace (012450.KS, Overweight), followed by Korean Aerospace Industries (047810.KS, Overweight) and LIG Nex1 (079550.KS, Equal-weight).

Exhibit 31: Price target and bull/bear scenario analysis

Company	Ticker	Rating	Share Price				Valuation (25E P/E)			
			Current Price	Target price	Bull case	Bear case	Current	Target price	Bull case	Bear case
Hanwha Aero	012450.KS	OW	910,000	1,250,000	1,400,000	755,000	31.20	42.86	48.00	25.88
KAI	047810.KS	OW	92,200	110,000	120,000	75,000	29.75	35.50	38.72	24.20
LIGNex1	079550.KS	EW	593,000	600,000	675,000	450,000	45.30	45.84	51.57	34.38

Source: FactSet, Morgan Stanley Research estimates. Note: Pricing data as of the market close on July 23, 2025.

Exhibit 32: Korea Defense: Annual share price performance vs. consensus net profit (NP) revisions

		2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025 YTD
Hanwha Aerospace	Share Price	49.1%	22.2%	-18.2%	-8.6%	7.8%	-18.7%	68.4%	53.3%	69.2%	136.1%	185.2%
	Consensus NP Revision	-85.0%	244.9%	-61.3%	-84.2%	97.5%	-28.7%	76.3%	18.7%	191.0%	39.4%	65.6%
LIG Nex1	Share Price	NA	-22.2%	-25.7%	-38.2%	-14.5%	-3.5%	124.9%	34.4%	41.5%	69.0%	185.7%
	Consensus NP Revision	NA	-31.0%	-45.6%	-90.9%	-40.2%	15.7%	68.0%	76.9%	-11.2%	0.0%	18.6%
KAI	Share Price	96.2%	-14.2%	-29.2%	-32.8%	6.7%	-24.2%	25.8%	56.9%	-1.8%	9.8%	66.7%
	Consensus NP Revision	32.1%	-8.5%	-131.6%	-61.6%	64.3%	-31.5%	-32.6%	54.0%	-14.3%	11.4%	-20.4%

Source: FactSet, Morgan Stanley Research.

Hanwha Aerospace (012450.KS, Overweight)

Why we are Overweight: Hanwha group is the largest defense contractor in Asia with Hanwha Aerospace as its mainstay and Hanwha Ocean also under Hanwha Aerospace's umbrella. Hanwha Aerospace's key product lines are in heavy demand across the globe and the group's vertically integrated manufacturing structure is a key advantage.

- While Hanwha Aerospace managed to grow its top line by 42% YoY in 2024, backed by a strong order backlog mainly consisting of K9 and Chunmoo from its land systems, we believe it will see more opportunities in new markets based on 3R demand. We are witnessing increased investment for its foreign production capabilities in Europe and the US, which we believe enhances the chance of winning new orders in the Western market.
- Light infantry vehicles will be a key growth area for land military vehicles over the next decade, and Hanwha Aerospace's Redback should be competitive in the global market.
- Upside optionality linked to Hanwha Ocean showing positive progress on naval vessel production and MRO services.
- Corporate governance reform should be positive for Hanwha group as a whole as there had been legacy issues surrounding corporate governance and actions.

Korea Aerospace Industries (047810.KS, Overweight)

Why we are Overweight: We believe Korea Aerospace Industries' KF-21 and FA-50 fighter aircraft can satiate growing demand for cost-effective air power. We see the warplanes segment having sizable growth potential in 4- to 4.5-generation warplanes as countries are looking to reinforce their air power and replace aging aircraft. So far only the US, China, and Russia have been able to independently develop 5th-generation fighter platforms, but the KF-21 block 2 (2028-32) and block 3 (2032~) are expected to be almost fully self-sufficiently developed 5th-generation fighter platforms, in our view.

- We expect to see more FA-50 orders as customers continue to be satisfied with the build quality, compatibility, and relatively low MRO costs of Korea Aerospace Industries' products. ASEAN countries have been the main purchasers of T-50/FA-50 planes, but we expect potential buyers to come from other markets such as MENA, Eastern Europe and/or LATAM.
- KF-21 is still in the final testing phase, but it is getting increasing interest from various countries as the testing process is showing strong performance results. KF-21 Block 1 will be strictly an air superiority combat aircraft, with 40 planes being delivered to the Korean Air Force, but Block 2 will be a multi-role fighter and 80 planes are set be delivered to the Korean Air Force, according to ROKAF (Republic

of Korea Air Force). Ultimately, Block 3 KF-21 EX will be a 5 generation multi-role fighter. Generally, jet fighter programs take a long time to reach completion, meaning KF-21 should be competitive in the global market for years to come once the program is in full flight.

LIG Nex1 (079550.KS, Equal-weight)

Why we are Equal-weight: LIG Nex1 has been very successful in selling its MSAM-II surface-to-air missile systems, with the UAE and Saudi Arabia making sizable new orders, which in turn has led to a strong upward trend in the share price. We believe the positives are priced in for now and we await more positive news on the new order front.

- We anticipate further new orders to come from the MENA market, mainly from countries looking to diversify away from US reliance, but there is currently low visibility on when these new orders will come out.
- We also see potential from Europe as the MSAM-II could address European countries' requirement for rapid delivery, especially when alternatives such as PAC-3 suffer from long lead times (+7 years). However, whether this leads to actual new orders will have to be seen.

What are the key risks?

There will be specific risks for each of the defense stocks, but overall we believe the risks are quite similar.

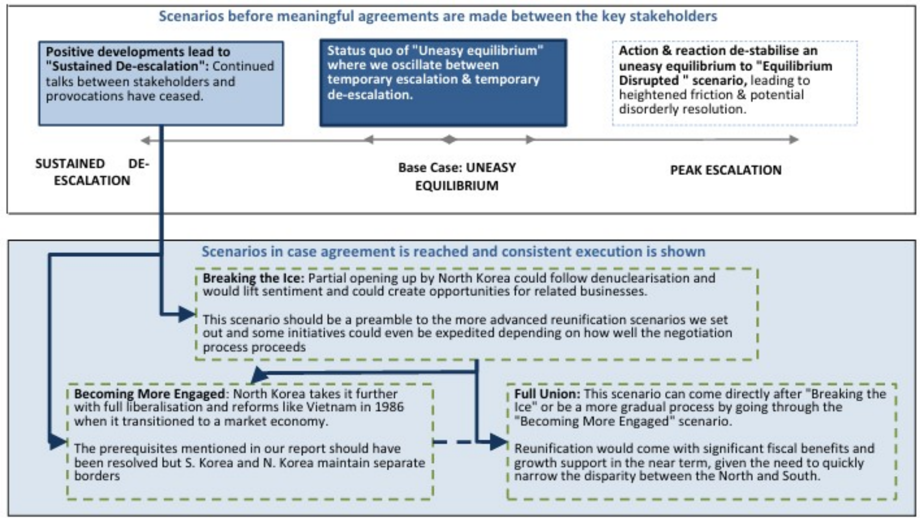
(1) A reversal by EU/NATO countries and/or other regions such as MENA and ASEAN regarding their policy path to increase defense budgets. This could come about from many angles, but the biggest risk would be if the US takes a different tack on allies' defense spending.

(2) More intense competition as other defense contractors aggressively move into new markets and push to upgrade existing equipment and/or develop new ones.

(3) Tensions with North Korea ease significantly, leading to a scenario of 'Breaking The Ice' or further where there is meaningful economic cooperation between the two Koreas.

For more details, see South Korea/China Strategy & Economics: Base Case of Sustainable De-escalation; But What If? (26 Feb 2019).

Exhibit 33: North Korea-South Korea Relations Scenario Analysis



Source: Morgan Stanley Research.

Defense Industry Outlook

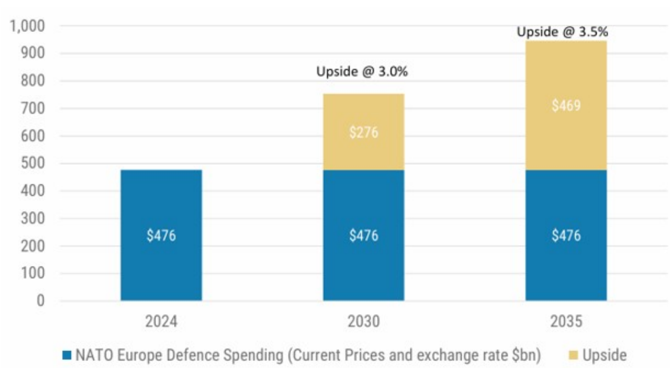
Multipolar world and demand by US to share defense costs boosting demand for equipment

One of our key research thematic is the Multipolar World, with a key sub-theme being the rise in defense spending. Governments are adjusting to a more uncertain security environment and the US is pushing for allies to be more self-reliant on defense.

We expect demand from the largest defense markets, i.e. the US and Europe, to continue to rise.

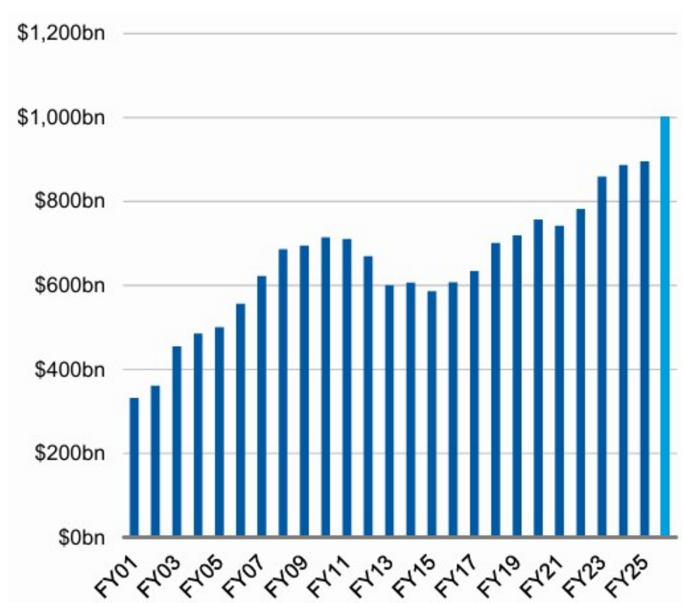
- US defense spending rose at a 4% CAGR historically, and a proposed US$1 trillion budget in FY26 would represent 12% y/y growth. We see FY26 budgetary details that have emerged in recent weeks as generally positive for the cohort (>20% y/y Modernization growth /13% y/y topline growth), and not priced at current levels. For more details, see Aerospace & Defense: Flyby: A&D 2Q25 Earnings Week 1 — Previewing Next Week's Prints (18 Jul 2025) and Defense: FY26 Defense Budget Request: Passing the $1 Trillion Mark (2 May 2025).
- Europe has underspent on defense for 30+ years. Europe would need to spend US $2tn just to make up for the underspend (vs 2% of GDP commitment). NATO leaders have agreed to a new defense spending commitment of 5% of GDP (3.5% core + 1.5% related spending) by 2035. For more details, see Aerospace & Defence: NATO Leaders Approve New Spending Target in Hague Declaration (25 Jun 2025) and Aerospace & Defence: The European Defence Dilemma Part IV: Mapping the Top Down Opportunity (16 Jul 2025).

Exhibit 34: Upside scenarios under increased NATO Europe defence spending – as % of GDP



Source: NATO, Morgan Stanley Research estimates.

Exhibit 35: US annual defense budget



Source: SIPRI, US Government, Morgan Stanley Research. Note: FY26 based on reports of US$1 trillion budget.

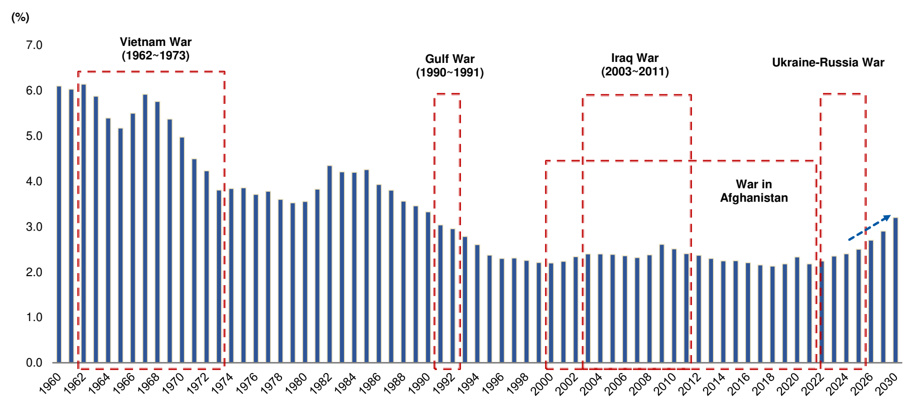
Whether or not conflicts subside, we believe demand for military equipment will remain strong considering the 3Rs below.

(1) **R**estocking demand (Central/Eastern Europe) – global military equipment/munitions restocking

(2) **R**einforcement (Central/Eastern Europe, MENA, ASEAN) – multiple countries/regions increasing defense budgets including Korea

(3) **R**eplacement (Central/Eastern Europe, MENA, ASEAN) – upgrading + advancement of equipment.

Exhibit 36: Global Defense Expenditure vs. GDP



Source: SIPRI, US Government, Morgan Stanley Research forecasts.

Can traditional arms suppliers meet demand?

There are questions as to whether traditional arms suppliers such as the US, France, Russia, China, Germany, Italy, and the UK have the products and capacity to satiate rising demand. Russia in particular has been showing a sharp decline in exports (2020-24: 7.8% of global arms exports, 2015-19: 21%, 2010-14: 27%) due to the conflict with Ukraine draining its ability to export.

Exhibit 37: The 15 largest exporters of major arms (2020–24)

Rank	Exporter	Share of global arms exports (%)			% change from 2015–19 to 2020–24
		2020-24	2015-19	2010~2014	
1	United States	43%	35%	31%	21%
2	France	10%	8.6%	4.8%	11%
3	Russia	7.8%	21%	27%	-64%
4	China	5.9%	6.2%	5.5%	-5.4%
5	Germany	5.6%	5.7%	5.3%	-2.6%
6	Italy	4.8%	2.0%	2.7%	138%
7	United Kingdom	3.6%	3.6%	4.6%	-1.4%
8	Israel	3.1%	3.2%	1.8%	-2.0%
9	Spain	3.0%	2.3%	2.9%	29%
10	**South Korea**	**2.2%**	**2.1%**	**0.9%**	**4.8%**
11	Turkiye	1.7%	0.8%	0.5%	103%
12	Netherlands	1.2%	1.9%	2.0%	-36%
13	Poland	1.0%	0.0%	NA	3878%
14	Sweden	0.9%	0.5%	1.8%	73%
15	Norway	0.8%	0.3%	0.6%	188%

Source: SIPRI Arms Transfer Database, March 2025, Morgan Stanley Research.

Korea has the motor and mettle to become a top 5 global defense manufacturer

We expect Korea to become a top 5 global military equipment/munitions manufacturing powerhouse within the next five years. Korea is currently a top 10 defense manufacturer. President Lee stated in his election campaign that the goal is make to make Korea a top 5 powerhouse in the global defense industry.

While market sentiment toward Korean defense contractors has turned quite favorable, we believe that the market is still underestimating the growth potential for the Korean defense sub-sector.

(1) The market seems to overestimate that the nature of military warfare will change due to technological advancement and improved logistical capabilities, which would imply that demand for more traditional conventional weapons such as howitzers, tanks and 4th- to 4.5th-generation warplanes will decline. We believe that demand for conventional military equipment will be pick up due to the multipolar world dynamics and hence there is sizable potential for the Korean defense sub-sector to grow exports.

(2) When the Gulf War ended in 1991, there was a relatively long period of peace aside from US conflicts with Afghanistan and Iraq, which were somewhat isolated to the US and a few allies. The need for military development in most parts of the world waned, but the situation for South Korea was different as the country officially still in a state of armistice with North Korea. South Korea began its journey to become self-sufficient in military equipment in the early 1990s, and it has now almost completed the development of a 4.5th- to 5th-generation jet fighter. We believe Korea has made considerable strides in advancing military technology and there is a strong incentive for the country to push the envelope of technological advancement.

Korea defense export outlook

Drones, mechanized equipment, and AI are expected to become more mainstream on the military side, but this does not mean demand for self-propelled howitzers, missile systems, and manned vehicles will become obsolete. Adjustments will likely have to be made and with Korea still being in a state of armistice, we believe Korean contractors will evolve into more advanced players.

Over the past decade, South Korea's defense exports have surged by 12%, making it the world's 10th major supplier in terms of arms transfer volume based on SIPRI data (as of 2024), and Korea's market share in the global arms export market has now reached beyond 2%. For the Korean defense sub-sector, we believe there should be more upside to growth from the following areas:

(1) Artillery and missile systems: As cost-effective defense systems, self-propelled howitzers have been gaining ground, especially in Europe, with multiple countries purchasing K-9 self-propelled howitzers from Hanwha Aerospace. We are seeing increasing demand for surface-to-air defense systems, which include missiles, aircraft, and drones. Since 2022, LIG Nex1 has been selling its MSAM-II to multiple MENA countries including UAE, Saudi Arabia, and Iraq.

According to Hanwha Aerospace, it has the capacity to continue to receive new orders and deliver K-9 self-propelled howitzers within a relatively quick schedule compared to competitors.

In MENA, recent conflicts have seen heavy use of surface-to-air defense systems. We expect demand for cost-efficient systems that can target drones and smaller flying equipment to rise considering the increasing use of UAMs (Urban Air Mobility). Korean defense players, notably Hanwha Aerospace and LIG Nex1, have become leading players in this field with potential for more orders, but near-term visibility is lacking.

Exhibit 38: Artillery and missile system comparison

Weapon	Company	Annual production volume (units)	Current order backlog (units)	Planned Capex Expansion (units)	Unit price (US$)
Hanwha Aerospace					
155mm Howizter					
K9	Hanwha Aero	~240	NA		~3 mn
PzH 2000	KMW	~15 (Not consistent)	~41 (as of Oct 2024)		~18 mn
M109A6	BAE Systems	60 - 70	NA		~1.7 mn
CAESAR	Nextar	~144	~313 (as of Sep 2023)		~6 mn
MRLS					
Chunmoo	Hanwha Aero				
M270 HIMARS	Lockheed Martin				
M142 HIMARS	Lockheed Martin				
LIG Nex1					
Mid-range Surface to Air Missile (M-SAM)					
Chungoong	LIG Nex-1	NA	>28 batteries	NA	~1.1 mn per missile ~320 - 347.5 mn per battery
PAC-3	Lockheed Martin	~300 missiles	~1000 missiles	~650 missiles (by 2027)	~6 - 10 mn per missile ~2.4 - 2.5 bn per battery
PAC-2	Lockheed Martin	~240 missiles	~1500 missiles	~360 - 420 missiles (by 2027)	~2 - 3 mn per missile ~0.8 - 1 bn per battery
Aster 30	Eurosam	~230 missiles	>220 missiles	~270 missiles (by 2025 end)	~2 mn per missile ~500 mn per battery

Source: Company data, Morgan Stanley Research.

(2) Warplanes: We see the warplanes segment as having sizable growth potential in 4th- to 4.5th-generation warplanes as countries are looking to reinforce their air power and replace aging aircraft. Except for the US, China, and Russia, development of 5th-generation fighters has been somewhat limited, with the US restricting exports of its mainstay 5th-generation fighters to very close allies, while Russia has less room to export warplanes and China has been less active in the export market compared to US and EU. KF-21 development will be moving to 4.5th-generation (Block 2) by 2032 and then 5th-generation (Block 3). We believe the key markets for Korean warplanes are ASEAN and MENA, although new markets such as LATAM may open up.

- The F-16 (4th-generation fighter) is the best-selling jet fighter in modern history with over 4,600 units manufactured since its launch in 1976 and around 2,000 in service globally. There have been continued upgrades to the F-16, the most recent being the F-16V (Block 70/72), which was jointly developed by Lockheed Martin and Taiwan's AIDC. While F-16s will likely remain in service, we see demand to replace them as time passes.
- Currently, there are around 862 F-15s (4th-generation) still in service with the configurations being F-15A, F-15C, F-15E, and F-15EX. The F-15 first came into service in 1976. The fighters were first delivered to the US Air Force and later to multiple countries including Saudi Arabia (210 in service), Japan (155), Korea (59), Singapore (40), and Israel (33).

Exhibit 39: Air Force Power Comparison (2022-23)

	MENA			Oceania	
	Saudi Arabia	UAE	Qatar	Australia	New Zealand
Warplanes	1105	560	NA	262	44
Combat planes	Typhoon (EU, 71)	Dassault Mirage (Fr, 59)	Dassault Rafale (Fr, 36)	F/A-18 E/F (US, 24)	
	Tornado (EU, 81)	Dassault Rafale (Fr, 1)	Typhoon (EU, 24)	F-35II (US, 72)	
	F-15 (US, 211)	F-16E Bl 60 (US, 55)	F-15X Eagle II (US, 37)	BAE Hawk (UK, 33)	
		F-16F Bl 60 (US, 21)		Pilatus PC-21(Swiss, 49)	
Trainer aircraft	BAE Hawk (UK, 81)	Aermacchi MB-339 (Italy, 12)	BAE Hawk (UK, 9)		T-6 TexanII (US, 11)
	Pilatus PC-21 (Swiss, 55)	BAE Hawk (UK, 12)	M-346 (Italy, 6)		Super King Air (US, 4)
	Cirrus SR22 (US, 25)	Grob G 115 (Ger, 12)	PAC Super (Pakistan, 8)		
	PAC MFI-17 (Pakistan, 20)	Pilatus PC-7 (Swiss, 31)	Pilatus PC-21 (Swiss, 24)		
		Pilatus PC-12 (Swiss, 25)	Pilatus PC-24 (Swiss, 2)		

	ASEAN + South Asia				Europe	
	Philippines	Indonesia	Thailand	India	Poland	Romania
Warplanes	222	244	674	1962	261	
Combat planes	T-50 (Korea, 11+12)	F-16 (US, 16)	Alpha Jet (EU, 18)	Dassault Mirage 2000 (Fr, 46)	F-16 (US, 48)	F-16 (US, 102)
	Embraer EMB 314 (Brazil, 6+6)	Su-27 (Russia, 27)	F-5II (US, 13)	Dassault Rafale (Fr, 46)	F-35 II (US, 6+32)	F-35 II (US, +32)
		Su-30 (Russia, 30)	F-16 Bl 15 (US, 45)	HAL Tejas (India, 37+85)	FA-50 (Korea, 12)	
		T-50i (Korea, 50)	JAS 39 Gripen (Sweden, 11+12)	MiG-29 (Russia, 59)	MiG-29 (Russia, 21)	
		EMB 314 (Brazil, 31)		Su-30MKI (Russia/India, 259)	Su-22 (Russia, 18)	
				MiG-21 (Russia/India, 36)		
				SEPECAT Jaguar (UK/India, 112)		
Trainer aircraft	Cessna T-41 (US, 16)		Pilatus PC-9 (Swiss, 19)		Alenia M-346 (Italy, 15)	IAR 99 (Romania, 21)
	Marchetti SF.260 (Italy, 19)		T-50 (Korea, 14+2)			IAR 316 (Romania, 7)
	Marchetti S.211 (Italy, 3)		T-6 TexanII (US, +12)			Yak-52 (Romania, 14)

Source: Wikipedia, Morgan Stanley Research. Note: Data as of end-2024.

Exhibit 40: Comparison of Training and Combat Planes (2022-24)

Weapon	Company	Annual production volume (units)	Current order backlog (units)	Unit price (US$)	Cost per hour of flight (US$)
4th Generation Light Combat Jet					
FA-50	KAI	~27	~118	~35 - 60 mn	NA
F-16	Lockheed Martin	~36	~117	~25 - 70 mn	US$27k
M-346FA	Leonardo	~24	~50	~25 - 65 mn	NA
4-5th Generation Combat Jet					
KF-21	KAI	~40*	40	~100 mn	< US$20k
Rafale	Dassult Aviation	~36	220 (as of Dec 2024)	~125 - 225 mn	US$16k
Typhoon	Eurofighter	~14	~112	~117 mn	US$65k
JAS 39 Gripen	Saab	~35	NA	~100 - 243 mn	US$8k

Source: Company reports, Wikipedia, Morgan Stanley Research.

- Korea's mainstay warplane exports have been T-50, FA-50, and KT-1. The FA-50 is currently competing with various foreign competitors, but it has been successful in penetrating the ASEAN market, and Korea Aerospace Industries expects to expand its customer base going forward.
- We expect KF-21 to also join the fray once final testing is completed and initial delivery is made to the Korea Air Force. KF-21 block 2 will be moving to a 5th-generation platform using AI technology to optimize pilot functionality. Block 3 will be moving towards a 6th-generation platform with full stealth functionality and linkage with a system to coordinate with drones (KUSX). We believe the market is underestimating KF-21's potential in the overseas market.

(3) Tanks and light armored infantry vehicles: Recently, we have seen growth in main battle tank (MBT) orders from Poland, and we believe there should be interest coming from other countries as K2 Black Panther tanks gain more recognition. The infantry fighting vehicles (IFV) segment has also received new orders from Australia for Hanwha Aerospace's Redback light armored vehicles. According to a survey made by a London-based data analytics firm, GlobalData, the global military land vehicle market will rise at a 4.3% CAGR until 2034. IFVs and MBTs are expected to be the main markets, taking 30.9% and 27.1% market share, respectively, within the military land vehicle market from 2025-34.

According to Global Defence Technology's Issue 151 "Armoured revival - Tanks and IFVs lead the way":

- Analysis conducted by GlobalData in 2024 projected that the global military land vehicles market, valued at US$31.6bn in 2024, was projected to expand at a 4.3% CAGR in 2025-34. It expects the market to reach US$48.3bn by 2034, with a cumulative value of US$431.2bn over the forecast period.

- Among geographic segments, GlobalData projects Europe to dominate the defense sub-sector with a share of 43.2%, followed by Asia-Pacific (20.6%) and North America (20.1%).

(4) Naval vessels: Warships are not a major export area for Korea compared to land systems, but we see considerable upside as demand for effective maritime defense is rising across the globe. Korean naval vessels are gaining prominence as effective warships with corvettes, frigates, and submarines having been already sold to countries in South Asia (e.g. Bangladesh) and ASEAN (e.g. Indonesia, Philippines) and with potential buyers likely to come from MENA, Canada, and other countries. MRO is another area where we think Korea shipyards have revenue opportunities going forward.

- Korea's mainstay naval vessels are the King Sejong class destroyers (made by Hyundai Heavy Industries), Incheon/Daegu class frigates (Hanwha Ocean and Hyundai Heavy Industries), and Dosan Ahn Chang class submarines (Hanwha Ocean).

According to Global Market Insights, the global naval vessels market was valued at US $110.2bn in 2024 and is estimated to rise at a 6.7% CAGR in 2025-34. Security threats have increased in many parts across the globe, and such threats have raised the need for offensive as well as defensive weapon systems.

Exhibit 41: Korean defense sub-sector exports are diversifying (for delivery after 2024)

	Korea	US	France	Russia	China	Italy	UK	Spain	Israel
Combat aircraft	142	1071	233	78		52	8		18
Combat helicopters		390	1			31			
Major warships	6	8	20	5	25	8	32	1	
SAM systems	10+	35	2	16+	37+				30+
Tanks	972	561		464	241	98			19
Other armoured vehicles	609+	2848+	498	8	1314	1757	20	558	45+
Artillery	1233+	718	141		31			12	95+

Source: SIPRI (as of March 2025), Morgan Stanley Research. Note: Selected major arms on order or preselected for future orders from the 10 largest arms exporters, for delivery after 2024.

Exports by geography

In our view, demand for Korean military equipment looks to be strongest in Central/ Eastern Europe, ASEAN and MENA, which have many Korea's key export partners. However, we expect demand to expand to new markets including Canada, LATAM, India, as well as traditional strongholds such as the US and Western Europe. Our scenario assumptions for global military expenditure are as follows.

- **Base case:** Global military expenditure to GDP ratio reaches 3% by 2027, with Korea’s export market share being 2.7%
- **Bull case:** Global military expenditure to GDP ratio reaches 3.3% by 2027, with Korea’s export market share being 3.2%
- **Bear case:** Global military expenditure to GDP ratio reaches 2.7% by 2027, with Korea’s export market share being 2.3%

Below is a list of some key countries Korea has been exporting military equipment to in the past 10 years, and with the Multipolar World we expect the list to increase. We note that Korea has been a key provider of other industrial services including infra/ downstream/power generation facilities construction, nuclear power plants, shipbuilding, together with manufacturing platforms across many countries. This has enabled Korea to gain many strategic partners especially in Eastern Europe, MENA and ASEAN markets.

Financing support is also an important element for defense exports, and policy financial institutions such as Korea Export Import Bank (KEXIM) and Korea Trade Insurance Corp. (KSure) have been key supporters. In February 2024, KEXIM raised its legal capital cap from W15tn to W25tn to expand its capacity to provide financial support. According to the press (*New Daily*, July 16, 2025), defense companies have set up collaborative relationships with commercial banks including NH Bank, Shinhan Bank, Hana Bank, and Woori Bank for additional financing support. The government also plans to inject around W85tn to support the defense, nuclear power, and infrastructure export industries until 2028.

Exhibit 42: Share of global arms imports: Strong trade partners and potential partners with Korea

Importer		Share of global arms imports (%) 2020-24	2015-19
India	Share of global arms imports	8.3%	9.1%
Key import countries	#1	Russia (36%)	Russia (56%)
	#2	France (33%)	Israel (14%)
	#3	Israel (13%)	France (12%)
Qatar	Share of global arms imports	6.8%	3.0%
Key import countries	#1	US (48%)	US (50%)
	#2	Italy (20%)	France (34%)
	#3	UK (15%)	Germany (9.2%)
Saudi Arabia	Share of global arms imports	6.8%	11.0%
Key import countries	#1	US (74%)	US (73%)
	#2	Spain (10%)	UK (13%)
	#3	France (6.2%)	France (4.3%)
Australia	Share of global arms imports	3.5%	4.8%
Key import countries	#1	US (81%)	US (68%)
	#2	Spain (15%)	Spain (21%)
	#3	Norway (1.3%)	France (3.6%)
UAE	Share of global arms imports	2.6%	3.3%
Key import countries	#1	US (42%)	US (68%)
	#2	France (17%)	France (11%)
	#3	Turkiye (11%)	Netherlands (3.4%)
Poland	Share of global arms imports	2.4%	0.4%
Key import countries	#1	US (45%)	US (29%)
	#2	Korea (42%)	Germany (18%)
	#3	Italy (3.5%)	Italy (14%)
Norway	Share of global arms imports	1.5%	1.1%
Key import countries	#1	US (91%)	US (77%)
	#2	Korea (4.2%)	Italy (7.9%)
	#3	Italy (2.4%)	Korea (7.2%)
Philippines	Share of global arms imports	0.9%	0.6%
Key import countries	#1	Korea (33%)	Korea (32%)
	#2	Israel (27%)	Indonesia (21%)
	#3	US (20%)	US (19%)
Indonesia	Share of global arms imports	0.9%	1.7%
Key import countries	#1	US (33%)	US (20%)
	#2	France (15%)	Netherlands (18%)
	#3	Korea (12%)	Korea (16%)
New Zealand	Share of global arms imports	0.7%	NA
Key import countries	#1	US (76%)	
	#2	Korea (10%)	
	#3	UK (9.1%)	
Thailand	Share of global arms imports	0.6%	1.2%
Key import countries	#1	China (43%)	Korea (21%)
	#2	US (14%)	China (21%)
	#3	Korea (12%)	Ukraine (14%)

Source: SIPRI Arms Transfer Database, March 2025, Morgan Stanley Research.

Exhibit 43: Key new orders by company

Company	Year	Country	Weapon	Value (USDmn)	Unit
Hanwha Aero	2001	Türkiye	K-9	1,000	280
	2017	Norway	K-9	383	24
	2017	India	K-9	650	100
	2017	Finland	K-9	160	48
	2018	Estonia	K-9	91	24
	2021	UAE	Chunmoo	810	10
	2021	Australia	K-9	1,100	30
	2022	Saudi Arabia	Chunmoo	800	10
	2022	Poland	Chunmoo	3,600	218
	2022	Poland	K-9	2,400	212
	2022	Norway	K-9	167	4
	2022	Finland	K-9	138	5
	2022	Egypt	K-9	1,700	216
	2023	Australia	IFV	2,400	129
	2023	Poland	K-9	2,700	152
	2023	Estonia	K-9	36	12
	2024	Poland	Chunmoo	1,700	72
	2024	Romania	K-9	750	54
KAI	2001	Indonesia	KT-1	60	17
	2007	Türkiye	KT-1	400	40
	2010	Indonesia	FA-50	400	16
	2012	Peru	KT-1	200	20
	2013	Iraq	FA-50	1,100	24
	2014	Philippines	FA-50	400	12
	2015	Thailand	FA-50	110	4
	2016	Senegal	KT-1	37	4
	2017	Thailand	FA-50	259	8
	2018	Indonesia	KT-1	24	3
	2021	Thailand	FA-50	80	2
	2021	Indonesia	FA-50	238	6
	2022	Poland	FA-50	3,000	48
	2023	Malaysia	FA-50	920	18
	2024	Iraq	Surion	100	2
LIG Nex1	2022	UAE	MSAM-II	3,500	
	2024	Saudi Arabia	MSAM-II	3,200	
	2024	Iraq	MSAM-II	2,800	

Source: Company data, Morgan Stanley Research.

Exhibit 44: Military Expenditure for Key Markets and Potential Korea Arms Transfer Market Forecasts

(USDbn)	2025E	2026E	2027E	2028E	2029E	2030E
Military expenditure by country, in billions of US$ at current prices and exchange rates,						
India	90.4	95.0	100.0	105.3	111.0	117.1
Saudi Arabia	83.9	88.6	93.7	99.1	105.0	111.3
Australia	34.5	35.2	36.0	36.8	37.7	38.6
New Zealand	3.0	3.1	3.1	3.1	3.2	3.2
Philippines	6.6	7.1	7.7	8.4	9.1	9.8
Indonesia	11.7	12.4	13.2	14.0	14.9	15.9
Thailand	5.6	5.6	5.7	5.8	5.9	6.0
Total	235.8	247.2	259.4	272.6	286.7	302.0
Annual Growth Rate	4.4%	4.8%	4.9%	5.1%	5.2%	5.3%
Military Expenditure for Arms Purchase	72.3	76.1	80.1	84.5	89.3	94.4
Scenario analysis						
Korea Arms Transfer Value (8% M/S)	5.1	5.3	5.6	5.9	6.3	6.6
Korea Arms Transfer Value (10% M/S)	7.2	7.6	8.0	8.5	8.9	9.4
Korea Arms Transfer Value (15% M/S)	10.8	11.4	12.0	12.7	13.4	14.2

Source: Morgan Stanley Research estimates.

Exhibit 45: EU Region Military Expenditure and Potential Korea Arms Transfer Market Forecasts

MS Forecast for EU Military Expenditure

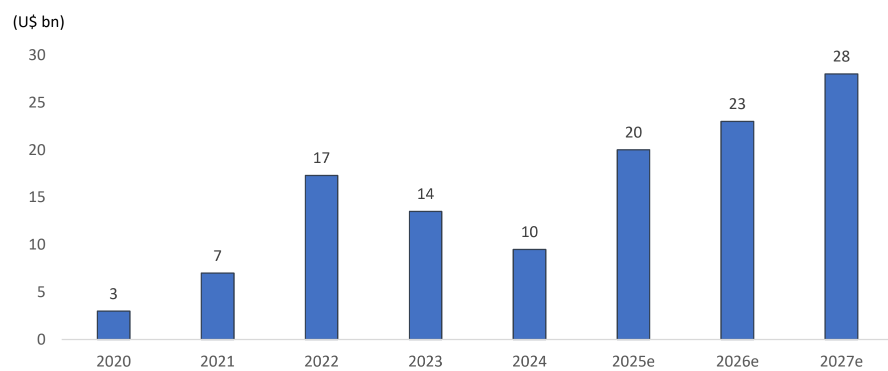
Our base case remains for Europe to reach 3% of GDP in aggregate in 2030, underpinned by Germany

EU	2024	2025E	2026E	2027E	2028E	2029E	2030E
GDP growth (Y/Y)	0.80%	1.15%	0.80%	0.80%	0.80%	0.80%	0.80%
GDP (USDtn)	15.78	15.96	16.09	16.22	16.35	16.48	16.61
Military Expenditure (USDtn)	0.35	0.38	0.40	0.42	0.45	0.47	0.50
% of GDP	2.2%	2.4%	2.5%	2.6%	2.7%	2.9%	**3.0%**

(Wtn)	M/S	2025E	2026E	2027E	2028E	2029E	2030E	6yr Average
Assuming 1.5% Market Share								
Military Exp - Attributed to Korea		8.06	8.57	9.08	9.59	10.12	10.65	9.34
Hanwha Aero	50%	4.03	4.28	4.54	4.80	5.06	5.32	4.67
KAI	15%	1.21	1.28	1.36	1.44	1.52	1.60	1.40
LIG Nex1	15%	1.21	1.28	1.36	1.44	1.52	1.60	1.40
Others	20%	1.61	1.71	1.82	1.92	2.02	2.13	1.87
Total	100%	8.06	8.57	9.08	9.59	10.12	10.65	9.34
Assuming 1.75% Market Share								
Military Exp - Attributed to Korea		9.41	9.99	10.59	11.19	11.80	12.42	10.90
Hanwha Aero	50%	4.70	5.00	5.29	5.60	5.90	6.21	5.45
KAI	15%	1.41	1.50	1.59	1.68	1.77	1.86	1.64
LIG Nex1	15%	1.41	1.50	1.59	1.68	1.77	1.86	1.64
Others	20%	1.88	2.00	2.12	2.24	2.36	2.48	2.18
Total	100%	9.41	9.99	10.59	11.19	11.80	12.42	10.90
Assuming 2% Market Share								
Military Exp - Attributed to Korea		10.75	11.42	12.10	12.79	13.49	14.20	12.46
Hanwha Aero	50%	5.38	5.71	6.05	6.40	6.74	7.10	6.23
KAI	15%	1.61	1.71	1.82	1.92	2.02	2.13	1.87
LIG Nex1	15%	1.61	1.71	1.82	1.92	2.02	2.13	1.87
Others	20%	2.15	2.28	2.42	2.56	2.70	2.84	2.49
Total	100%	10.75	11.42	12.10	12.79	13.49	14.20	12.46

Source: Morgan Stanley Research estimates.

Exhibit 46: Korea defense export growth trend and forecasts



Source: Ministry of Defense, Defense Procurement Agency, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Hanwha Aerospace (012450.KS): Asia's Leading Vertically Integrated Defense Player; OW

We initiate coverage on Hanwha Aerospace as our preferred pick. Our W1,250,000 price target is derived from SOTP valuation backed by an economic profit model, which implies 27.9x 2026E P/E – on par with global peers. We believe this is appropriate given (1) strong order/export momentum backed by its competitive and comprehensive product portfolio potentially penetrating into Western countries, (2) earnings growth momentum backed by strong order backlog mainly from its land systems business, and (3) the company's commitment to becoming a top-tier global defense player through structural change.

Investment thesis: While the company managed to grow its top line by 42% in 2024, backed by a strong order backlog mainly consisting of K9 and Chunmoo from its land systems, we believe it has more opportunities in new markets going forward based on 3R demands. We are witnessing increased investment for its foreign production capabilities in Europe and the US, which we believe enhances its chances of winning new orders in the Western market.

Earnings outlook: We expect Hanwha Aerospace to deliver strong topline growth, leveraging its leading market share in self-propelled artillery (K9 M/S: 50%), rapid delivery capabilities, diversified portfolio, and pricing power. We see overseas sales as the key engine, with export revenue within the land defense division projected to reach 69.6% in 2025, and we expect this favorable sales mix to be sustained through at least 2027. Our forecast for overseas new orders is W7.8tn in 2026, as we anticipate further contract momentum from Poland (third-phase), Norway, and the Middle East. We are also constructive on additional Chunmoo orders, given rising interest from Middle Eastern markets, including Egypt, and Peru. We see substantial upside to order inflow going forward. As more of these high-margin export contracts are delivered from 2026, we expect the overseas revenue contribution to increase further, resulting in operating profit margin expansion to 14% in 2026.

Exhibit 47: Hanwha Aerospace: Key financial metrics

Hanwha Aerospace (012450.KS)	
S.Korea Industrials | S.Korea	
Stock Rating	**Overweight**
Industry View	Attractive
Price Target (W)	**1,250,000**
Up/Downside to Price Target (%)	37%
Share Price (Jul 23, 2025)	910,000
Mkt. Cap (Bn)	46.9

Fiscal Year Ending	12/24	12/25e	12/26e	12/27e
EPS (W)	44,843	29,168	44,839	55,840
EPS (W)*	47,738	34,678	44,745	56,363
New Order (Wbn)	11,506	11,927	13,592	15,011
Backlog (Wbn)	32,400	35,835	39,989	43,613
Backlog/Sales	2.9%	1.3%	1.3%	1.3%
Sales (Wbn)	11,241	27,094	29,713	33,354
OP (Wbn)	1,731	3,274	4,113	5,032
Sales Growth	42.5%	141.0%	9.7%	12.3%
OPM	15.4%	12.1%	13.8%	15.1%
ROE	45.6%	15.3%	19.7%	20.0%

Source: *=Refinitiv Consensus, Morgan Stanley Research. e = Morgan Stanley Research estimates,

Valuation: Hanwha Aerospace is currently trading at 20.3x our 2026 EPS estimate. We believe that Hanwha Aerospace's valuation is still not challenging despite the strong rally in 2024 and YTD. The company had been trading below its 10-year average 12m forward P/E of 16.4x during 2019-21. We believe the previous valuation discount was mainly due to its high dependency on domestic demand. However, the company has entered a new phase where exports are expanding, coupled with increased conventional weapons demand. We expect its valuation gap with global peers should narrow. Rheinmetall is the major competitor, recording US$10.5bn of sales with a 14.5% OPM in 2024 (vs. Hanwha Aerospace: US$8.2bn of sales with 15.4% OPM). Rheinmetall is currently trading at 47x 12m forward P/E, implying a 52% discount to Hanwha Aerospace.

Key downside risks: (1) Order delays, (2) further transition to self-procurement in EU, (3) slowing military expenditure, and (4) increasing revenue-sharing program (RSP) losses from the geared turbofan (GTF) engine business.

Hanwha Aerospace: Financial Summary

Exhibit 48: Hanwha Aerospace: Financial summary

Hanwha Aerospace (012450.KS)

Won billion; Years Ending December 31

Key Metrics	2023	2024	2025e	2026e	2027e
New Order	**12,256**	**11,506**	**11,927**	**13,592**	**15,011**
Backlog	**27,900**	**32,400**	**35,835**	**39,989**	**43,613**
Backlog/Sales	**3.5**	**2.9**	**1.3**	**1.3**	**1.3**

Income Statement	2023	2024	2025e	2026e	2027e
Revenue	**7,889**	**11,241**	**27,094**	**29,713**	**33,354**
Gross Profit	**1,430**	**2,680**	**5,566**	**6,557**	**7,821**
SG&A	735	897	2,059	2,220	2,528
R&D	99	51	233	224	261
Operating Profit	**596**	**1,731**	**3,274**	**4,113**	**5,032**
Other Non-Operating gains	108	1,472	196	131	131
Other Non-Operating losses	351	552	375	200	200
Financial Income	1,131	371	333	172	138
Financial Expenses	367	485	763	436	394
Gains/Loss on Subsidiaries	16	105	(17)	-	-
Pre-tax Profit	**1,132**	**2,642**	**2,649**	**3,780**	**4,708**
Income tax	232	125	522	745	928
Minority Interest	159	241	646	759	945
Net Profit	**741**	**2,277**	**1,481**	**2,276**	**2,835**
EPS (W)	**14,596**	**44,843**	**29,168**	**44,839**	**55,840**

Cash Flow (Wbn)	2023	2024	2025e	2026e	2027e
Operating cash flow	**1,390**	**1,393**	**(3,045)**	**2,948**	**2,344**
Net profit	741	2,277	1,481	2,276	2,835
Depreciation	261	259	466	501	541
Change in WC	1,075	(2,425)	(3,979)	851	(2,007)
Others	(686)	1,282	(1,012)	(680)	975
Investing cash flow	**(3,029)**	**(1,367)**	**(1,311)**	**(1,167)**	**(1,334)**
Acquisition of PP&E	(453)	(578)	(1,042)	(1,167)	(1,334)
Others	(2,576)	(789)	(269)	-	-
Financing cash flow	**368**	**1,066**	**3,278**	**(581)**	**(289)**
Change in ST debt	404	927	(715)	(950)	(950)
Change in LT debt	439	938	712	(30)	(30)
Dividend	(51)	(120)	(203)	(228)	(254)
Others	(425)	(679)	3,484	628	945
Net changes in cash	**(1,271)**	**1,091**	**(1,077)**	**1,200**	**721**

Balance Sheet (Wbn)	2023	2024	2025e	2026e	2027e
Cash & Equivalents	1,899	3,291	2,183	3,383	4,104
Trade Receivables	2,119	8,884	4,129	4,125	4,914
Inventories	2,869	6,290	8,809	7,831	8,392
Other current assets	2,306	4,403	11,751	11,746	12,758
Current assets	**9,192**	**22,868**	**26,873**	**27,086**	**30,168**
Investment in Associates & JVs	3,204	964	982	982	982
PP&E	3,410	8,320	8,958	9,624	10,418
Intangible assets	2,203	7,744	7,640	7,520	7,345
Others assets	1,534	3,441	3,173	3,173	3,173
Total assets	**19,543**	**43,337**	**47,626**	**48,385**	**52,086**
ST Debt	1,374	5,153	4,537	3,537	2,537
Curr portion of LT Debt	1,165	965	949	999	1,049
Trade payables	1,879	4,270	5,008	4,871	5,227
Other current liabilities	7,659	15,128	14,023	13,223	14,023
Current liabilities	**12,077**	**25,516**	**24,517**	**22,631**	**22,836**
LT Debt	759	2,458	2,579	2,549	2,519
Bonds	645	1,713	2,186	2,186	2,186
Other long-term liabilities	1,378	2,285	1,954	1,954	1,954
Total liabilities	**14,859**	**31,973**	**31,238**	**29,321**	**29,496**
Shareholders' equity	**4,684**	**11,364**	**16,388**	**19,064**	**22,590**
Controlling Stake equity	**3,528**	**4,995**	**9,667**	**11,584**	**14,165**
Total liabilities & equity	**19,543**	**43,337**	**47,626**	**48,385**	**52,086**

Ratios	2023	2024	2025e	2026e	2027e
Margins					
GPM	18.1%	23.8%	20.5%	22.1%	23.4%
EBITDA Margin	11.9%	18.3%	14.3%	15.9%	17.2%
OPM	7.6%	15.4%	12.1%	13.8%	15.1%
Net Margin	9.4%	20.3%	5.5%	7.7%	8.5%
Growth					
Sales	11.7%	42.5%	141.0%	9.7%	12.3%
GP	-5.3%	87.4%	107.7%	17.8%	19.3%
Net Profit	340.7%	207.2%	-35.0%	53.7%	24.5%
Returns					
ROE	21.0%	45.6%	15.3%	19.7%	20.0%
Gearing					
Net Debt/Equity	44%	62%	49%	31%	19%
Total Debt/Equity	84%	91%	63%	49%	37%
Interest Cover	4.0	7.3	6.5	9.4	12.8
BVPS (Won)	69,500	98,391	190,416	228,174	279,014
P/B	17.99	12.70	6.56	5.48	4.48

Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Risk Reward – Hanwha Aerospace (012450.KS)

Asia's Leading Vertically Integrated
Defense Player

PRICE TARGET W1,250,000

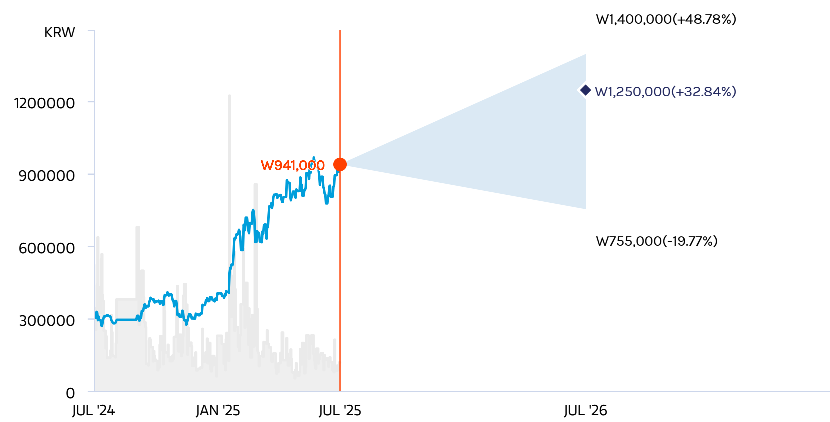
Base case, derived from SOTP valuation, wherein: (1) we use economic profit model (EPM) to value its core businesses (land systems and aerospace), with a cost of equity of 8.8% and a terminal growth rate of 3.5%; (2) we apply a 25% discount to listed subsidiaries' market values, and (3) we use equity book value for the unlisted affiliates.

Consensus Price Target Distribution W472,659.00 — W959,505 — W1,320,000.00 — MS PT

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART



Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- Hanwha Group is the largest defense contractor in Asia, with Hanwha Aerospace as its mainstay and Hanwha Ocean also under its umbrella. The group's vertically integrated manufacturing structure is a key advantage.
- Light infantry vehicles look set to be a key growth area over the next decade. We believe the Redback will be competitive in the global market.
- We are OW relative to our coverage as we expect rising export orders to bring both earnings growth and multiple re-rating.

Consensus Rating Distribution

96%	Overweight
4%	Equal-weight
0%	Underweight

● MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Contrarian: *Positive*

View descriptions of Risk Rewards Themes here

BULL CASE W1,400,000

Improved product and geographical breadth

Higher-than-expected growth in global military expenditure, combined with robust penetration into the Western market to provide strong upside to order and earnings momentum. We assume a 9% new order CAGR, 2024-33e, with a 16% sales CAGR 2024-33e, from core businesses (land systems and aerospace).

BASE CASE W1,250,000

Key beneficiary of rise in global defense spending

As the largest defense contractor in Asia, we expect order momentum to remain robust, particularly for the K9 and Chunmoo within its land systems division, supported by the group's vertically integrated manufacturing structure. We assume a 9% new order CAGR, 2024-33e, with a 16% sales CAGR, 2024-33e, from core businesses (land systems and aerospace).

BEAR CASE W755,000

Geopolitical tensions ease; lower defense spending

Slower-than-expected growth in defense spending, as well as a potential shortfall in new orders stemming from EU's self-procurement, could work as headwinds to the company's order and earnings momentum. We assume a 8% new order CAGR, 2024-33e, with a 14% sales CAGR 2024-33e, from core businesses (land systems and aerospace).

Risk Reward – Hanwha Aerospace (012450.KS)

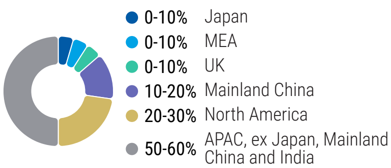
KEY EARNINGS INPUTS

Drivers	2024	2025e	2026e	2027e
Land Systems New Order (Wbn) (W, bn)	11,506	11,927	13,592	15,011
Land Systems Backlog (Wbn) (W, bn)	32,400	35,835	39,989	43,613
Land Systems Revenue/Backlog (%) (%)	32.2	31.9	31.6	34.3
Overseas Land Systems Revenue Growth (%) (%)	158.9	50.3	9.3	26.2
Gross Profit Margin (%) (%)	23.8	20.5	22.1	23.4

INVESTMENT DRIVERS

- Overseas order momentum for key product (K-9, IFV, Chunmoo)
- Global military expenditure

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

4/5 **MOST**	3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

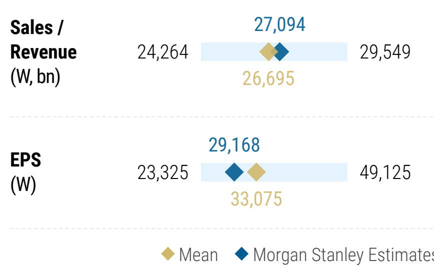
- Stronger-than-expected order momentum from other regions (MENA, Europe, APAC)
- Increase in aircraft engine parts orders driven by recovery in Aerospace business
- Significant increase in global military expenditure

RISKS TO DOWNSIDE

- Order delays
- Increased move for EU's self-procurement
- Slowing military expenditure
- Increasing revenue-sharing program (RSP) losses from the geared turbofan (GTF) engine business

MS ESTIMATES VS. CONSENSUS

FY 2025e



Source: Refinitiv, Morgan Stanley Research

Hanwha Aerospace: Key Beneficiary of Global Demand for Land Systems

#1. Strong earnings growth from overseas orders; geopolitical disputes turned its weaknesses into strengths

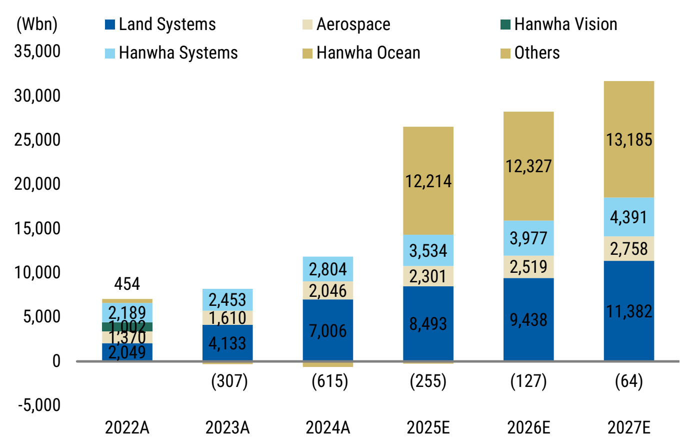
We expect Hanwha Aerospace to maintain its earnings growth driven by expanding exports, followed by the winning of large-scale overseas orders from Australia, Poland, and Romania over the past four years.

The increase in exports is positive in two aspects: (1) higher profitability compared to domestic production (OPM of less than 10% for domestic production vs. exports in the mid-teens to 20% range), and (2) it allows for high utilization rates even after domestic needs are met.

We expect the overseas sales proportion in land systems to be 69.6% in 2025, with an OPM of 24.4%. Orders from Poland will continue to drive performance this year, with more than 70 K9 self-propelled howitzers and 50 Chunmoo systems expected to be delivered. Moreover, orders from Egypt and Australia are set to attribute to earnings from 2H25. We expect additional orders to come not only from Europe but also from Asia and the ME.

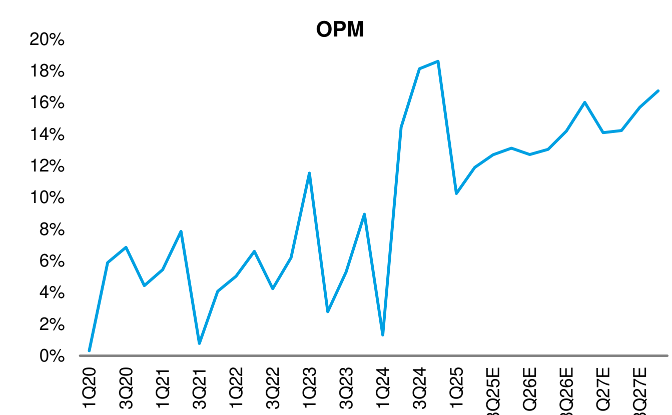
For the Redback, which has already exported 129 units to Australia, we see imminent orders from the Middle East and Romania. Vietnam has also expressed interest in introducing the K9, indicating sufficient potential demand.

Exhibit 49: Revenue by Division



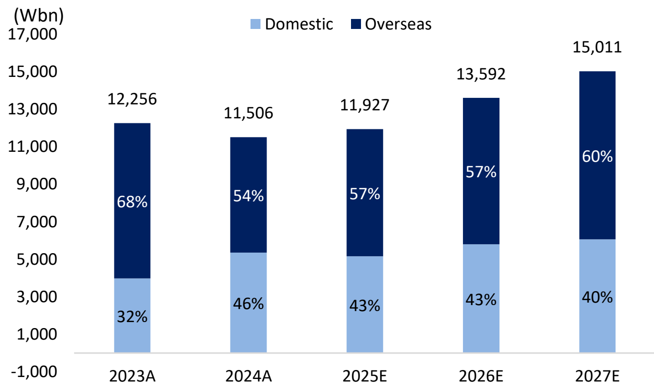
Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Exhibit 50: OPM Forecasts



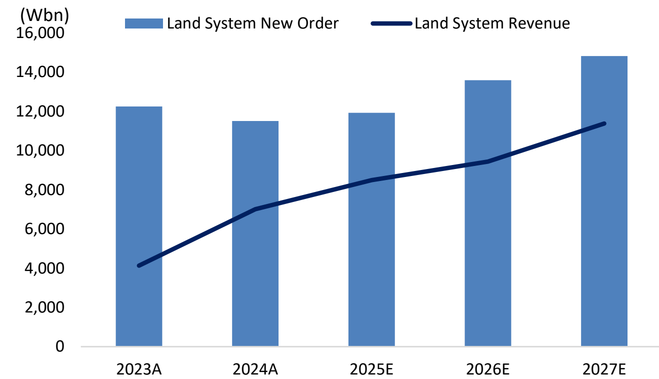
Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Exhibit 51: Land Systems: New Orders



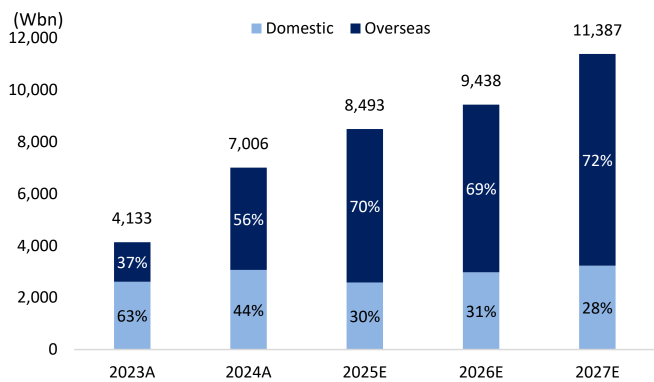
Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Exhibit 52: Land Systems: New Orders vs. Revenue



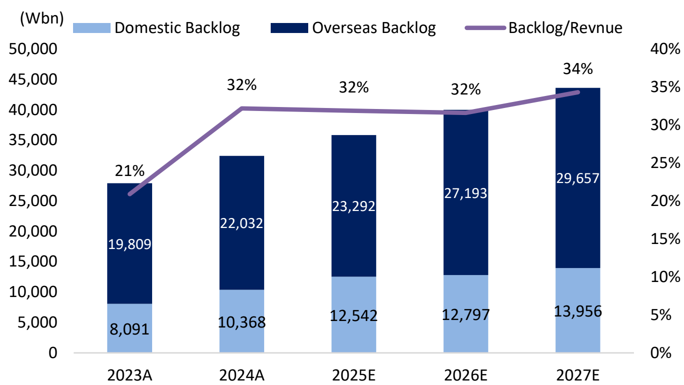
Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Exhibit 53: Land Systems: Revenue



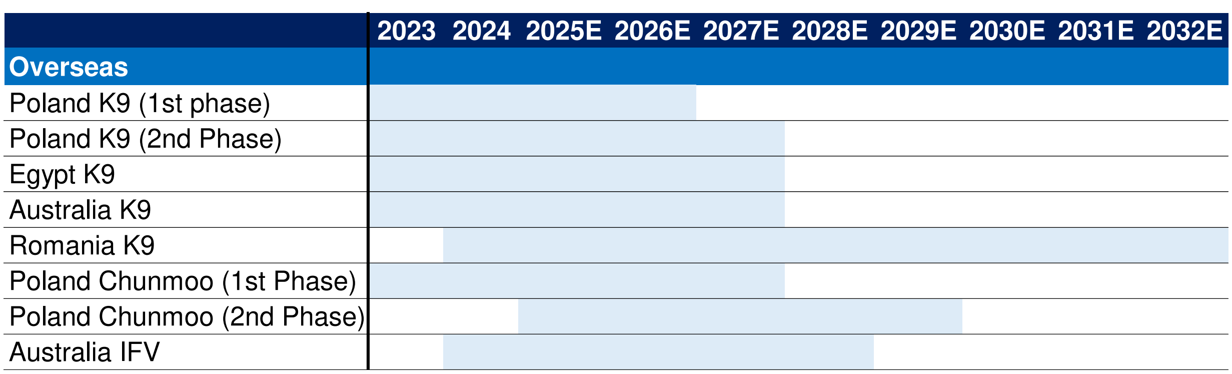
Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Exhibit 54: Land Systems: Backlog vs. Backlog/Revenue



Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Exhibit 55:
Project Timeline



Source: Company reports, Morgan Stanley Research. e = Morgan Stanley Research estimates.

#2. Key beneficiary of heavy weapons linked to reinforcement and replacement demand as it diversifies its production and customer base

We expect the surge in investment, driven by the government, will be an opportunity for Hanwha Aerospace, whose main strength is land systems.

European restocking and rearmament demand will lead Hanwha Aerospace to solidify its status in Europe, we believe, which has experienced the fastest depletion of land system weapons and has the most urgent need. Moreover, the company's efforts to expand local production should allow it to capture new orders from Europe. Hanwha Aerospace is currently planning to utilize overseas bases centered on Eastern Europe (Poland, Romania), and we expect the recent capital raise to be used as a resource for future cooperation in Europe. We expect Hanwha Aerospace to be able to continue to diversify its customer base as it provides local production and technology transfers.

Exhibit 56: Overseas Order Pipeline

Weapon	Country	Amount (Wtn)	Unit.	Date	Note
K9	India	0.4	100	Apr-17	
	Finland	0.4	96	Mar-17	
	Norway	0.2	24	Dec-17	
	Estonia	0.2	36	Jun-18	
	Australia	1.0	30	Jun-18	
	Egypt	2.4	200	Feb-22	
	Poland (1st)	3.2	212	Aug-22	Poland plans to introduce 672 units of K9, and has ordered total 364 unit
	Poland (2nd)	3.4	152	Dec-23	
	Romania	1.4	54	Jul-24	
	Poland		308	2026E	3rd contract to introduce K9
	Estonia			2025E	
	Norway			2025E	
	Vietnam	1	108		Expressed interest in introducing K9
	India	1	100		Interested in introducing additional K9
	Saudi Arabia	5	200		Potential orders are expected from Saudi
Chunmoo	UAE	0.7	12	2017	
	Poland	5.0	218	Nov-22	
	Poland (2nd)	2.3	72	Apr-24	
	Romania				Contracted MOU between HA and Romania
	Egypt				Negotiation between HA and Egypt is underway
	Peru				Highly interested in introducing MLRS
	Norway				Plan to introduce 18 of MLRS and interested in Chunmoo
	Finland Malaysia Estonia				Interested in introducing MLRS
IFV (Redback / K-21)	Australia	3.2	129	Dec-23	
	Romania	3	264		Romania is planning to modernize its army by introducing new IFVs to replace aging Soviet era vehicles
	Poland	4			IFV is not Poland's top priority, but limited demestic capacity will lead to future orders
	ME (UAE, Saudi)				Potential orders are expected

Source: Various media articles, company reports, Morgan Stanley Research.

Exhibit 57: Heavy Weapons: NATO stocks vs. units sent to Ukraine

	HOWITZERS (155/152mm)			MLRS		
Country	**NATO (excl. EU)**	**EU**	**To Ukraine**	**NATO (excl. EU)**	**EU**	**To Ukraine**
Total	**4,130**	**2,740**	**705**	**828**	**824**	**82**
% Committed			10.3%			5.0%
Greece		672			145	
Poland					179	
Germany						
Romania		351			188	
Italy		357				
Others (EU)		1,360			312	
United States	2,538			635		
Turkiye	1,385			146		
United Kingdom	117			35		
Norway						
Others (NATO)	90			12		
To be delivered to Ukraine			189			17
Delivered to Ukraine			516			65

Source: KIEL Institute. Note: This figure shows weapon stock estimates of tanks, howitzers, and MLRS compared with donations to Ukraine between January 24, 2022, and December 31, 2024. The data on stock is taken from IISS Military Balance (2022) and considers only ready-to-use weapons.

Exhibit 58: MLRS Allocated to Ukraine

Donor	Unit	Value (USDmn)	mlrs_list
United States	39	588	HIMARS 142 multiple rocket launcher (39)
Czechia	12	20	RM-70 multiple rocket launcher (12)
Norway	11	35	M270 weapon system (Norway) (11)
Germany	8	51	Mars II (5), HIMARS 142 multiple rocket launcher (3)
United Kingdom	6	82	M270 weapon system (6)
France	4	54	M270 weapon system (4)
Italy	2	27	M270 weapon system (2)

Source: KIEL Institute. Note: Number and value ($) allocated from January 24, 2022 to December 31, 2024.

Exhibit 59: IFV Allocated to Ukraine

Donor	Unit	Value (USDmn)	ifv_list
United States	352	669	Bradley infantry fighting vehicle (348), Bradley fire support team vehicle (4)
Netherlands	269	90	YPR-765 infantry vehicle (269)
Germany	175	254	infantry fighting vehicle MARDER (160), Caracal air assault vehicle (15)
Czechia	131	33	BVP-1 infantry fighting vehicle (131)
Denmark	100	314	BMP-2 (80), CV9035 MKIIIC (20)
Sweden	70	574	CV90 combat vehicle (50), CV9035 MKIIIC (20)
Greece	60	34	BMP-1 (60)
Poland	42	24	BMP-1 (42)
Slovenia	35	25	M80A infantry fighting vehicle (35)
Croatia	30	21	M80A infantry fighting vehicle (30)
Slovakia	30	17	BMP-1 (30)

Source: KIEL Institute. Note: Number and value ($) allocated from January 24, 2022 to December 31, 2024.

Exhibit 60: Howitzer (152/155mm) Allocated to Ukraine

Donor	Unit	Value (USDmn)	howitzers_list
United States	201	977	M777 howitzer (142), M109 A3GM 155mm howitzer (18), 155mm howitzer (41)
United Kingdom	94	925	M109 A3GM 155mm howitzer (28), AS-90 howitzer (66)
Germany	92	1085	Panzerhaubitze 2000 (44), Zusana-2 howitzer (5), RCH-155 self propelled howitzer (36), M109 155mm howitzer (7)
Italy	76	196	FH70 155mm howitzer (10), Panzerhaubitze 2000 (6), M109 A3GM 155mm howitzer (60)
France	69	420	CAESAR artillery howitzer (54), TRF1 howitzer (15)
Denmark	42	200	Zusana-2 howitzer (5), CAESAR artillery howitzer (19), Bohdana self-propelled howitzer (18)
Norway	27	72	M109 A3GM 155mm howitzer (22), Zusana-2 howitzer (5)
Estonia	24	21	FH70 155mm howitzer (24)
Netherlands	23	207	Panzerhaubitze 2000 (8), DITA howitzer (15)
Poland	18	230	AHS Krab self-propelled howitzer (18)
Czechia	13	20	Dana Howitzer (13)
Sweden	8	32	Archer self-propelled artillery system (8)
Australia	6	31	M777 howitzer (6)
Latvia	6	10	M109 A3GM 155mm howitzer (6)
Canada	4	21	M777 howitzer (4)
Luxembourg	1	6	CAESAR artillery howitzer (1)

Source: KIEL Institute. Note: Number and value ($) allocated from January 24, 2022 to December 31, 2024.

Exhibit 61: Comparison of the military capabilities of NATO and Russia (as of July 2025)

		NATO	Russia	vs.
Air Force	Total Aircraft	22,377	4,292	18,085
	Fighters/interceptors	3,312	833	2,479
	Grount attack aircraft	1,163	689	474
	Total helicopters	9,141	1,651	7,490
	Combat helicopters	1,416	557	859
Gound Combat vehicles	Main battle tanks	11,495	5,750	5,745
	Armored vehicles	971,280	131,527	839,753
	Self-propelled artillery	3,985	5,168	(1,183)
	Towed artillery	6,325	8,505	(2,180)
	Self-propelled rocket launchers	1,977	3,005	(1,028)

Source: Statista, Morgan Stanley Research.

Exhibit 62:

US-made Weapons Used in Europe

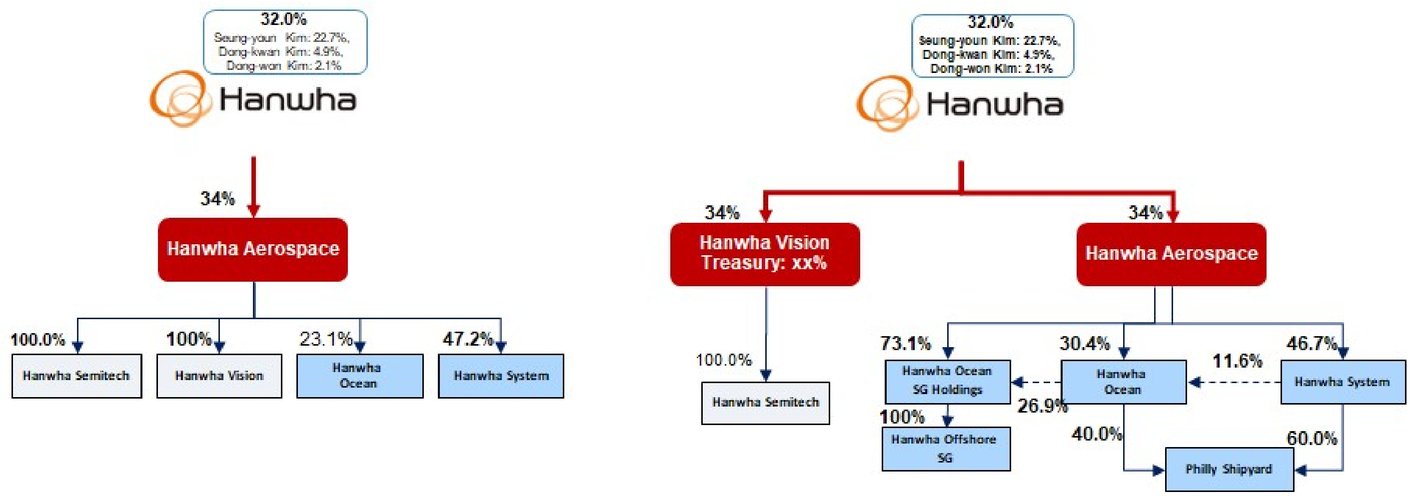
Name	Type	Company	Operating countries	Substitutions	Company	Characteristics
HIMARS	MLRS	Lockheed Martin	Poland, Italy, Romania, Estonia, Lithuania, Latvia, Croatia	**K239 Chunmoo**	**Hanwha Aerospace**	Chunmoos is a high-quality MLRS with (1) interoperability, and (2) cheap price

Source: Morgan Stanley Research.

#3. Becoming a pure defense player – Hanwha Aerospace's roadmap to becoming a global top-tier defense company

Hanwha Aerospace completed its business restructuring centered on land defense and aerospace after finishing the spin-off of non-defense subsidiaries Precision Machinery and Vision on September 1, 2024. Also, the company has expanded its focus on naval defense and completed the acquisition of Hanwha Ocean, which has been consolidated as a subsidiary starting from 1Q25.

Exhibit 63: Hanwha Aerospace: Structural Change



Source: Company report. Note: Data as of 1Q25.

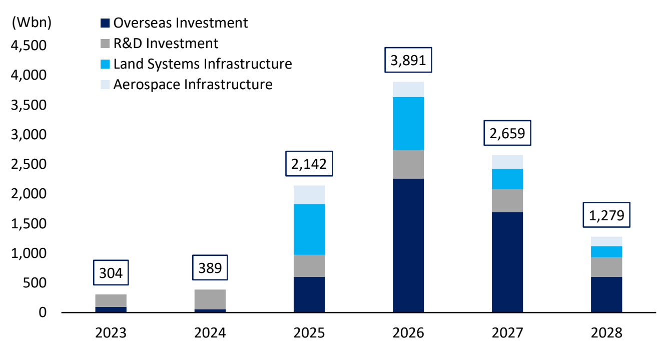
Hanwha Aerospace is also planning to expand its proactive investments as part of its ambition to become a global top-tier defense company. Through its value-up plan announcement, the company unveiled a W11tn investment plan – funded by a W4tn rights offering and around W7tn of operating cash flow, issuance of corporate bonds, and net borrowings – with around W6tn allocated to overseas land systems. These initiatives are designed to capitalize on growing trends toward localization in response to intensifying global defense industry competition. Hanwha Aerospace aims to seize new growth opportunities by investing in overseas defense joint ventures, establishing a smart ammunition factory in the US, and more.

Exhibit 64: 2025-28 Investment Plan

Investment Plan	Value (Wtn)	Defense	Shipyard
Overseas Investment for Sales Increase	6.3	- Invest in a JV for Chunmoo MLRS in Eastern Europe - Invest in a JV with Saudi Ministry of Defense - Invest in a smart ammunition factory in the United States - Establish unmanned aerial vehicle systems and engine facilities - Invest in European hubs for missile, ammunition, and ground equipment	- Invest in overseas shipyard (W0.8tn) - Invest in eco-friendly shipping (W0.3tn) 1) Secure energy transport capacity (e.g., LNG, Crude Oil) 2) Promote JV in shipping with overseas energy companies
R&D Investment to enter new markets	1.6	- Invest in the development of export-oriented ground equipment and guided missile system - Invest in advanced technologies such as ESS, advanced aircraft engines, AI, and unmanned aerial vehicles (UAV)	Under consideration for new investment 1) Invest in North American LNG liquefaction terminals 2) Enter LNG trading business 3) Establish JV in shipping industry 4) Invest in offshore wind installation vessel projects
Land Systems Infrastructure Investment	2.3	- Invest in MSC smart factories - Invest in production capacity and smart manufacturing infrastructure	
Aerospace Infrastructure Investment	1.0	- Invest in expanding aircraft engine production and aerospace infrastructure including propellant tank factories for space launch vehicles	

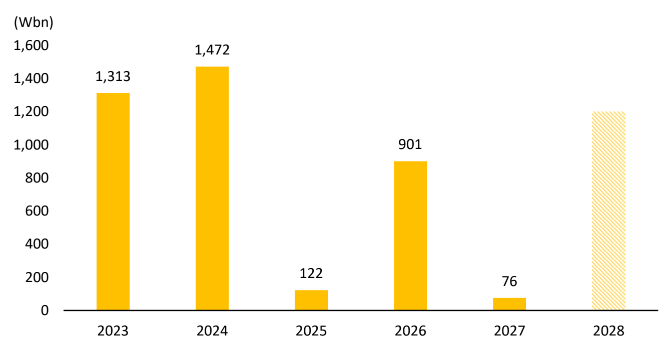
Source: Company report, Morgan Stanley Research.

Exhibit 65: Investment Plan in Defense



Source: Company report, Morgan Stanley Research.

Exhibit 66: Investment Plan in Shipyards



Source: Company report, Morgan Stanley Research.

Hanwha Aerospace is currently pursuing plans to establish local production bases in Poland, Romania, and Germany. In Australia, the company has already completed its first overseas plant and intends to finish the second phase of construction by 2026, enabling local production and mass manufacturing of the Redback IFV. Additionally, Hanwha Aerospace has signed an MOU with Saudi Arabia's Ministry of National Guard to support Vision 2030's localization goal.

Exhibit 67: Hanwha Aerospace: Offshoring Investment Cases

Region	Country	Contents
Europe	Poland	The company plans to establish defense R&D center in Poland - A joint venture with WB group was established for local production of MLRS for the Chunmoo system, with Hanwha Aerospace holding 51% and WB Electronics 49%
	Romania	The company plans to set up a comprehensive facility (incl. manufacturing, R&D, testing, and MRO)- with the new plant expected to be completed by the end of 2027 - Currently, site selection is finished and permitting with local authorities is underway
	Germany	The company announced plans to build a local production base in Germany to further strengthen its manufacturing presence there
APAC	Australia	The company established the first oversea local production site, H-ACE (Hanwha Armored Vehicle Centre of Excellence in Geelong, Australia in August 24. It will manufacture Redback IFV, which was exported to Australia in 2023, and self-propelled howitzer (K9) - AS9 SPH and AS10 ammunition supply vehicles production has begun in 2H24, and IFV will commence its production from 2026
MENA	Saudi Arabia	The company plans to establish JV for defense cooperation; Vision 2030: Saudi Arabia is actively promote defense industry, targeting to locally manufacture 50% of military supplies - The company has already signed a MOU with Saudi's MNG (Ministry of National Guard)

Source: Company reports, Morgan Stanley Research.

Investment positives and concerns

Investment positives

Hanwha Group is the largest defense contractor in Asia with Hanwha Aerospace as its mainstay and Hanwha Ocean also under its umbrella. Hanwha Aerospace's key product lines are in heavy demand across the globe and the group's vertically integrated manufacturing structure is a key advantage.

(1) While the company managed to grow its top line by 42% YoY in 2024, backed by a strong order backlog mainly consisting of K9 and Chunmoo from its land systems, we believe it will see more opportunities in new markets based on 3R demand. We are witnessing increased investment for its foreign production capabilities in Europe and the US, which we believe enhances its prospects of winning new orders in the Western market.

(2) Light infantry vehicles will be a key growth area for the next decade and Hanwha Aerospace's Redback should be competitive in the global market.

(3) Upside optionality linked to Hanwha Ocean showing positive progress on naval vessel production and MRO (Maintenance, Repair and Overhaul).

(4) Corporate governance reform should be positive for Hanwha group as a whole as there had been legacy issues surrounding corporate governance and actions.

Investment concerns

(1) A reversal by EU/NATO countries and/or other regions such as MENA and ASEAN, which are Hanwha Aerospace's main customers, regarding their policy path to increase defense budgets. This could come about from many angles but the biggest risk would be if the US takes a different tack on allies' defense spending.

(2) Competitors are able to expand their manufacturing capacities and/or become more aggressive in marketing their products to Hanwha Aerospace's key markets.

(3) Hanwha Aerospace's subsidiaries such as Hanwha Ocean and/or Hanwha Systems do not perform to management expectations due to adverse external conditions and/or operational issues.

(4) Hanwha group carries out unfavorable corporate actions toward minority shareholders that directly or indirectly impact corporate value.

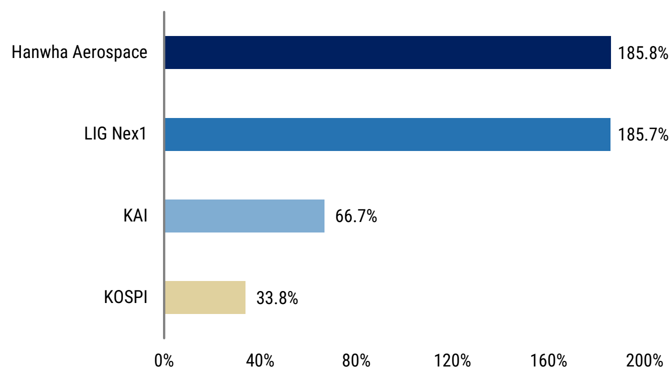
Share price analysis

Hanwha Aerospace’s share price has strongly outperformed the market since 2024. The share price was up 162% (vs. KOSPI: -9.6%) in 2024 and is up 183% YTD (vs. KOSPI: 33%) as of July 23. The strong performance was driven by growing interest in defense names due to ongoing geopolitical disputes and top-line growth on the back of export visibility. Compared to domestic peers, Hanwha Aerospace stands out with its well positioned weapons line-up, largely exposed to land systems.

However, the share price has corrected since March 21, 2025, when the company announced an unexpected capital raise of US$2.5bn (11.6% EPS dilution) to expand its global weapons and maritime production plants and also purchase stakes in related facilities. However, we see this as positive in terms of long-term growth, providing global opportunities to Hanwha Aerospace to capture potential opportunities in the EU, including initiating local production in Poland and Romania as part of its next steps.

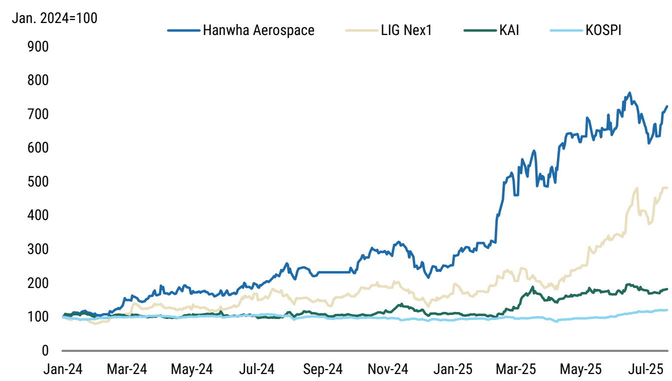
We believe that Hanwha Aerospace will be a key beneficiary of global demand, and potential new orders will work as a catalyst.

Exhibit 68: YTD Share Price Performance



Source: FactSet, Morgan Stanley Research. Note: Data as of July 21, 2025.

Exhibit 69: Share Price Performance Since 2024



Source: FactSet, Morgan Stanley Research.

Product overview: Diversified product line-up

The competitive advantage of Hanwha Aerospace's weapons lies in the company's (1) proven track record, (2) cheap prices, and (3) fast delivery and readily available capacity.

Hanwha Aerospace has two main business units, land systems and aerospace, with land systems accounting for 69% of sales in 2024. We expect this proportion will increase further, as the company spun off its non-defense businesses in 2024. Land systems is further divided into four parts, (1) armored vehicles, (2) artillery systems, (3) air defense systems, and (4) advanced ammunition. Within the land systems business, Hanwha Aerospace's main export products are (1) Redback IFV (infantry fighting vehicle), (2) K9 self-propelled howitzer, and (3) Chunmoo MLRS (multiple launch rocket system). The aerospace business accounted for 15% of sales in 2024 and includes creating engines for both rotary-wing and fixed-wing aircraft. Its largest customers in this business are Korea Aerospace Industries and Boeing.

- **K9 is a self-propelled howitzer** with specifications similar to that of its competitor, the German-made PzH2000. The K9 has achieved economies of scale, as it is less expensive than its peers (50%>). Its low cost and operating leverage has resulted in good profit margins (OPM of mass production: high single to 10% vs. export: mid-teens to 20%). We note that 90% of the K9's parts are locally sourced, allowing for fast delivery. Hanwha Aerospace currently exports the K9 to nine countries including Poland, Norway and Romania and we expect additional orders from Europe and the Middle East in the near term.

- **Chunmoo is a multiple launch rocket system (MLRS)** that is also cheaper than its competitors such as Lockheed Martin's HIMARS. We expect re-order demand to rise amid the Russia-Ukraine conflict.
- **Redback is an IFV (infantry fighting vehicle)** that is exported to Australia. We see potential opportunities for armored vehicles as the importance of infantry vehicles and their defense capabilities has increased amid the Russia-Ukraine conflict. Although the Redback has not yet achieved economies of scale, we expect additional contracts to come from Europe, Asia and the Middle East, considering replacement demand of aging and Russian-made armored vehicles. We believe armored vehicles will be the main growth driver in the next five years.
- **Propellant is one of Hanwha Aerospace's main export products**. We expect Hanwha Aerospace to win more propellant orders, considering its export capability. Its propellant met US and NATO specifications in 2019 and 2023, respectively. Moreover, Europe's ammunition supplies have been swiftly depleted due to the Russia-Ukraine war, leading to an increase in propellant orders.

Exhibit 70: Product Overview

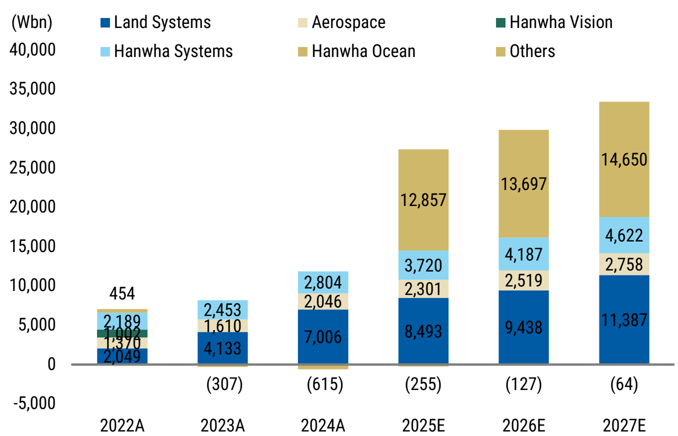
Product	Company	Operating Countries	Specifications	Characteristics
Artillery Systems - Self-propelled Howitzer			Speed/Rate of fire	
K9	Hanwha Aerospace	Poland, Norway, Türkiye, Finland, Estonia, Romania	Max. 67km / 6-8 rounds per min	- Cheaper than competitors on the back of economies of scale and 90% localization of core components - Production capability; 200 units annually - 50% of global self-propelled howitzer M/S
PzH 2000	KMW	Germany, Netherlands, Greece, Lithuania, Qatar, Hungary, Ukraine	Max. 60km / 10 rounds per min	production rate has slowed
M109A6	BAE Systems	United States	Max. 61km / 4 rounds per min	- M109A7 is the most recent version American self-propelled howitzer, developed in 1960s, is still in service in the US - US produces 60-70 units annually
CAESAR	Nexter	France, Saudi Arabia, Indonesia, Thailand, Czech	Max. 90km / 6 rounds per min	- Produce 6 units per month
Armored Vehicles - Infantry Fighting Vehicle			Speed	
RedBack	Hanwha Aerospace	Australia	Max 65km	-Cheap price / Equipped with advanced active protecting system (APS)
KF41 (Lynx)	Rheinmetall	Germany, Hungary (planned to), Ukraine	Max 70km	
CV90	BAE Systems	Sweden, Norway, Finland, Denmark	Max 70km	
Ajax		United Kingdom	Max 70km	
Air Defense Systems - MRLS			Maximum Range	
K239 Chunmoo	Hanwha Aerospace	Korea, Poland, UAE, Saudi Arabia	80km / 12 rockets per min	- It has double ammunition capacity and can fire up to 12 rounds of 230mm guided missiles. It boasts mobility with a top speed of 80km/h and rapid response capability, able to fire the first round within 7 minutes of arriving at the firing position - More compatible than its peers
M142 HIMARS	Lockheed Martin	US, Singapore, UAE, Romania	6 rockets per min	
M270 MLRS	Lockheed Martin	US, France, Korea, Germany, United Kingdom	70km / 12 rockets per min	
BM-30	Russia		90km / 12 rockets per min	

Source: Company report, various media articles, Morgan Stanley Research.

Hanwha Aerospace: Earnings Outlook

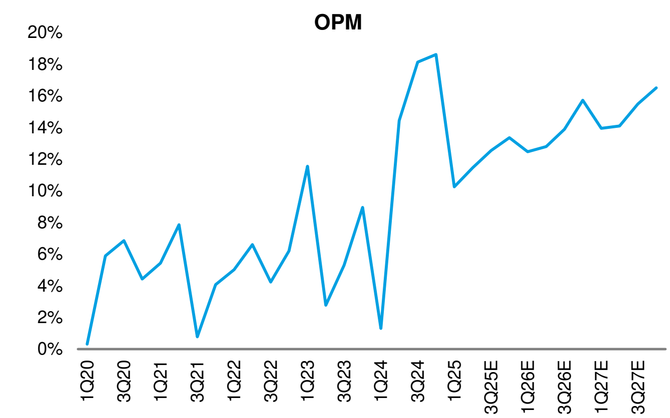
We expect Hanwha Aerospace to deliver strong top-line growth, leveraging its leading market share in self-propelled artillery (K9's global self-propelled artillery market share as of May 2025 was 50%), rapid delivery capabilities, diversified portfolio, and pricing power. We see overseas sales as the key engines, with export revenue within the land defense division projected to reach 69.6% in 2025, and we expect this favorable sales mix to be sustained through at least 2027. Our forecast for new overseas orders in 2026 is W7.7tn, as we anticipate further contract momentum from Poland (third-phase), Norway, and the Middle East. We are also constructive on additional Chunmoo orders, given rising interest from Middle Eastern markets and Peru. We see substantial upside to order inflow going forward. As more of these high-margin export contracts are delivered from 2026, we expect the overseas revenue contribution to increase further, resulting in operating profit margin expansion to 14% in 2026.

Exhibit 71: Hanwha Aerospace: Revenue by division



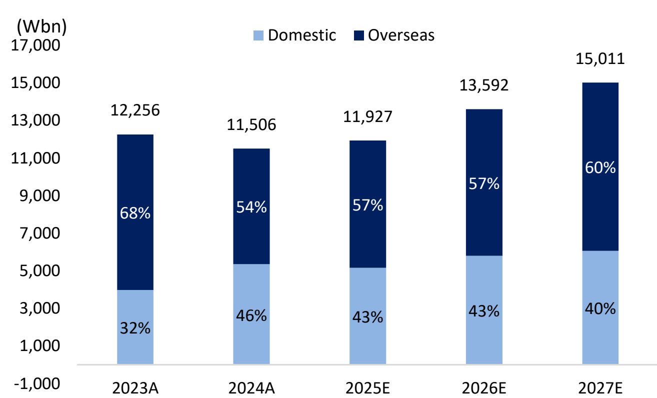
Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Exhibit 72: Hanwha Aerospace: Land system revenue



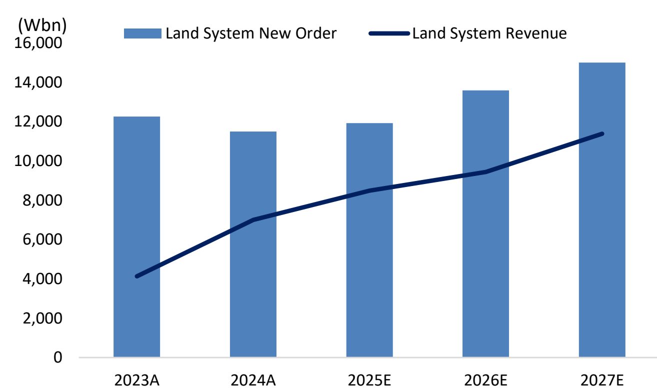
Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Exhibit 73: Hanwha Aerospace: New land system orders



Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Exhibit 74: Hanwha Aerospace: Land system backlog vs. backlog/revenue



Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Hanwha Aerospace: Valuation Methodology

Our price target (which is also our base case scenario value) of W1,250,000 is derived from SOTP valuation model, where (1) the value of core businesses (land systems and aerospace) is derived from EPM, assuming a cost of equity of 8.8% and a terminal growth rate of 3.5%, (2) we apply a 25% discount to its listed subsidiaries taking into the account the tax factor, and (3) we use equity book value for unlisted affiliates. Our scenario analysis reflects different cases in (1) % of global military expenditure to GDP and (2) the global market share of Korean defense companies.

Our price target implies 27.9x 2026E P/E, which is slightly higher than its global peer average of 27.2x. We think this valuation is manageable considering (1) strong order/export momentum backed by its competitive and comprehensive product portfolio potentially penetrating into Western countries, (2) earnings growth momentum backed by a strong order backlog mainly from its land systems business, and (3) the company's commitment to transforming into a pure defense play through structural change.

Exhibit 75: Hanwha Aerospace base case: SOTP valuation

(W bn)				EPM EV
Operating Segment (Land Systems & Aerospace)				**58,807**
	No. of Shares (mn)	**Latest Price**	**Market Value**	**Discounted BV**
				25%
Hanwha Systems	88.3	54,900	4,847	3,635
Hanwha Ocean	93.3	83,500	7,789	5,841
Listed Affiliates				**9,477**
				Latest BV
Unlisted Affiliates and Others				**2,179**
Total EV				**70,463**
(-) Net debt				6,998
Equity Value				63,465
No of shares (Mil)				51
Price Target (Won)				**1,250,000**
Latest Share Price				910,000
Implied Upside/Downside				26%

Source: Morgan Stanley Research estimates.

Bull case: Our bull case scenario value is 12% higher than our base case. This scenario assumes higher growth in global military expenditure, combined with robust penetration into Western market to provide strong upside to order and earnings momentum. In this scenario, we assume consolidated sales rise at a 16% CAGR (2024-33e), with land systems overseas sales expanding at a 23% CAGR (2024-33e). We estimate an average OPM of 22% for 2028-33e. We assume 3.3% of global military expenditure to GDP, vs. 3% in our base case, leading to higher overseas revenue, higher revenue and margins vs. our base case. Under this scenario, our bull case value per share comes out to W1,400,000, implying 31x 2026E P/E.

Exhibit 76: Hanwha Aerospace bull case: SOTP valuation

(W bn)				EPM EV
Operating Segment (Land Systems & Aerospace)				**66,442**
	No. of Shares (mn)	**Latest Price**	**Market Value**	**Discounted BV**
				25%
Hanwha Systems	88.3	54,900	4,847	3,635
Hanwha Ocean	93.3	83,500	7,789	5,841
Listed Affiliates				**9,477**
				Latest BV
Unlisted Affiliates and Others				**2,179**
Total EV				**78,098**
(-) Net debt				6,998
Equity Value				71,100
No of shares (Mil)				51
Price Target (Won)				**1,400,000**
Latest Share Price				910,000
Implied Upside/Downside				41%

Source: Morgan Stanley Research estimates.

Bear case: Our bear case scenario value is 40% lower than our base case. This scenario assumes slower growth in defense spending, as well as a potential shortfall in new orders stemming from EU self-procurement. Here we assume 2.7% of global military expenditure to GDP, vs. 3% in the base case, which leads to lower overseas land systems revenue, and results in lower revenue and margins, vs. our base case. Under this scenario, consolidated sales rise at a 14% CAGR (2024-33e), with overseas sales at a 20% CAGR (2024-33e). Our bear case value per share comes out to W755,000, implying 16.8x 2026E P/E.

Exhibit 77: Hanwha Aerospace bear case: SOTP valuation

(W bn)				EPM EV
Operating Segment (Land Systems & Aerospace)				**33,679**
	No. of Shares (mn)	**Latest Price**	**Market Value**	**Discounted BV**
				25%
Hanwha Systems	88.3	54,900	4,847	3,635
Hanwha Ocean	93.3	83,500	7,789	5,841
Listed Affiliates				**9,477**
				Latest BV
Unlisted Affiliates and Others				**2,179**
Total EV				**45,335**
(-) Net debt				6,998
Equity Value				38,337
No of shares (Mil)				51
Price Target (Won)				**755,000**
Latest Share Price				910,000
Implied Upside/Downside				-24%

Source: Morgan Stanley Research estimates.

Korea Aerospace Industries (047810.KS): Poised to Fly; OW

We initiate coverage on Korea Aerospace Industries at Overweight with a price target of W110,000, implying 19% upside to the current price. Our price target is derived from economic profit valuation and implies 25.0x 2026E P/E, at a premium to its domestic defense peers' average multiple of 23.4x given (1) our expectations of double-digit earnings growth in 2025E/26E, backed by gradual deliveries of export orders for FA-50 (signed in 2023-24), (2) further export opportunities to Europe/the Middle East/Asia with its products' cheap prices and compatibility with Lockheed Martin (F-16), and (3) attractive valuation vs peers.

Investment thesis: The company has three main businesses, of which (1) 46.9% of its total sales come from fixed-wing aircraft, (2) 19.3% from rotary-wing aircraft, and (3) 23.7% from the aerostructure business. We expect its total sales and OP to grow 13%, and 34% YoY in 2025, respectively, largely driven by growing sales contribution from FA-50 exports and mass production of KF-21. The company has also stated that it set to export its training jet and rotary-wing aircraft, which thinks is evidence of its competitiveness and proven track record.

Earnings overview: In 2024, both sales and operating profit decreased due to base effects following one-time FA-50 deliveries to Poland in 2023. We forecast significant sales growth in 2026 and 2027, underpinned by the ramp-up in domestic production of KF-21 and large-scale deliveries of FA-50 to Poland. We expect the export revenue mix to rise from 33% in 2025 to 37% in 2026 and 2027 each, allowing the company to sustain strong operating margins. We also expect further order wins for the FA-50, driven by its superior performance and the cost-advantage rollout of the KF-21 program, which should result in incremental domestic order volume. Accordingly, we estimate new orders to reach W8.7tn in 2026 and W10tn in 2027.

Exhibit 78: Korea Aerospace Industries: Key financial metrics

Korea Aerospace Industries (047810.KS)	
S.Korea Industrials | S.Korea	
Stock Rating	**Overweight**
Industry View	Attractive
Price Target (W)	**110,000**
Up/Downside to Price Target (%)	19%
Share Price (Jul 23, 2025)	92,200
Mkt. Cap (Bn)	9.0

Fiscal Year Ending	12/24	12/25e	12/26e	12/27e
EPS (W)	1,765	3,099	4,406	5,261
EPS (W)*	1,765	2,517	3,933	5,180
New Order (Wbn)	4,902	7,568	8,688	9,888
Backlog (Wbn)	24,699	28,136	31,291	34,837
Backlog/Sales	6.8%	6.9%	5.7%	5.5%
Sales (Wbn)	3,634	4,103	5,533	6,343
OP (Wbn)	241	324	479	581
Sales Growth	-4.9%	12.9%	34.9%	14.6%
OPM	6.6%	7.9%	8.7%	9.2%
ROE	10.1%	15.9%	19.0%	19.0%

Source: *=Refinitiv Consensus, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Valuation: Korea Aerospace Industries is currently trading at 31.3x and 22.2x P/E on our 2025 and 2026 EPS estimates, respectively. The company is trading at 26.7x 12M Fwd P/E, which is higher than its peers' average of 21.4x as well as its own 10-year average of 23.5x. We believe that its valuation premium is appropriate considering the development of KF-21 and potential export opportunities with growing demand for 4.5 generation aircraft, which we believe would work as a key catalyst.

Key downside risks: (1) order delays or cancellations, (2) delays in commercial aircraft delivery for Boeing and Airbus, and (3) pushback in purchase plans due to priorities for other weapon systems over aircraft.

Korea Aerospace Industries: Financial Summary

Exhibit 79: Korea Aerospace Industries: Financial summary

Korea Aerospace Industries (047810.KS)

Won billion; Years Ending December 31

Key Metrics	2023	2024	2025e	2026e	2027e
New Order	**4,632**	**4,902**	**7,568**	**8,688**	**9,888**
Domestic	2,617	2,139	3,574	3,800	4,200
Export	1,240	178	3,002	3,500	4,300
Aerostructure	775	2,585	1,322	1,388	1,458
Backlog	**21,776**	**24,699**	**28,136**	**31,291**	**34,837**
Domestic	9,067	9,113	10,945	12,434	13,741
Export	5,329	5,292	6,905	8,376	10,355
Aerostructure	7,379	10,295	10,321	10,562	10,873
Backlog/Sales	**5.7**	**6.8**	**6.9**	**5.7**	**5.5**

Income Statement	2023	2024	2025e	2026e	2027e
Revenue	**3,819**	**3,634**	**4,103**	**5,533**	**6,343**
Gross Profit	**464**	**481**	**630**	**858**	**1,015**
SG&A	**216**	**241**	**306**	**378**	**434**
Operating Profit	**248**	**241**	**324**	**479**	**581**
Other Non-Operating gains	17	9	7	9	9
Other Non-Operating losses	21	27	6	2	2
Financial Income	112	121	148	94	101
Financial Expenses	88	144	110	66	76
Gains/Loss on Subsidiaries	0	(2)	(4)	(4)	(4)
Pre-tax Profit	**269**	**197**	**359**	**510**	**609**
Income tax	47	26	66	93	111
Minority Interest	(3)	(1)	(9)	(13)	(15)
Net Profit	**224**	**172**	**302**	**429**	**513**
EPS (W)	2,298	1,765	3,099	4,406	5,261

Balance Sheet (Wbn)	2023	2024	2025e	2026e	2027e
Cash & Equivalents	658	115	561	702	811
Trade Receivables	422	428	541	605	689
Inventories	1,736	2,359	2,519	2,668	2,863
Other current assets	1,864	2,537	3,408	3,695	4,075
Current assets	**4,681**	**5,439**	**7,028**	**7,669**	**8,439**
Investment in Associates & JV	13	15	11	7	3
PP&E	815	913	958	1,004	1,065
Intangible assets	804	846	854	823	776
Others assets	827	814	775	775	775
Total assets	**7,140**	**8,026**	**9,627**	**10,278**	**11,058**
ST Debt	21	414	392	607	772
Curr portion of LT Debt	379	29	26	26	26
Trade payables	688	982	1,565	1,518	1,548
Other current liabilities	3,312	3,490	3,844	3,954	4,094
Current liabilities	**4,400**	**4,915**	**5,826**	**6,104**	**6,440**
LT Debt	60	45	66	91	116
Bonds	150	550	1,050	1,050	1,050
Other long-term liabilities	910	789	772	772	772
Total liabilities	**5,520**	**6,298**	**7,714**	**8,017**	**8,377**
Shareholders' equity	**1,620**	**1,727**	**1,913**	**2,261**	**2,681**
Controlling Stake equity	**1,597**	**1,706**	**1,901**	**2,262**	**2,697**
Total liabilities & equity	**7,140**	**8,026**	**9,627**	**10,278**	**11,058**

Cash Flow (Wbn)	2023	2024	2025e	2026e	2027e
Operating cash flow	**(700)**	**(728)**	**420**	**118**	**170**
Net profit	224	172	302	429	513
Depreciation	91	82	87	91	96
Change in WC	(1,166)	(829)	(206)	(538)	(589)
Others	151	(153)	236	136	150
Investing cash flow	**(108)**	**(189)**	**(400)**	**(136)**	**(156)**
Acquisition of PP&E	(78)	(161)	(101)	(136)	(156)
Others	(30)	(28)	(299)	-	-
Financing cash flow	**(546)**	**369**	**426**	**159**	**97**
Change in ST debt	3	390	(22)	215	165
Change in LT debt	(492)	64	517	25	25
Dividend	(24)	(49)	(58)	(68)	(78)
Others	(33)	(36)	(10)	(13)	(15)
Net changes in cash	**(1,355)**	**(548)**	**445**	**141**	**110**

Ratios	2023	2024	2025e	2026e	2027e
Margins					
GPM	12.1%	13.3%	15.4%	15.5%	16.0%
EBITDA Margin	9.6%	9.5%	10.5%	10.9%	11.4%
OPM	6.5%	6.6%	7.9%	8.7%	9.2%
Net Margin	5.9%	4.7%	7.4%	7.8%	8.1%
Growth					
Sales	37.0%	-4.9%	12.9%	34.9%	14.6%
GP	24.1%	3.8%	30.8%	36.2%	18.3%
Net Profit	89.1%	-23.2%	75.5%	42.2%	19.4%
Returns					
ROE	14.0%	10.1%	15.9%	19.0%	19.0%
Gearing					
Net Debt/Equity	-9%	53%	51%	47%	43%
Total Debt/Equity	38%	60%	80%	78%	73%
Interest Cover	8.5	7.2	5.5	7.3	7.7
BVPS (Won)	16,388	17,500	19,499	23,205	27,666
P/B	6.71	6.29	5.64	4.74	3.98

Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

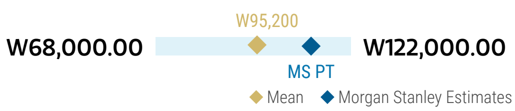
Risk Reward – Korea Aerospace Industries Ltd (047810.KS)

Poised to Fly

PRICE TARGET W110,000

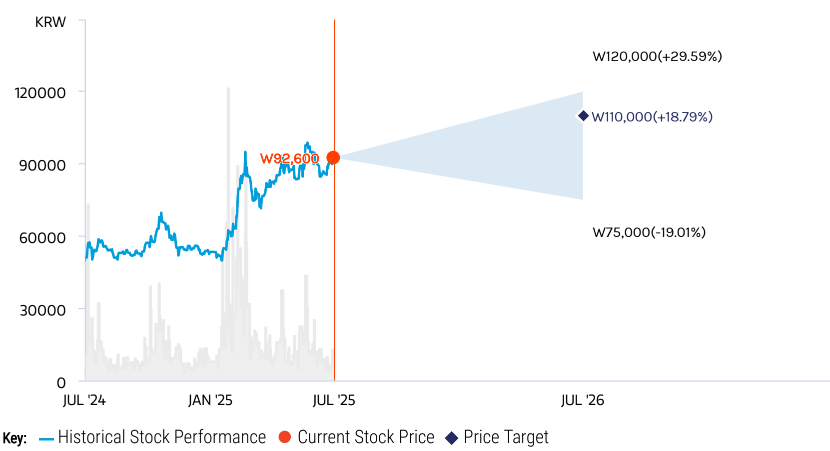
Base case, derived from an economic profit model (EPM) valuation, assuming a cost of equity of 8% and a terminal growth rate of 3.5%. Our scenario analysis reflects different cases in (1) % of global military expenditure to GDP, and (2) global market share of Korean defense companies.

Consensus Price Target Distribution



Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART

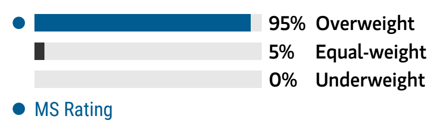


Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- We are OW Korea Aerospace Industries as we believe the KF-21 and FA-50 should satiate growing demand for cost-effective air power. We see upside potential in the warplanes segment, particularly in 4th- to 4.5th-generation warplanes, as countries look to reinforce their air power and replace aging aircraft.
- We expect more FA-50 new orders as its customers continue to be satisfied with the build quality, compatibility, and relatively low MRO costs.
- KF-21 is still in the final testing phase, but there is increasing interest from various countries as the testing process is showing strong performance results.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Contrarian: *Positive*

View descriptions of Risk Rewards Themes here

BULL CASE W120,000

Able to soar on KF-21 orders

Higher-than-expected growth in global military expenditure, combined with further order momentum from FA-50, provide strong upside to order and earnings momentum. We assume a 14.2% new order CAGR, 2024-33e, with a 12.1% sales CAGR, 2024-33e.

BASE CASE W110,000

Set to take flight

KF-X for KF-21 combat jet program proceeds smoothly and additional FA-50 orders from the more diverse regions support earnings momentum. We expect a 12.8% new order CAGR, 2024-33e, with 11.6% sales CAGR, 2024-33e.

BEAR CASE W75,000

Competition tougher than expected

Delay in the KF-X program's upgrade of the KF-21, combined with slower global defense spending, could dampen earnings momentum. We assume a 13% new order CAGR, 2024-33e, with 10.3% sales CAGR, 2024-33e.

Risk Reward – Korea Aerospace Industries Ltd (047810.KS)

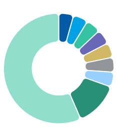
KEY EARNINGS INPUTS

Drivers	2024	2025e	2026e	2027e
New Order (Wbn) (W, bn)	4,902	7,568	8,688	9,888
Backlog (Wbn) (W, bn)	24,699	28,136	31,291	34,837
Revenue/Backlog (%) (%)	24.1	20.6	25.3	27.4
Overseas Revenue Growth (%) (%)	(38.8)	113.4	48.5	14.4
Gross Profit Margin (%) (%)	13.3	15.4	15.5	16.0

INVESTMENT DRIVERS

- Overseas order momentum for key product (FA-50)
- Domestic mass production of KF-21 combat jet + overseas orders
- Global military expenditure

GLOBAL REVENUE EXPOSURE



- 0-10% India
- 0-10% Japan
- 0-10% Latin America
- 0-10% MEA
- 0-10% Mainland China
- 0-10% North America
- 0-10% UK
- 10-20% Europe ex UK
- 60-70% APAC, ex Japan, Mainland China and India

Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

5/5 **MOST** — 3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

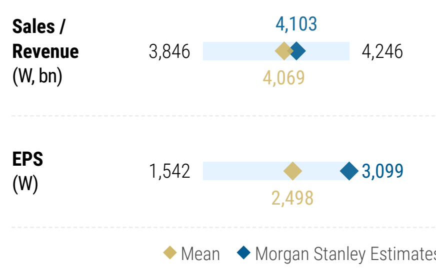
- Increased domestic delivery production and overseas exports
- Increase in airframe components driven by recovery in commercial aviation demand

RISKS TO DOWNSIDE

- Order delays or cancellations
- Delays in commercial aircraft delivery for Boeing and Airbus
- Pushback in purchase plans due to priorities for other weapon systems over aircrafts

MS ESTIMATES VS. CONSENSUS

FY 2025e



Source: Refinitiv, Morgan Stanley Research

Korea Aerospace Industries: Poised to Fly

#1. Expected earnings growth backed by strong backlog

We expect sales and OP to grow 13% and 34% YoY in 2025, respectively, largely driven by (1) domestic mass production of LAH and KF-21, and (2) FA-50 exports to Poland and Malaysia, which should begin to contribute to sales starting in 2025. KF-21, Korea Aerospace Industries' next-generation fighter jet, was developed in 2001 and completed operating tests in 2024. The KF-21 won its first mass production order from the Republic of Korea Air Force (ROKAF) last year, expected to be delivered 2026. Meanwhile, the company won W5.5tn of FA-50 orders from Poland in 2022 and Malaysia in 2023, which will largely contribute to sales from 2025. We expect Korea Aerospace Industries' export sales to contribute 33% in 2025 and 37% in 2026, with OPM reaching 7.9% in 2025 (vs. 6.6% in 2024).

Exhibit 80: Korea Aerospace Industries: Order Pipeline

	Unit	Value (Wtn)	2024	2025E	2026E	2027E	2028E	2029E	2030E
T-50 Series									
Indonesia	6								
FA-50 Series									
Poland	48	4.2							
Malaysia	18	1.3							
Philippines	12	1							
Uzbekistan	14	1.1							
Malaysia (2nd-phase)	18								
Saudi									
Egypt	36-100	5.0							
UAE									
Peru	20-24	1.0							
Slovakia	10	0.6							
KF-21									
Domestic	**40**	**3**							
Domestic (Block2)	**80**	**6**							
Indonesia, Philippines, UAE, Saudi									
KUH									
Iraq	2	0.1							
UAE	TBD								

Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Exhibit 81: Korea Aerospace Industries: Imminent Orders

FA-50 Series	
Egypt	Plans to to purchase 36-100 units of light combat jet and interested in FA-50
Peru	Highly interested in acquiring light combat jet (FA-50) to modernize its air force (W1tn)
Slovakia	Slovakia government reviews to acquire FA-50 to modernize its advanced training jet (L-39)
KUH	
UAE	KAI began to negotiate with UAE in 2023, but has not finalized yet

Source: Various articles (Global Economy, April 2025), company data, Morgan Stanley Research.

#2. Export opportunities to Europe, the Middle East, and Asia given its products' cheap prices and high compatibility with other jets

We expect Korea Aerospace Industries to win overseas orders for T-50, FA-50 in the near term and orders for the Korean utility helicopter (KUH) and KF-21 in the mid-to-long term. After securing its first KUH export order from Iraq, we think Korea Aerospace Industries is set to expand exports with imminent orders from UAE, which has historically focused on Korean defense names.

- **T-50/FA-50 has been** exported to Asia and Europe including Malaysia, the Philippines, and Poland. We expect the company to win further orders from the Middle East, Europe, and Latin America with imminent orders from Uzbekistan, where negotiations are currently underway. The competitiveness of Korea Aerospace Industries' FA-50 lies in its high compatibility with Lockheed Martin's F-16. It allows F-16 operators to significantly reduce transition costs. We discuss key imminent orders below;
  - Malaysia's project involves 36 units, with 18 units (US$1bn) being signed in 2023, and remained 18 units (US$1bn) to be signed by 2026, in our view.
  - Egypt's project involves 36 units (US$2bn), in which major competitors include the L-15 from China and M-345 from Italy. Korea Aerospace Industries' advantage lies in Egypt's existing operation of F-16 fighters (240 units). The Egypt contract is crucial for Korea Aerospace Industries' growth, in our view, as additional projects involving over 100 units, in the near term.
- **KF-21:** The KF-X project to develop a domestic fighter jet began in 2001 and culminated in its first mass production contract in 2024. The KF-21 has achieved localization of 65% parts and Korea Aerospace Industries plans to achieve 100% localization by 2030. We think this would enhance Korea Aerospace Industries' price competitiveness, unlike the FA/T-series (localization rate of 45-60%), which depends heavily on US-made components.
  - The initial production batch includes 40 units; a contract for 20 units was signed in 2024, with an additional order for the remaining 20 units singed in June this year. The total planned production volume of the Korea Air Force's KF-21 program is 120 units, including the 40 units already ordered, with deliveries scheduled from 2026 through 2032. While current deliveries are focused on the domestic market, early discussions to secure export orders from UAE, Peru, and the Philippines are underway, including facility investment-tailored solutions to meet production deadlines. Although we expect near-term momentum from FA/T series exports, establishing a domestic track record would open up long-term growth opportunities for Korea Aerospace Industries, in our view.
- **KUH:** Exports of KUH helicopters to Iraq marks the first time that South Korea has exported rotary-wing aircraft. We see this as a milestone that strengthens Korea Aerospace Industries' position for further overseas expansion. KUH's key competitiveness lies in its swift deliveries on the back of sufficient production capacity exceeding 30 units per year. We expect Korea Aerospace Industries to see KUH exports, with imminent orders from UAE.

Exhibit 82: Korea Aerospace Industries: Product Line-up

Name	Company	Type	Full loaded (kg)	Max Speed (Mach)	Stealth Technology	Price (USDmn)
FA-50 (T-50)	KAI	Advanced training & Combat jet	4,500	1.5	NA	30~35
T-7A	US	Training jet	3,000	1.2	NA	29
M-346FA	Italy	Advanced training & combat jet	3,000	1.2	NA	25~30
L-15B	China	Combat jet	3,500	1.4	NA	25
KF-21	KAI	4.5th generation combat jet	7,700	1.8	limited availability	65
F-35	Lockheed Martin	5th generation combat jet	8,160	1.6	O	150~200
F-16	US	Combat jet	7,700	2	limited availability	80~90
F-22	Lockheed Martin	5th generation combat jet	8,200	2.25	O	143
Rafale		4.5th generation combat jet	9,500	1.8	limited availability	115~124

Source: Company data, Morgan Stanley Research.

Exhibit 83: Korea Aerospace Industries: Upgrade Roadmap for the KF-21

Type	Timeline	Planned Mass Production Volume	Armament
Block 1	2026-2028	40	Air-to-Air Missiles
Block 2	2028-2032	80	Addition of Air-to-Ground and Anti-Ship Munitions
Block 3	2032~	-	Integration of indigenous short-and medium range Air-to Air-Missiles

Source: Company data, Morgan Stanley Research.

Exhibit 84: Korea Aerospace Industries KF-X Project: Key Participants

Area	Company	Details
System Integration	KAI	Overall program management and final assembly
Electronic Warfare	LIG Nex1	EWX, RWR, CMDS
Guided munitions		Development of air-to-air, air-to-ground, and anti-ship missiles
Engine	Hanwha Aerospace	Domestic licensed production of GE F414
Optical sensors and radar	Hanwha Systems	AESA Radar, IRST, EOTGP

Source: Company data, Morgan Stanley Research.

Investment positives and concerns

Investment positives

We see the warplanes segment having sizable growth potential in 4-4.5 generation warplanes as countries are looking to reinforce their air power and also replace aging aircraft. Only the US, China and Russia have been able to independently develop 5th-generation fighter platforms but KF-21 block 2 and block 3 are expected to be almost fully self-developed 5th gen fighter platforms.

(1) We expect to see more FA-50 orders as customers remain satisfied with the build quality, compatibility, and relatively lower MRO (Maintenance, Repair, and Operations) costs. ASEAN countries have been the main purchasers of T-50/FA-50 planes but we believe potential buyers may come from other markets such as MENA, Eastern Europe and/or LATAM.

(2) KF-21 is still in the final testing phase, but it is getting increasing interest from various countries as the testing process is showing strong performance results. KF-21 Block 1 will be strictly an air superiority combat aircraft with 40 planes being delivered to the Korean Air Force but Block 2 will be a multi-role fighter and 80 planes are to be delivered to the KAF. Ultimately, Block 3 KF-21 EX will be a 5-5.5th gen multi-role fighter. Generally, jet fighter programs take a long time to reach completion meaning that KF-21 should be competitive in the global market.

(3) Revenue growth becomes more sustainable as strong export sales leads to continued growth in MRO related revenue stream while in the future there would be potential for upgrade related requests from existing buyers, which again leads to revenue realization.

Investment concerns

(1) A reversal by Eastern European countries and/or other regions such as MENA and ASEAN, which are Korea Aerospace Industries' main customers, regarding their policy path to increase defense budgets. This could come about from many angles but the biggest risk would be if the US takes a different tack on allies' defense spending.

(2) Replacement cycle for existing warplanes gets pushed back due to various issues such as key customer nations experience budgeting difficulties, political instability and/or other factors.

(3) Upgrades to existing warplanes such as F-16s (upgrade version F-16V) and F-15s (F-15EX) are deemed to be better options for potential buyers of KF-21.

(4) Domestic engine and/or cooperative engine development experiences hitches leading to delayed self-sufficiency for KF-21.

Share price analysis

Korea Aerospace Industries was not a part of the Korean defense names' share price rally in 2024 (Korea Aerospace Industries: + 9.8% vs Korean Defense: +71.9%), mainly due to weaker order momentum and muted earnings improvement due to the base effect in 2023. However, its share price has surged since January 2025, up 67.9% YTD following the announcement of Europe's rearmament plans indicating significant investments. We expect its sales and OP to grow 13% and 34% YoY in 2025, respectively, as (1) FA-50 exports to Poland and Malysia begin to contribute to sales from 2025, and (2) mass production of KF-21, a 4.5th-generation Korean fighter jet, begins. Furthermore, we expect Korea Aerospace Industries to win new orders for its training/fighter jet (T-50, FA-50) and helicopter, especially from the Middle East and Asia on the back of its competitiveness in terms of price and specifications.

Product overview

Korea Aerospace Industries has three main business units: (1) fixed-wing aircraft, (2) rotary-wing aircraft, and (3) aerostructure, which accounted for 46.9%, 19.3%, and 10.2% of sales in 2024, respectively. Key products in fixed-wing aircraft include training jets (T-50, FA-50) and its fighter jet (KF-21).

- **T-50** is a basic training jet that was co-developed with Lockheed Martin. FA-50 is a fighter jet version of the T-50, which supports multi-roles as both an advanced training jet and a light combat jet. The competitive edge of FA-50 lies in (1) high compatibility with Lockheed Martin jet (F-16, F-35) by sharing a lot of parts, and (2) cheap price. On the back of these strengths, FA-50 is attractive to those companies that operate the F-16 series and need high-performance aircraft on a limited budget.
- **KF-21** is a combat fighter jet developed by Korea Aerospace Industries and ROKAF. The KF-21 won its first mass production order from ROKAF in 2H24. We expect sales of KF-21 to be booked from 2026.

- **Rotary-wing aircraft includes Surion** (KUH or korean utility helicopter), LAH (light armed helicopter), and LCH (,ight civil helicopter). We expect more export opportunities for KUH following the first overseas order from Iraq in 2024.
- **Aerostructure** unit includes airframe parts and wings for commercial aircraft. Boeing and Airbus are Korea Aerospace Industries' main clients.

Exhibit 85: Korea Aerospace Industries: Product Line-up

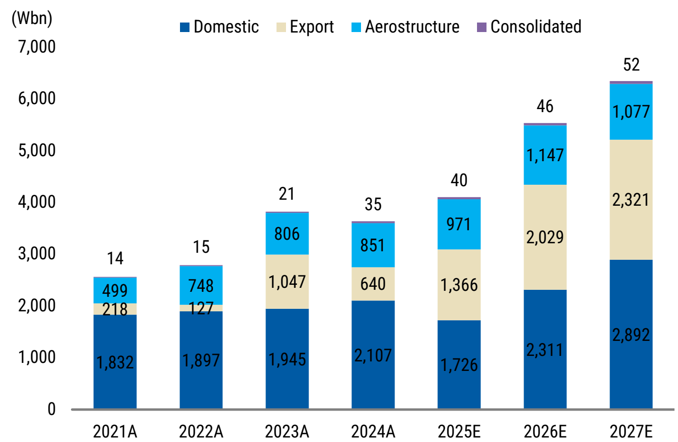
Type	Product	Price	Operating Country	Note	
Fixed Wing Aircraft	KT-1			Basic training jet	
	T-50		South Korea, Indonesia, Iraq, Thailand	Advanced training jet after basic training Co-developed with Lockheed Martin in 2005	Key strengths are (1) high compatibility with Lockheed Martin jet (F-16, F-35) and (2) proven performance
	FA-50		South Korea, Philippines, Poland	Combat jet version of T-50 Multi role as both training / combat jet FA50 shares a lot of parts with Lockheed Martin - export license must be obtained from the US government, limiting sales to countries with stale diplomatic relations with the US However, the KF-21 fighter jet being developed in Korea and already localized core parts	Key strength is cheap price. Despite the inferior specifications than Lockheed Martin xx, it would be a good choice for countries that need similar fighter jet with cheaper price rather than performance
	KF-21		South Korea	5th generation of combat fight jet	
Rotary Wing Aircraft	LAH/LCH		South Korea		
	Surion (KUH)		South Korea, Iraq	Transport helicopter operated by National Police Agency and etc	Won the first oversea project from Iraq in 2024 and expected to expand export orders

Source: Company data, Morgan Stanley Research.

Korea Aerospace Industries: Earnings Outlook

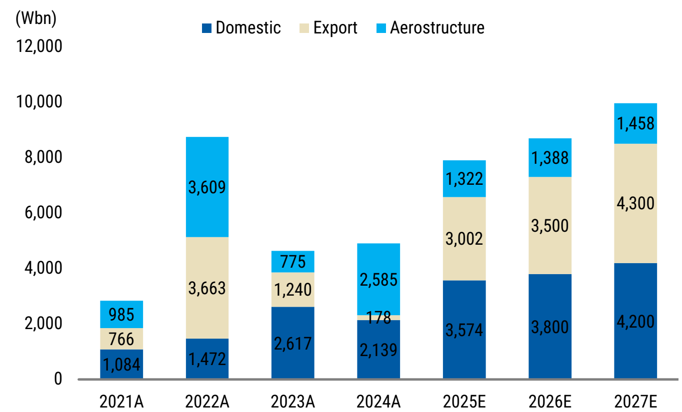
We forecast significant sales growth in 2026 and 2027, underpinned by the ramp-up in KF-21 domestic mass production and large-scale FA-50 deliveries to Poland. In our view, the export revenue mix rises from 33% in 2025 to 37% in both 2026 and 2027, allowing the company to sustain strong operating margins. We expect further order wins for the FA-50, driven by its superior performance and cost advantage rollout of the KF-21 program to result in incremental domestic order volume. Accordingly, we estimate new orders at W8.7tn in 2026 and W9.9tn in 2027.

Exhibit 86: Korea Aerospace Industries: Revenue by division



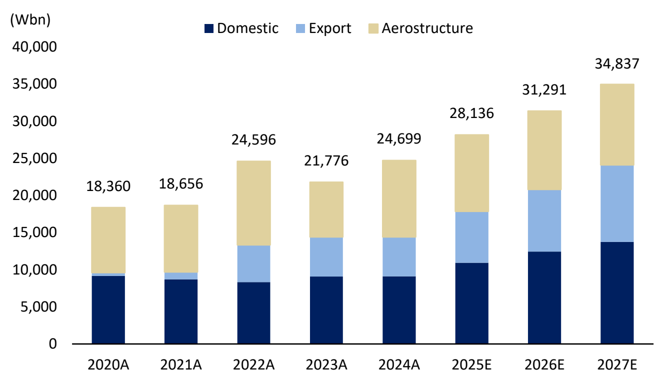
Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Exhibit 87: Korea Aerospace Industries: New orders by division



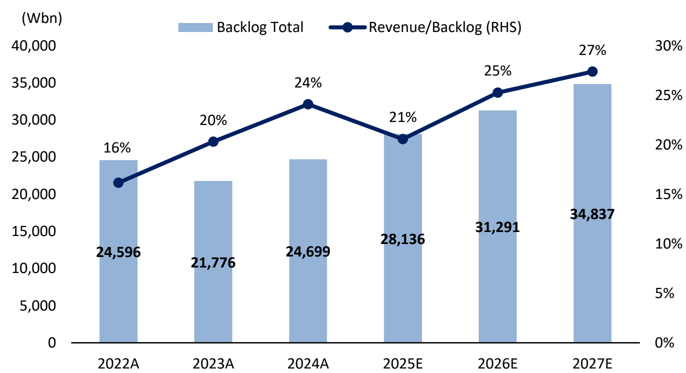
Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Exhibit 88: Korea Aerospace Industries: Backlog by division



Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Exhibit 89: Korea Aerospace Industries: Backlog vs. backlog/ revenue



Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates

Korea Aerospace Industries: Valuation Methodology

Our price target (which is also our base case scenario value) of W110,000 is derived from economic profit (EPM) valuation, assuming a cost of equity of 8% and a terminal growth rate of 3.5%. Our scenario analysis reflects different cases in (1) % of global military expenditure to GDP (3.3% vs. 3% vs. 2.7% for Bull, Base, Bear case scenarios, respectively), and (2) global market share of Korean defense companies.

Our price target implies 25.0x 2026E P/E, which is higher than its peer average of 23.8x. We believe this valuation is justified by (1) our expectations of double-digit earnings growth in 2025E/26E, backed by gradual deliveries of export orders for FA-50 (signed in 2023-24), and (2) further export opportunities to Europe/the Middle East/Asia with its products' cheap prices and F-16 compatibility.

Bull case: Our bull case scenario value is 9% higher than our base case. In this scenario, we assume consolidated sales rise at a 12.1% CAGR (2024-33e), with overseas sales growing at a 22.4% CAGR (2024-33e). We estimate an average OPM of 12.5% for 2028-33e. We assume 3.3% of global military expenditure to GDP, vs. 3% in our base case, leading to higher overseas revenue vs. our base case. Under this scenario, our bull case value per share comes out to W120,000, which implies 27.2x 2026E bull case P/E.

Bear case: Our bear case scenario value is 32% lower than our base case. Here we assume 2.7% of global military expenditure to GDP, vs. 3% in the base case, which leads to lower overseas revenue and results in lower revenue and margins vs. our base case. Under this scenario, consolidated sales rise at a 10.3% CAGR (2024-33e), with overseas sales at a 20.4% CAGR (2024-23e). Our bear case value per share comes out to W75,000, which implies 17.0x 2026E bear case P/E.

Exhibit 90: Bull/Bear/Base Case Scenario Analysis

	Value	P/E	EPS	Implied P/E	BVPS	Implied P/E	Implied P/B
Bull Price	**120,000**		4,807	25.0	21,272	27.2	6.2
Base Price	**110,000**	25.0	4,406	25.0	19,499	25.0	5.6
Bear Price	**75,000**		3,004	25.0	13,295	17.0	3.8

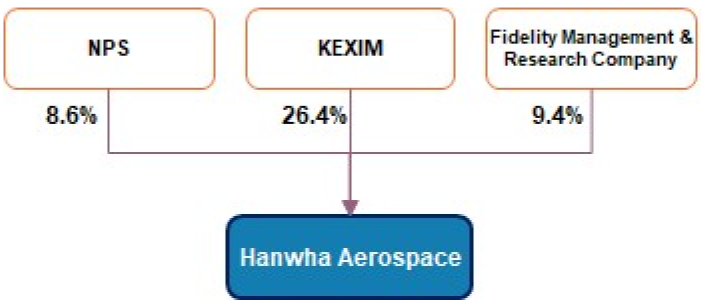
	Base	Bull	Bear
PV of Forecast Period	**2,013**	**2,240**	**1,793**
PV of Continuing Value	7,913	8,671	4,724
Equity Value (W bn)	11,654	12,638	8,244
- Net debt (W bn)	923	923	923
Equity Value (W bn)	10,731	11,715	7,321
No of shares (Mil)	**97**	**97**	**97**
Price Target (Won)	**110,000**	**120,000**	**75,000**

Source: Morgan Stanley Research estimates.

Korea Aerospace Industries: Company Background

Korea Aerospace Industries is South Korea's sole aircraft design and manufacturing company. It was established in 1999 through the integration of the aerospace divisions of Samsung Aerospace Industries, Daewoo Heavy Industries, and Hyundai Space & Aircraft. Its business areas include complete aircraft exports (53% of total sales), and airframe components manufacturing (24% of total sales). Major customers include South Korea's Defense Acquisition Program Administration (DAPA), Airbus, Boeing, and Lockheed Martin. Military products include T-50/ FA-50 series jets, KF-21 fighters, KUH/LUH helicopters, and MRO services. Civilian products comprise Korean-style medium-sized aircraft and structural components.

Exhibit 91: Korea Aerospace Industries: Major Shareholders



Source: Company report, Morgan Stanley Research. Note: Data as of 1Q25.

LIG Nex1 (079550.KS): Waiting for the Next Boost; EW

We initiate coverage on LIG Nex1 at Equal-weight with a price target at W600,000, implying 1% upside from the current price. Our price target derived on economic profit model (EPM) valuation and implies 32.6x 2026E P/E, in line with domestic peers but lower than its current P/E of 38x.

Investment thesis: LIG Nex1 has received many orders for its MSAM-II (surface-to-air missile system), with UAE and Saudi Arabia making sizable new orders. This has led to a strong upward trend in its share price. We believe positives are priced in for now and we await more positive news on the new order front.

- We anticipate further new orders to come from MENA, mainly from countries looking to diversify away from the US, although visibility on when these new orders will come out is low at the moment.
- We see potential orders coming from Europe, as the MSAM II could address European countries' requirement for rapid delivery, especially when alternatives such as the PAC-3 are suffering from long lead times (+7 years). However, whether this leads to actual new orders will have to be seen.

Earnings overview: We believe LIG Nex1 will continue to see top-line growth as revenue contribution from exports of M-SAM II to UAE and Saudi Arabia meaningfully kick in starting next year. In terms of orders, we anticipate that heightened air defense needs in the Middle East – largely from countries looking to diversify away from the US will drive demand for LIG Nex1's M-SAM II. As a result, we estimate new orders to rise to W8.3tn in 2026 and W9tn in 2027. As M-SAM II exports ramp up, we expect LIG Nex1's operating profit margin to improve to 15.1% in 2025 and further to 16.2% in 2026, with delivery recognition ramping up for Egypt, UAE and Saudi Arabia.

Exhibit 92: LIG Nex1: Key financial metrics

LIG Nex1 (079550.KS)	
S.Korea Industrials | S.Korea	
Stock Rating	**Equal-weight**
Industry View	Attractive
Price Target (W)	**600,000**
Up/Downside to Price Target (%)	1%
Share Price (Jul 23, 2025)	593,000
Mkt. Cap (Bn)	13.0

Fiscal Year Ending	12/24	12/25e	12/26e	12/27e
EPS (W)	10,173	13,090	18,429	23,489
EPS (W)*	10,078	13,980	18,497	25,062
New Order (Wbn)	3,737	8,000	8,300	9,000
Backlog (Wbn)	20,053	23,962	27,378	30,825
Backlog/Sales	6.1%	5.9%	5.6%	5.6%
Sales (Wbn)	3,277	4,091	4,884	5,553
OP (Wbn)	230	347	467	616
Sales Growth	42.0%	24.8%	19.4%	13.7%
OPM	7.0%	8.5%	9.6%	11.1%
ROE	18.3%	20.7%	23.4%	23.8%

Source: *=Refinitiv consensus, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Valuation: LIG Nex1 is trading at 45.3x and 32.2x P/E on our 2025 and 2026 EPS estimates, respectively. The company is trading at 38.2x 12M Fwd P/E, which is higher than average of other domestic companies (23.3x). LIG Nex1 is also trading near its historical highs since 2016, and is currently positioned well above +1 standard deviation band (19.97x). We believe that much of the positives are already priced in, and new export orders will be necessary to drive further re-rating.

Key risks: Downside: (1) Delay in weapon system development and production, and (2) decrease in global defense spending and domestic defense capability enhancement budget. **Upside**: (1) increase of domestic defense budget and overseas exports of weapon systems, (2) surge in demand for air defense systems as the geopolitical environment deteriorates, and (3) greater increase in global defense spending and domestic defense capabilities' budgets.

LIG Nex1: Financial Summary

Exhibit 93: LIG Nex1: Financial summary

LIG Nex1 (079550.KS)

Won billion; Years Ending December 31

Key Metrics	2023	2024	2025e	2026e	2027e
New Order	**9,637**	**3,737**	**8,000**	**8,300**	**9,000**
Backlog	**19,593**	**20,053**	**23,962**	**27,378**	**30,825**
Backlog/Sales	**8.5**	**6.1**	**5.9**	**5.6**	**5.6**

Income Statement	2023	2024	2025e	2026e	2027e
Revenue	**2,309**	**3,277**	**4,091**	**4,884**	**5,553**
Gross Profit	**347**	**453**	**617**	**790**	**983**
SG&A	**160**	**223**	**271**	**323**	**367**
Operating Profit	**186**	**230**	**347**	**467**	**616**
Other Non-Operating gains	26	86	28	32	32
Other Non-Operating losses	21	50	29	20	40
Financial Income	10	28	9	16	22
Financial Expenses	15	85	9	8	8
Gains/Loss on Subsidiaries	0	0	1	1	1
Pre-tax Profit	**185**	**210**	**348**	**489**	**624**
Income tax	10	(7)	67	94	120
Minority Interest	-	(5)	(5)	(7)	(8)
Net Profit	**175**	**222**	**285**	**402**	**512**
EPS (W)	8,029	10,173	13,090	18,429	23,489

Balance Sheet (Wbn)	2023	2024	2025e	2026e	2027e
Cash & Equivalents	445	547	668	907	1,212
Trade Receivables	309	425	555	584	620
Inventories	229	364	449	424	445
Other current assets	1,553	2,560	2,779	2,779	2,779
Current assets	**2,536**	**3,895**	**4,450**	**4,693**	**5,055**
Investment in Associates & JV	1	3	6	7	9
PP&E	894	1,371	1,483	1,614	1,768
Intangible assets	188	629	636	638	610
Others assets	197	227	215	244	244
Total assets	**3,816**	**6,125**	**6,790**	**7,197**	**7,686**
ST Debt	108	163	170	170	170
Curr portion of LT Debt	75	41	59	79	79
Trade payables	460	727	1,158	1,238	1,300
Other current liabilities	2,007	3,686	3,707	3,707	3,707
Current liabilities	**2,651**	**4,617**	**5,095**	**5,194**	**5,256**
LT Debt	20	51	72	52	52
Bonds	-	137	137	137	137
Other long-term liabilities	93	83	82	82	82
Total liabilities	**2,763**	**4,888**	**5,385**	**5,465**	**5,527**
Shareholders' equity	**1,052**	**1,237**	**1,405**	**1,732**	**2,159**
Controlling Stake equity	**1,052**	**1,211**	**1,380**	**1,714**	**2,150**
Total liabilities & equity	**3,816**	**6,125**	**6,790**	**7,197**	**7,686**

Cash Flow (Wbn)	2023	2024	2025e	2026e	2027e
Operating cash flow	**467**	**952**	**376**	**586**	**642**
Net profit	175	222	285	402	512
Depreciation	59	68	79	90	98
Change in WC	173	551	(35)	76	5
Others	60	112	46	18	26
Investing cash flow	**(97)**	**(978)**	**(229)**	**(272)**	**(252)**
Acquisition of PP&E	(59)	(497)	(174)	(222)	(252)
Others	(38)	(481)	(55)	(51)	-
Financing cash flow	**(177)**	**122**	**(26)**	**(74)**	**(85)**
Change in ST debt	107	51	28	20	-
Change in LT debt	(200)	153	30	(20)	-
Dividend	(33)	(43)	(63)	(68)	(76)
Others	(51)	(40)	(20)	(7)	(8)
Net changes in cash	**194**	**96**	**120**	**239**	**305**

Ratios	2023	2024	2025e	2026e	2027e
Margins					
GPM	15.0%	13.8%	15.1%	16.2%	17.7%
EBITDA Margin	11.1%	9.4%	10.8%	11.8%	13.4%
OPM	8.1%	7.0%	8.5%	9.6%	11.1%
Net Margin	7.6%	6.8%	7.0%	8.2%	9.2%
Growth					
Sales	4.0%	42.0%	24.8%	19.4%	13.7%
GP	6.2%	30.5%	36.4%	28.0%	24.4%
Net Profit	42.3%	26.7%	28.7%	40.8%	27.5%
Returns					
ROE	16.6%	18.3%	20.7%	23.4%	23.8%
Gearing					
Net Debt/Equity	-23%	-13%	-16%	-27%	-36%
Total Debt/Equity	19%	32%	31%	25%	20%
Interest Cover	19.3	19.2	51.3	62.3	82.0
BVPS (Won)	48,286	55,550	63,323	78,651	98,640
P/B	12.43	10.80	9.48	7.63	6.08

Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

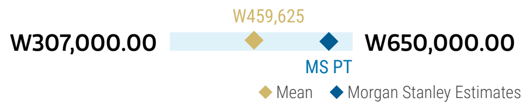
Risk Reward – LIG Nex1 (079550.KS)

Waiting for the Next Leg Up

PRICE TARGET W600,000

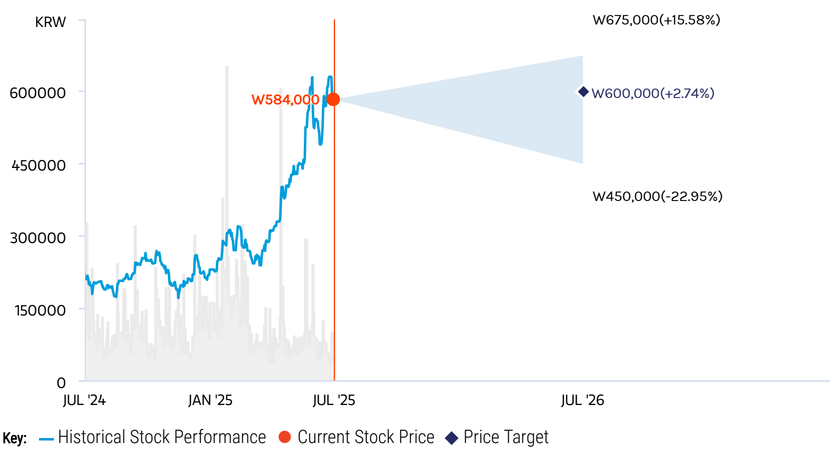
Base case, derived from a economic profit model (EPM) valuation, assuming a cost of equity of 8% and a terminal growth rate of 3.5%. Our scenario analysis reflects different cases in (1) % of global military expenditure to GDP, and (2) global market share of Korean defense companies.

Consensus Price Target Distribution



Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART

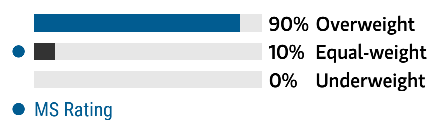


Source: Refinitiv, Morgan Stanley Research

EQUAL-WEIGHT THESIS

- Orders for LIGNex1's MSAM II (surface-to-air missile system), particularly from UAE and Saudi Arabia, have caused its share price to surge in 2025. We believe the positives are priced in for now, and we await more visibility on the new order front.
- We anticipate further new orders to come from MENA, mainly from countries looking to diversity away from the US, although visibility on when these new orders will come out is low at the moment.
- LIG Nex1's MSAM II could address European countries' requirement for rapid delivery, especially when alternatives such as PAC-3 are suffering from long lead times (+7 years). However, whether this leads to actual new orders will have to be seen.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Contrarian: *Positive*

View descriptions of Risk Rewards Themes here

BULL CASE W675,000

Demand for missile system proliferates

Higher-than-expected growth in global military expenditure, combined with further order momentum from MENA and European markets provide strong upside to order and earnings momentum. We assume an 11.57% new order CAGR, 2024-33e, with a 12.47% sales CAGR in 2024-33e.

BASE CASE W600,000

Waiting for the next leg up

Domestic mass production program proceeds smoothly and additional MSAM orders from the MENA region support earnings momentum. We forecast a 10.5% new order CAGR, 2024-33e, with a 11.59% sales CAGR, 2024-33e.

BEAR CASE W450,000

Easing of global tensions leads to lower demand

Slower global defense spending leads to weaker incremental MSAM export orders and delays in LSAM mass production, thus dampening the earnings momentum. We assume a 9.8% new order CAGR, 2024-33e, with 9.76% sales CAGR, 2024-33e.

Risk Reward – LIG Nex1 (079550.KS)

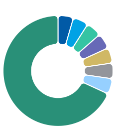
KEY EARNINGS INPUTS

Drivers	2024	2025e	2026e	2027e
New Order (Wbn) (W, bn)	3,737	8,000	8,300	9,000
Backlog (Wbn) (W, bn)	20,053	23,962	27,378	30,825
Revenue/Backlog (%) (%)	22.2	22.5	24.5	23.9
Overseas Revenue Growth (%) (%)	116.1	27.0	31.9	33.8
Gross Profit Margin (%) (%)	13.8	15.1	16.2	17.7

INVESTMENT DRIVERS

- Overseas order momentum for key product (M-SAM)
- Korean government's military force budget
- Global military expenditure

GLOBAL REVENUE EXPOSURE



- 0-10% Europe ex UK
- 0-10% India
- 0-10% Japan
- 0-10% MEA
- 0-10% Mainland China
- 0-10% North America
- 0-10% UK
- 70-80% APAC, ex Japan, Mainland China and India

Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

5/5 MOST	3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

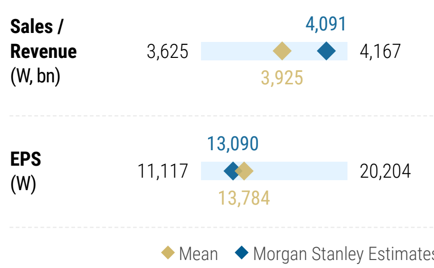
- Increase of domestic defense budget and overseas exports of weapon systems
- Surge in demand for air defense systems as the geopolitical environment deteriorates
- Greater increase in global defense spending as well as domestic defense budgets

RISKS TO DOWNSIDE

- Delays in weapon systems development and production
- Decrease in global defense spending and domestic defense capabilities' budgets

MS ESTIMATES VS. CONSENSUS

FY 2025e



Source: Refinitiv, Morgan Stanley Research

LIG Nex1: Waiting for the Next Boost

#1. Key beneficiary of government-led growth in advanced weapons:

Under the leadership of the Korean government, we expect sustained growth in the guided missile and unmanned-manned integrated systems sector. While the Russia-Ukraine conflict has temporarily increased demand for traditional ground weapons, the investment direction over the long term for the defense industry is shifting towards advanced weaponry such as guided missiles and electronic warfare systems. This trend is reflected in S.Korea's 2024-2028 Mid-Term Defense Plan, which shows an increasing share of the defense budget allocated to capability improvements and R&D, with a strong focus on missile defense capabilities. We expect these factors will work as a solid foundation for LIG Nex1's growth over the mid to long term.

Exhibit 94: S.Korea's 2024-28 Mid-Term Defense Plan: Key Capabilities Enhancement

Target	Note
Prioritize **(1) multi-layered missile defense** and **(2) unmanned system countermeasures**	
Korean Air and Missile Defense (KAMD)	- Deployment of Cheongung-2, PAC-3 missiles, and L-SAM systems to strengthen upper/lower-tier interception - Development of long-range artillery interceptors, M-SAM Blcok-3, and L-SAM Block 2 for integrated, layered defense
Counter-UAS Systems	- Operationalization of detection/identification systems with soft-kill capabilities (e.g., electronic warfare) to neutralize hostile drones

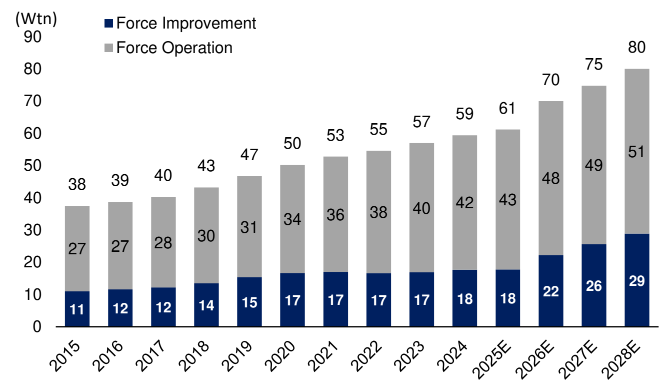
Source: Defense Acquisition Program Administration, Morgan Stanley Research.

Exhibit 95: S.Korea's 2024-2028 Mid-Term Defense Plan and Defense Budget Trend

(Wtn, %)	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025E	2026E	2027E	2028E
Total Defense Budget	37.5	38.7	40.3	43.2	46.7	50.2	52.8	54.6	57	59.4	61.2	70.0	74.8	80.0
Force Improvement	**11**	**11.6**	**12.2**	**13.5**	**15.4**	**16.7**	**17**	**16.6**	**16.9**	**17.7**	**17.8**	**22.3**	**25.6**	**28.9**
Force Operation	26.5	27.1	28.1	29.7	31.3	33.5	35.8	38	40.1	41.7	43.4	47.7	49.2	51.1
% of Total														
Force Improvement	29.3%	30.0%	30.3%	31.3%	33.0%	33.3%	32.2%	30.4%	29.6%	29.8%	29.1%	31.9%	34.2%	36.1%
Force Operation	70.7%	70.0%	69.7%	68.8%	67.0%	66.7%	67.8%	69.6%	70.4%	70.2%	70.9%	68.1%	65.8%	63.9%
Y/Y		3.2%	4.1%	7.2%	8.1%	7.5%	5.2%	3.4%	4.4%	4.2%	3.0%	14.4%	6.9%	7.0%
Force Improvement		5.5%	5.2%	10.7%	14.1%	8.4%	1.8%	-2.4%	1.8%	4.7%	0.6%	25.3%	14.8%	12.9%
Force Operation		2.3%	3.7%	5.7%	5.4%	7.0%	6.9%	6.1%	5.5%	4.0%	4.1%	9.9%	3.1%	3.9%

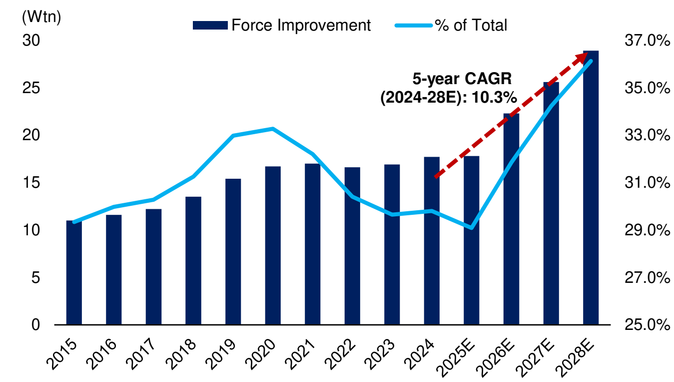
Source: Defense Acquisition Program Administration, Morgan Stanley Research. e = Korea Defense Acquisition Program Administration's Targets.

Exhibit 96: S.Korea Defense Budget Trend



Source: Defense Acquisition Program Administration, Morgan Stanley Research. e = Korea Defense Acquisition Program Administration's Targets.

Exhibit 97: S.Korea Force Improvement Budget Trend



Source: Defense Acquisition Program Administration, Morgan Stanley Research. e = Korea Defense Acquisition Program Administration's Targets.

#2. M-SAM contracts as a potential catalyst for L-SAM export opportunities

LIG Nex1 has won substantial orders for its medium-range surface-to-air missile (M-SAM), Cheongung, from UAE, Saudi Arabia and Iraq. In S.Korea, mass production of L-SAM system, designed to provide upper-tier air defense, has recently commenced following the completion of its development. A key aspect of air defense is having a dedicated missile system for each altitude layer, as integrating different missile systems into a single defense network can be challenging. Therefore, we expect that countries that have introduced M-SAM systems will likely adopt L-SAM.

#3. Strong order backlog supports future earnings growth

As of 2024, the company's order backlog stood at around W20tn, with exports accounting for approximately W10tn. Of this, W7tn is tied to Cheongung orders from UAE and Saudi Arabia, while W3tn is from other orders including for Bigung systems. If we include the additional W3.7tn Cheongung order from Iraq won in September 2024 but not included in 2024 new orders, the company's backlog would be robust (23.96tn in 2025E), in our view. Considering the typical lead time of 1-1.5 years between securing export contracts and recognizing revenue, we expect the full impact of sales to the Middle Eastern to begin to materialize from 2026, driving strong earnings momentum, particularly as margins improve. We expect LIG Nex1's OP to grow 35% and 32% in 2026 and 2027, respectively.

Exhibit 98: Overseas Backlog: Project Timeline

Weapon	Country	Date	Value (Wtn)	2024	2025E	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E
Overseas														
M-SAM (Chungoong)	UAE	1/18/2022	2.6											
M-SAM (Chungoong)	Saudi	2/7/2024	4.3											
M-SAM (Chungoong)	Iraq	9/20/2024	3.7											

Source: Company reports, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Investment positives and concerns

Investment positives

LIG Nex1 has become one of the top global players in the missile systems space, especially surface-to-air systems, owing to its strong and cost-effective product lineup.

(1) LIG Nex1 has been successful in selling its MSAM-II surface-to-air missile systems, with the UAE and Saudi Arabia making sizable new orders, which has led to a strong upward trend in the share price. We expect there to be follow-up orders from existing buyers and also for new buyers to emerge.

(2) Aside from M-SAM, other guided weapon systems include the portable SAM Shingung, anti-tank missile Hyeongung, anti-ship missile Bigung, ship-to-air missile Haegung and the high-altitude missile interception system (L-SAM) are expected to be exported.

(3) Aside from the PGM segment, the company expects to see revenue growth on the C4I unit owing to (1) second-phase mass production of the TMMR tactical radio system for domestic use, and (2) overseas contracts such as network infrastructure deployment in Indonesia.

Investment concerns

(1) A reversal by countries mainly in MENA and ASEAN, which are LIG Nex1's main customers, regarding their policy path to increase defense budgets. This could come about from many angles but the biggest risk would be a de-escalation of tensions in the respective regions.

(2) Competitors are able to expand their manufacturing capacities and/or become more aggressive in marketing their products to LIG Nex1's key markets.

(3) Low earnings visibility stems from the critical nature of missiles as defense systems and the frequent confidentiality of supply contracts, limiting the precision of quarterly earnings forecasts.

Share price analysis

LIG Nex1’s share price has outperformed the market since 2024. Its share price was up 69% in 2024 (vs. KOSPI: -10%) and is up 169% YTD (vs. KOSPI: 33%) as of July 23, 2025. The strong performance was driven by growing interest on defense names amid the current geopolitical landscape and top-line growth on the back of its competitive missile product line-up supported by (1) more attractive prices than peers (e.g., PAC-3) and (2) high quality.

LIG Nex1's subdued share price performance in 2024 compared to Hanwha Aerospace (+153.5% in 2024) stemmed from its reliance on offensive weapons, where opaque contracts and frequent small-scale orders (often undisclosed) limited major catalysts. This contrasts with peers such as Hanwha Aerospace and Hyundai Rotem, which secured large-scale contracts and saw substantial share price rallies. However, in 2025, the share price has surged 168.9% as of July 23, driven by anticipated earnings growth from M-SAM orders, with a typical lead time of 1-1.5 years between securing new contracts and the commencement of revenue recognition. Moreover, the Israel-Iran conflict in June initially fueled a sharp rise in the company's share price, although the subsequent stabilization following US intervention has brought valuations back to more reasonable levels, reflecting underlying business fundamentals.

The Israel-Iran conflict, characterized by long-range drone and missile strikes, heightened investor interest in missile manufacturers such as LIG Nex1. While geopolitical risks benefit defense contractors in guided weapons, systems such as L-SAM follow state-level procurement cycles, delaying earnings visibility. We recommend investors adopt a long-term view, considering L-SAM deployment and Korea's equivalent of Israel's Iron Dome. We believe the company's significant share price outperformance this year justifies our Equal-weight rating despite sector optimism.

Product overview

LIG Nex1 has four main business units, (1) Precision Guided Munitions (PGM), (2) Intelligence Surveillance Reconnaissance (ISR), (3) Avionics/Electronic Warfare (AEW), and (4) Command, Control, Communication, Computer, and Intelligence (C4I), which accounted for 47%, 14%, 12%, and 25% of revenue in 2024, respectively. Key export products in PGM include the mid-range surface-to-air missile system (M-SAM) Cheongung-II, with export contracts worth W10tn signed between 2022 and 2024 with Saudi Arabia, UAE and Iraq. Other guided weapon systems include the portable SAM Shingung, anti-tank missile Hyeongung, anti-ship missile Bigung, ship-to-air missile Haegung. Going forward, exports of high-altitude missile interception systems (L-SAM), which are entering mass production following the completion of system development in 2H25, are expected to target Middle Eastern countries, particularly those that have already imported M-SAM (Saudi Arabia, Iraq and UAE), in our view.

In addition to the PGM segment, the company is experiencing top-line growth in its C4I unit. Notable drivers include (1) second-phase mass production of the TMMR tactical radio system for domestic use, and (2) overseas contracts such as network infrastructure deployment in Indonesia. As defense systems become more digitalized and network-centric, the C4I segment is expected to continue expanding as a core contributor to the company’s long-term earnings growth, in our view.

Exhibit 99: LIG Nex1: Product Line-up

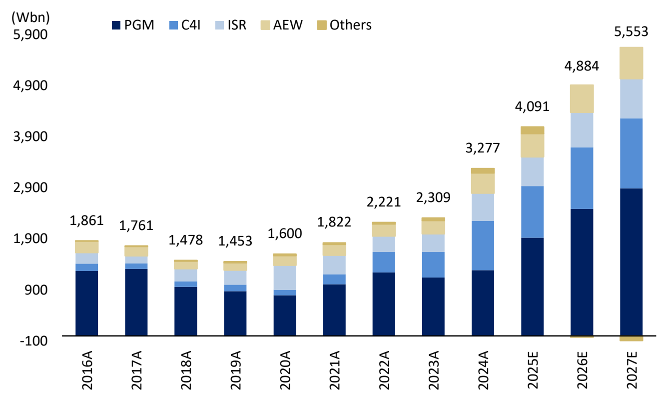
Weapon	Feature	Note	Exported to
PGM (Precision Guided Munition)			
M-SAM 2 (Chungoong)	Mid-range Surface to Air Missile Block-2)	Strengthened engagement and anti-ballistic capabilities	UAE, Saudi, Iraq
L-SAM	Long-range Air-to-Surface Missile	Mass production is scheduled to begin in 2025, with the goal of completing full operational deployment by 2032	Countries that have already introduced M-SAM system are expected to introduce L-SAM as well
K-SAAM	Surface-to-Air	Anti-aircraft weapon that simultaneously responds to various targets including aircraft, surface ships, as well as enemy anti-ship missiles	
Raybolt	Portable Anti-Tank Guided Weapon	A third-generation anti-tank missile for effective response to the threats of enemy tanks (portable/vehicle-mounted)	
Poniard	2-75 inch guided rocket	A guided weapon that accurately strikes multiple LCACs (Landing Craft Air Cushion) of an enemy landing on the coast at high speed / high performance at a lower competitive price	(Imminent) US
ISR (Intelligence Surveillance Reconnaissance)		Unmanned systems and future warfare, including micro-statellite system, long range radar, and dividion-level weapon locating radar	
AEW (Avionics / Electronic Warfare)		Combination of attack and defense technologies essential for electronic warfare, including KF-21 EW (electronic warfare) suite and LAH EW suite	
C4I (Command, Control, Communication, Computer, Intelligence)		Connection to battlefield to support fast and accurate judgment	Indonesia

Source: Company reports, Morgan Stanley Research.

LIG Nex1: Earnings Outlook

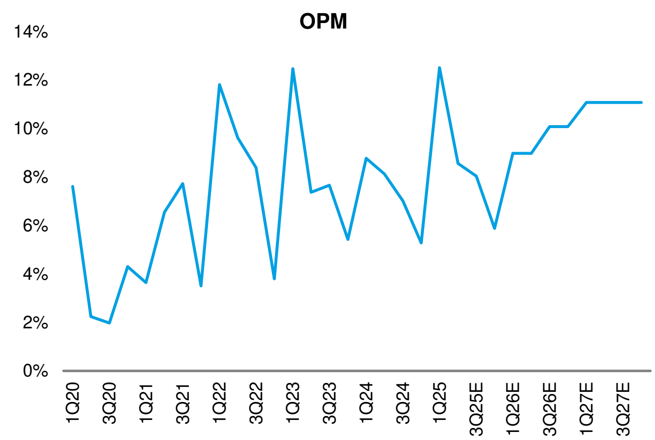
We believe the company will continue to see top-line growth as revenue contributions from UAE and Saudi M-SAM II exports meaningfully kick in starting next year. On the orders front, we anticipate that heightened air defense needs in the Middle East – largely from countries looking to diversify away from US reliance – will drive demand for M-SAM II. As a result, we estimate new orders to rise to W8.3tn in 2026 and W9tn in 2027. As M-SAM II exports ramp up, we expect operating profit margin to improve to 16.2% in 2026 and further to 17.7% in 2027, with delivery recognition ramping up for Egypt, UAE, and Saudi Arabia.

Exhibit 100: LIG Nex1: Revenue by Division



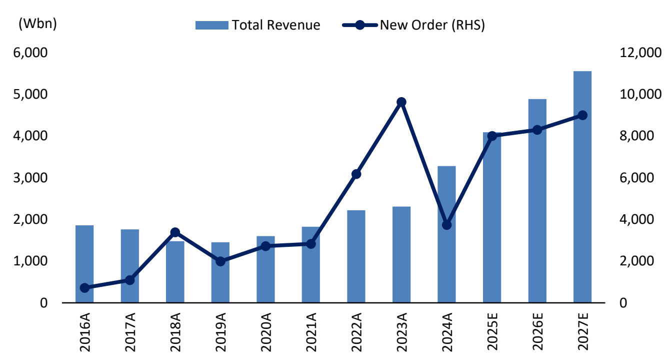
Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Exhibit 101: LIG Nex1: OPM Forecasts



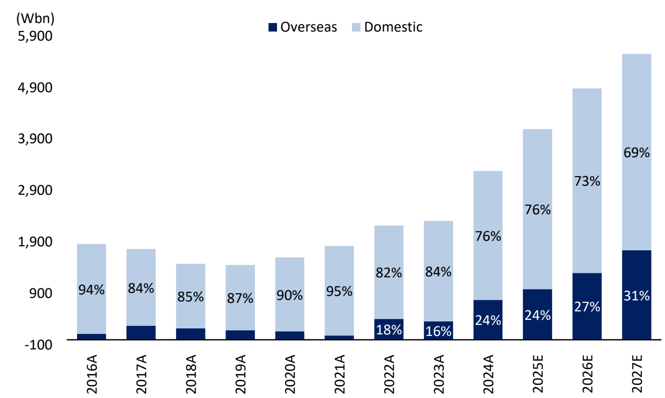
Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Exhibit 102: LIG Nex1: New Orders vs. Revenue



Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Exhibit 103: LIG Nex1 Revenue: Domestic vs. Overseas



Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

LIG Nex1: Valuation Methodology

Our price target (which is also our base case scenario value) of W600,000 is derived from EPM, assuming a cost of equity of 8% and a terminal growth rate of 3.5%. Our scenario analysis reflects different cases in (1) % of global military expenditure to GDP (3.3% vs. 3% vs. 2.7% for Bull, Base, Bear case scenarios analysis, respectively), and (2) global market share of Korean defense companies.

Our price target implies 32.6x 2026E P/E, which is lower than its current P/E of 38x, but higher than its peer average of 18.1x. We believe that upside potential remains from its successful MSAM-II overseas contracts. However, we believe much of this optimism is already priced in. Nevertheless, given expected future earnings growth and potential orders from MENA and Europe, trading at a premium vs. peers is justifiable, in our view.

Bull case: Our bull case scenario value is 13% higher than our base case. This scenario sees higher-than-expected growth in global military expenditure, combined with further order momentum from MENA and European markets. Here we assume consolidated sales rise at a 12.47% CAGR, 2024-33e, with overseas sales at a 18.1% CAGR, 2024-33e. We also assume (1) average OPM reaches 14% (2028-33E), and (2) 3.3% of global military expenditure to GDP, vs. 3% in base case, which leads to higher overseas revenue and results in higher revenue and margins vs. our base case. Under this scenario, our bull case value per share comes out to W675,000, which implies 36.6x 2026E bull case P/E.

Bear case: Our bear case scenario value is 25% lower than our base case. Here we assume 2.7% of global military expenditure to GDP vs. 3% in the base case, which leads to lower overseas land system revenue and results in lower revenue vs. our base case. Under this scenario, we assume consolidated sales rise at an 9.8% CAGR, 2024-33e, with overseas sales at a 13.8% CAGR, 2024-33e. Our bear case value per share comes out to W450,000, which implies 24.4x 2026E bear case P/E.

Exhibit 104: LIG Nex1: Bull/Bear/Base Case Scenario Analysis

	Value	26E P/E	EPS	Implied P/E	BVPS	Implied P/E	Implied P/B
Bull Price	**675,000**		20,732	32.6	71,238	36.6	10.7
Base Price	**600,000**	32.6	18,429	32.6	63,323	32.6	9.5
Bear Price	**450,000**		13,821	32.6	47,492	24.4	7.1

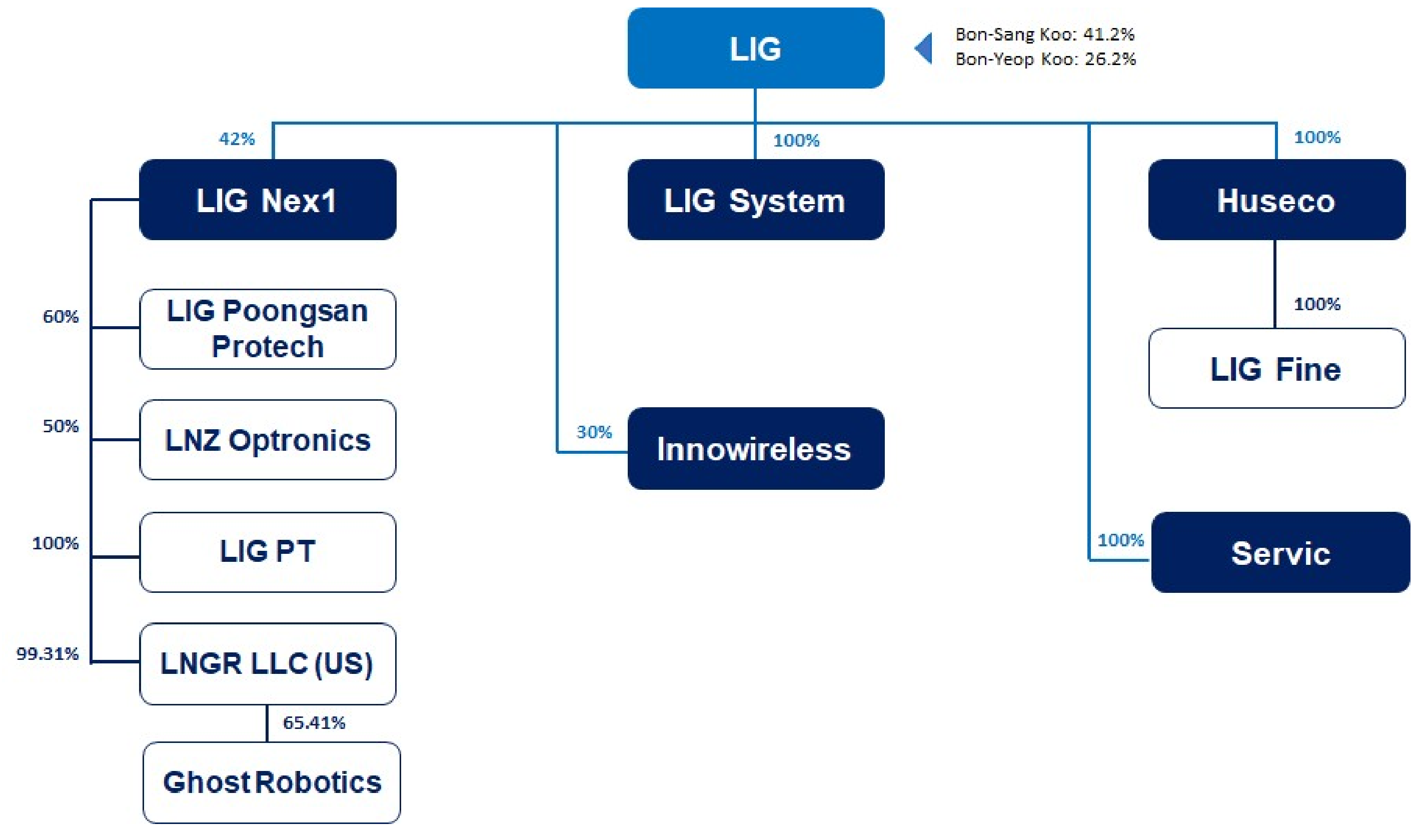
	Base	Bull	Bear
PV of Forecast Period	**2,947**	**3,367**	**2,406**
PV of Continuing Value	8,669	9,880	5,924
Equity Value (W bn)	12,853	14,485	9,567
- Net debt (W bn)	(230)	(230)	(230)
Equity Value (W bn)	13,083	14,715	9,798
No of shares (Mil)	**22**	**22**	**22**
Price Target (Won)	**600,000**	**675,000**	**450,000**

Source: Morgan Stanley Research estimates.

LIG Nex1: Company Background

LIG Nex1 was established in 2004 as a spin-off of LG Innotek's defense division under the name of Nex1 Future, and rebranded as LIG Nex1 in 2007. The company operates across four core segments: PGM, ISR, C4I, and AEW. As of 2024, exports accounted for approximately 24% of total revenue and an estimated 50% of order backlog.

Exhibit 105: LIG Nex1: Ownership Structure



Source: Company reports, Morgan Stanley Research. Note: Data as of 2024.

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Risk Reward Reference links

1. View explanation of Options Probabilities methodology - Options_Probabilities_Exhibit_Link.pdf

2. View descriptions of Risk Rewards Themes - RR_Themes_Exhibit_Link.pdf

3. View explanation of regional hierarchies - GEG_Exhibit_Link.pdf

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(as of June 30, 2025)

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1497	40%	372	45%	25%	702	41%
Equal-weight/Hold	1632	44%	380	46%	23%	776	45%
Not-Rated/Hold	4	0%	0	0%	0%	2	0%
Underweight/Sell	593	16%	79	10%	13%	233	14%
Total	3,726		831			1713	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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Joon Seok		
Hanwha Aerospace (012450.KS)	O (07/25/2025)	W941,000
Korea Aerospace Industries Ltd (047810.KS)	O (07/25/2025)	W92,600
LIG Nex1 (079550.KS)	E (07/25/2025)	W584,000

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